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SPECIFICATION — mdv6 (Markdown viewer, native macOS GUI + CLI launcher, Swift/SwiftUI)

- **Status:** v0.14.8 (2026-09-27). Every row is realised in this tree: the original’s post-v0.7 features (R-44...R-51) by the W10–W15 build, line citations (C-19) by W8; three rows carry *verification pending* for clauses only a person at an unlocked console can observe (§11). Changes per version, with their findings, are in the *Revision history* at the end. A plain §11 row is realised and verified; *not yet realised* marks specified work with no implementation; **open defect** marks known implemented behaviour that violates the normative row; *verification pending* marks a revised implemented path whose conformance has not yet been observed. Rows are never weakened to hide a defect.
- **Language / stack:** Swift 5.9 (SwiftPM, no Xcode project) | SwiftUI + AppKit | MarkdownUI 2.4.1 (cmark-gfm) · SwiftTreeSitter 0.25.0 (manifest floor 0.8.0) + eighteen vendored tree-sitter grammars · beautiful-mermaid-swift 1.0.4 (ELK layout) · SwiftMath 1.7.3 (vendored, patched) · MarkdownUI 2.4.1 vendored with one patch (I-016) · mermaid.js 11.4.1 in WebKit (WKWebView) for the diagram types the native library lacks (R-46) · SQLite (FTS5) | surfaces: macOS app bundle, bin/mdv6 shell launcher, make targets
- **Sources:** the original application, **tqbf/mdv** — its README and Help (user-facing behaviour), its NOTES (library gaps and their work-arounds), **TYPOGRAPHY.md** (theme conventions, carried into this repository), its code-block design notes, its vendored-SwiftMath README, its implementation, launcher, build script, Makefile, package manifest and CI workflow, and its git history through the v0.7 commit — and, for v0.14, through 68aa008 (its `Frontmatter.swift`, `FrontmatterView.swift`, `PrintController.swift`, `MermaidWebRenderer.swift`, `RawHTMLImages.swift`, `ScrollKeyMonitor.swift`, `DiffHighlighter.swift`, `Vendor/MarkdownUI/README.md`, the Help and README of that commit, and its `test-docs/` fixtures for those features); `SPEC_REVIEW_REPORT.md` (the sixth review, F-114..F-125; the five earlier reviews, F-001..F-093, were applied through v0.8 and are not carried); `reference/MDV-SCREEN.png` and `reference/MDV-ORIGINAL-SEVILLA.png` (the original application, normative for chrome structure — C-18); `reference/RECREATION-MDV6-SEVILLA.png` and `reference/RECREATION-MDV7-SEVILLA.png` (the two v0.8 recreations, shown as the gaps T-44..T-47 close). This repository is a from-scratch rebuild against this specification; none of the original’s source is used.
- **Scope of this document:** the observable behaviour of the mdv6 application and its launcher — file opening and closing, rendering (Markdown, frontmatter, code, diffs, Mermaid, LaTeX, tags), printing, navigation, find and search, history, bookmarks, persistence, theming, packaging and release. It does **not** specify the internals of the third-party renderers beyond the contracts mdv6 relies on, nor the colour and type values of individual themes (those live in **TYPOGRAPHY.md**). Since v0.9 it **does** specify the structure of the window chrome — title, toolbar, sidebar rows, inspector rows, badges and active states — because the v0.8 recreations showed that behaviour-only rows reproduce the function of the original and none of its look (§5.5, D-40).
- **Normative language:** MUST/MUST NOT/SHALL/SHALL NOT = normative; SHOULD = strong recommendation; MAY = optional.
- **Principle:** *Native and honest*. Every pixel is drawn by AppKit/SwiftUI/CoreText — no

WebView, no JavaScript bridge — with **one** bounded exception: a Mermaid diagram of a type the native library does not speak is drawn by a bundled, hash-pinned mermaid.js inside a WKWebView that loads nothing else and reaches no network (R-46, C-06.5, D-46). When a renderer cannot handle an input, the user sees the source, never a blank.

0. Intent and purpose

mdv6 is a macOS application for *reading* Markdown. It renders a .md file the way a good document viewer renders a PDF: typographically deliberate, fast to open, with the navigation aids a long technical document needs (table of contents, in-document find, cross-file full-text search, bookmarks, back/forward history) and without any editing surface. It is meant to be the default handler for .md files on a developer's Mac, reachable from Finder, from the terminal (mdv6 FILE), and from other Markdown files via links.

The renderer is a pipeline of native components: cmark-gfm (via MarkdownUI) for Markdown, tree-sitter for code syntax highlighting, an ELK-based layout library for Mermaid diagrams, and a CoreText math typesetter for LaTeX. Each of those libraries has gaps relative to what real documents contain (Mermaid.js syntax, amssymb,
 in labels, ...); mdv6 owns a **sanitising and repair layer** in front of each one so that documents written for GitHub or Mermaid.js render faithfully, and a **fallback rule** so that anything the layer cannot repair is shown as its source text with an explanation rather than dropped.

Non-goals. mdv6 does not edit Markdown (it hands off to an external editor); it does not render HTML beyond what cmark-gfm passes through as text, with the single exception of tags (R-51); it does not fetch remote images unless the user opts in; it does not sync anything off the machine; it is not sandboxed for the App Store (it reads arbitrary user files and installs a CLI symlink).

Trust boundary. Everything the application processes — file contents, link targets, image URLs, Mermaid and LaTeX source — is untrusted document content. The only runtime network activity is the user-enabled remote-image fetch governed by R-16 and C-16; the release pipeline (§10) and the first build's download of the pinned mermaid.js (C-13) are separate. The application never executes document content: the Mermaid web path (R-46) runs the bundled mermaid.js, which reads the diagram source as data under **securityLevel: 'strict'**, in a web view whose every load other than the generated page is refused (C-06.5).

1. Actors and goals

Actor	Goals
Reader (human, GUI)	Open Markdown files by any route, read them with good typography, move around them quickly, find text within and across files, keep places, and get back to files read before.
Terminal user (human, bin/mdv6)	Open one or more files, a directory, or stdin from a shell in the running application; query the version; install the launcher once.
Finder / LaunchServices (com.mdv6.app document-type registration)	Route double-clicks, drag-onto-icon, and open -a of .md/.markdown/.mdown/net.daringfireball.markdown/public.pl items into the application (C-01).
External editor (any macOS app the reader chooses)	Receive the current file on E; its saves are picked up by the live-reload watcher.
Document author (indirect)	Their GitHub-flavoured Markdown, Mermaid.js diagrams, and LaTeX math render as they would on GitHub / mermaid.live, or degrade visibly.
Release engineer (human, make dist, CI)	Produce a signed, notarised, stapled .zip from an exact vX.Y.Z tag; CI builds every push to main and publishes a rolling latest prerelease.

Actor	Goals
Persistence store (SQLite at ~/Library/Application Support/mdv6/mdv6.db, UserDefaults)	Durably hold history, the full-text index, bookmarks, per-file scroll positions, and preferences across launches.

2. Requirements (intent, high level)

Sources are cited as [Help §...], [README], [NOTES], or a file or symbol in the original's tree (**tqbf/mdv**) — where the behaviour was read from, not where it is realised here (§11 names this tree's files).

2.1 Opening and loading

ID	Statement
R-01	The application MUST open a Markdown file from every one of: File → Open... (O), Open in New Window... (O), a LaunchServices open event (Finder double-click, drag onto the icon, <code>open -a</code>), a file dropped onto the window, a Markdown link clicked inside a document (any local path, per R-19), a history-sidebar row, a search hit, a bookmark, the placeholder (R-28), and <code>←/ →</code> (R-18). Routes are of two kinds: adding routes — O, O, LaunchServices, drop, link, bookmark, placeholder, and the directory scan (R-02) — add the file’s history row or move an existing one to the top (R-20) and index it (R-26); selecting routes — a history-sidebar row, a search hit whose file is already in history, <code>←/ →</code>, Next / Previous File (R-48), and the delete-current-row transition (§3.1; swipe-delete or W, R-47) — display an entry the list already holds and MUST NOT reorder history or re-index. All routes MUST load the file into the target window’s content view only, where the target is the key window when it is a document window, else the frontmost visible document window in window order, else any document window (a <i>document window</i> is one hosting the article view; SwiftUI’s helper windows are not) — the key window is absent while the application is inactive, the usual case for <code>open -a</code> from a terminal or a Finder double-click; a second window (only O creates one) MUST NOT react to a menu command or an open event addressed to another window (E-26). A LaunchServices open event MUST NOT activate the application or order a window front: LaunchServices activates it for an ordinary open, and a background open (<code>open -g</code>) MUST leave the frontmost application frontmost while the file loads into the target window; only when the application is already active does the target window become key and move to the front. No document window (the last one closed by W or its close button, R-47; the application keeps running): an open event, File · Open.../Open in New Window..., a bookmark slot or row, O, and Help · mdv6 Help MUST create exactly one window — as a launch does, but without R-40’s history restoration — which is then the target and receives the file, and MUST create it only once the action has a file to load: O shows its open panel application-modal first and creates no window when the panel is cancelled; O and a bookmark check their target file first and, when there is no placeholder or the file is missing, beep (E-09, E-27) without creating a window (F-178); a click on the Dock icon MUST create one window that follows R-40; every other window-scoped command (W, P, F, D, E, <code>←/ →</code>, Next / Previous File, O, zoom) is disabled or, where §5.1 says it is always enabled, does nothing (E-38). When one open event carries several URLs (mdv6 A.md B.md, a multi-file <code>open -a</code>), each MUST be added to history in the order received and the last MUST be displayed. [Help \$Opening files; mdv6/mdv6App.swift application(_:open:), DocumentWindows.frontmost (commit 700d3de); ContentView.loadFile (adding), List(selection:) / openHit / applySnapshot / stepFile (selecting)]
R-02	When the opened path is a directory, the application MUST consider the directory’s non-hidden, readable files whose extension (case-insensitive) is <code>md</code>, <code>markdown</code>, or <code>mdown</code>, ordered by case-insensitive localized comparison of the filename; it MUST load README.md (case-insensitive match on the stem) if present, else the first file in that order, and MUST add every other such file to history as rows (not displayed, but indexed per R-26; primary first, the rest in that order) (R-20). [Help \$Opening files; ContentView.loadDirectory]
R-03	Only the first item of a drop is considered (further items are ignored without error); it MUST be accepted only when its extension (case-insensitive) is one of <code>md</code>, <code>markdown</code>, <code>txt</code>, <code>mdown</code>, <code>mkd</code>; other drops MUST be ignored without error. [ContentView.handleDrop]

ID	Statement
R-04	The file MUST be read and decoded as UTF-8 before any history change; a file that is not valid UTF-8, or that vanishes between the existence check and the read, is unreadable and the load MUST abort per E-03 — no history row, no selection change, no watcher re-arm. A file that decodes to zero bytes or only whitespace is readable: it MUST display an empty article with the file selected and watched, never the EMPTY panel (§3.1). One leading U+FEFF (a UTF-8 byte-order mark) MUST be removed after decoding and before C-02 runs, so a BOM-prefixed file splits, recognises its frontmatter (C-20.1) and counts lines (C-02 rule 8) like the same file without one. The document MUST be split into blocks once per load (C-02) and every per-frame consumer (rendering, find, TOC, bookmarks) MUST read the cached split, never re-parse. [<code>ParsedDocument</code>; commit 38df878; <code>ContentView.loadFile/loadCurrentEntry</code>]
R-05	While a file is displayed, the application MUST watch it by path (not by open file descriptor or inode) and reload its content when the file changes on disk — a plain write, an atomic save that renames a temporary file over it, or a delete-and-recreate MUST all trigger a reload of the new content. Change events are delivered with a 50 ms latency and no deferral: the first event of a burst is delivered at once and later events within 50 ms are batched into at most one further delivery, so a burst yields at most two reloads and the content read last is the content displayed (K-06). A reload MUST keep the reader’s scroll position (clamped to the new block count); any text selection is not preserved. A read that fails (file deleted, moved away, or not valid UTF-8 — a half-written save) MUST be ignored: the window keeps its content, shows no error, and keeps the watch armed for the path until a later event yields a readable file. A read that returns zero bytes while content is displayed MUST be treated as a truncate-then-write in progress: the page is kept and the file is re-read after 500 ms, and whatever that second read yields (including empty) is shown (E-21, D-18). [<code>FileWatcher</code> (FSEvents on the parent directory); <code>Help \$Editor integration</code>]
R-06	On load, the application MUST restore the reader’s last scroll position for that path (C-08) when the stored anchor still resolves (E-08); otherwise it MUST start at the top. It MUST persist the current position on window close, on quit, and before loading a different file into the window — except when the outgoing file’s history row is being removed (swipe-delete, <code>W</code>, <i>Close All</i>, cap eviction; R-26, R-47), which persists nothing, so no position outlives its row. [<code>ContentView.persistScrollPosition</code>]
R-47	File · <i>Close File</i> (<code>W</code>) MUST close the displayed file exactly as swipe-deleting its history row does (R-20, R-26, §3.1): the row, its index row and its C-08 scroll position are removed (bookmarks are kept), every snapshot of that path is dropped from both stacks (R-18), and the new first history row loads as a selecting route, or the window enters EMPTY when no rows remain. Closing is not a navigation: it MUST NOT push a back snapshot, and the closed file’s scroll position MUST NOT be written back after its row is gone. <i>Close File</i> is disabled while the window is EMPTY. File · <i>Close Window</i> (<code>W</code>) MUST close the target window (§3.1 CLOSED); the application keeps running. File · <i>Close All</i> (<code>W</code>) is disabled when history is empty; otherwise it MUST first ask for confirmation — title <i>Close all files?</i>, message <i>This clears the sidebar history and the search index.</i>, buttons <i>Close All</i> (destructive) and <i>Cancel</i> — and on <i>Close All</i> MUST remove every history row (with its index row and scroll position, as R-26 removes one; bookmarks are kept), clear both navigation stacks, and put every window into EMPTY (E-36); <i>Cancel</i> changes nothing. These three items replace the system File · <i>Close</i>. [<code>Help \$Opening files</code>; <code>ContentView.closeFile/requestCloseAll/closeAllFiles</code>; commit d263b15]

2.2 Rendering

ID	Statement
R-07	Markdown MUST be rendered as GitHub-flavoured Markdown (cmark-gfm: tables, task lists, strikethrough, autolinks, footnotes) using the active theme’s typography (R-29). [README; ThemeManager.markdownTheme]
R-08	Fenced code blocks MUST be syntax-highlighted with tree-sitter for the languages in K-05 (with the alias map in C-05), and MUST render as plain monospaced text — never an error — for any other or missing language hint, except the diff/patch fence words, which R-50 colours line by line. The block MUST show a language label, a hover-revealed toolbar (wrap toggle, copy), and a context menu; blocks whose fence word is in the prompt-aware set of C-05 and whose non-empty lines are at least half $\$/\#$ -prompted MUST additionally offer <i>Copy Without Prompts</i> , whose output is the block with the leading $\$$ or $\#$ removed from each prompted line and every other line copied unchanged (output lines are kept; line count is preserved). [CodeRenderer; CodeBlockChrome.copyWithWithoutPrompts]
R-09	A <code>```mermaid</code> fence MUST render as a diagram image drawn natively (C-06). The block MUST offer: a style menu (Document, Light, Dark, Tokyo Night, Catppuccin — the choice persisted document-wide in <code>mdv6.mermaid.style</code>), <i>Show Mermaid source</i> (toggles to a monospaced source view — no Mermaid grammar exists, so it is not syntax-coloured), <i>Export diagram as PNG</i> , copy source, and pinch-to-zoom between $0.5\times$ and $4\times$. The style menu, <i>Export diagram as PNG</i> and pinch-to-zoom apply to the native path only; a diagram dispatched to the web path (C-06.4, R-46) offers <i>Show Mermaid source</i> and copy, and its hover toolbar and context menu — the block’s menu, never WebKit’s (C-06.5) — omit the style menu and the export item (the web path honours only the theme’s light/dark choice, C-06.5). [Help \$Diagrams and math; MermaidCodeBlockChrome; commits b48a28d, a2368f0]
R-10	Before parsing, Mermaid source MUST be sanitised per C-06.1 so that the Mermaid.js constructs listed there render; after layout, the corrections in C-06.2 MUST be applied. A diagram is dispatched by its first directive line (C-06.4): the six families the library speaks go to the native pipeline, every other diagram type to the web path (R-46). A diagram that fails on its path — the library cannot parse or lay out a natively dispatched source, or mermaid.js rejects a web-dispatched one (C-06.5) — MUST render the fallback: the text “Mermaid diagram could not be rendered” and the source in monospace. [NOTES \$Mermaid; MermaidFallbackView]
R-11	A natively dispatched diagram (C-06.4) MUST be rasterised at the exact width from the §7.2 formula — its natural width or the available column width minus 36 pt, floored to whole points and bounded below by 1 pt — at the backing scale of the screen the window is on (as built: <code>NSScreen.main</code> , the screen of the key window). It MUST be re-rasterised when that width, committed zoom, or backing scale changes; the raster cache key includes all three. It MUST NOT be drawn wider than its natural width. [NOTES \$Resolution; MDVMermaidDiagramView]
R-12	LaTeX math delimited by $\$...\$$ (inline) and $\\\$...\$\\$ (display) MUST be typeset natively with SwiftMath in every block type — paragraphs, headings, list items, blockquotes, table cells — following the delimiter rules in C-07. Display math on its own paragraph MUST be centred and MUST offer <i>Copy LaTeX</i> in its context menu. [Help \$Diagrams and math; MathRenderer]
R-13	Math inside an ATX heading MUST be sized by that heading’s em factor (C-09); elsewhere by the body size times the zoom factor (R-30). Math colour MUST be the theme’s text colour. [MathMarkdown.rewrite]

ID	Statement
R-14	LaTeX that SwiftMath rejects MUST render as its source ($\dots$ delimiters included) in monospace; a display block MUST additionally show the parser’s message. Before typesetting, the command rewrites and symbol registrations of C-07.2 MUST be applied. [MathImageCache.typeset; MathSymbols]
R-15	A node label in a Mermaid flowchart or state diagram that is exactly one $\dots$ span MUST be typeset with SwiftMath and composited centred in the node at the same pixel weight as document math (I-009). Math mixed with text, and math in edge labels, MUST be rendered as the Unicode approximation of C-07.3. [NOTES $\S$ LaTeX in labels]
R-16	Images MUST resolve data: URIs inline and relative paths against the document’s directory. http(s) images MUST NOT be fetched unless View $\rightarrow$ <i>Load Remote Images</i> is on; when on, the fetch MUST obey C-16, and turning the preference off MUST cancel in-flight remote-image requests. When off, a clickable “Remote image blocked” placeholder MUST be shown instead. A missing local image MUST show an “image not found” placeholder naming the file. These rules apply alike to a block image (the image alone in its paragraph) and an inline image (one in a sentence, a list item, or a table cell): an inline relative path MUST resolve against the document’s directory and load from disk, never through the network loader (commit 68aa008), and an inline http(s) image is gated by the same preference. An inline image cannot carry its own click action inside a line of text, so a blocked inline remote image MUST draw, in its place, an inline image of the text <i>Remote image blocked</i> (secondary text colour, sized to the line’s font) and is not clickable — the View-menu reveal belongs to the block placeholder; an inline remote image that fails once loading is on draws <i>Image failed to load</i> the same way; a missing inline local image is left out of the line. Every local, data-URI, and remote image MUST obey K-14 and MUST NOT be scaled above its intrinsic size. [LocalImageProvider; mdv6App View menu]
R-17	When View $\rightarrow$ <i>Smart Typography</i> is on and the active theme allows it, prose blocks MUST be rendered with curly quotes, en/em dashes, and ellipses per C-10; fenced/inline code, GFM table blocks, thematic-break lines, link URLs, and $\langle \dots \rangle$ spans MUST be left verbatim. Math spans MUST be rewritten to image references <i>before</i> smartening so LaTeX is never altered. [SmartTypography.swift; ContentView.blockView]

ID	Statement
R-44	A document whose first line is exactly <code>---</code> or <code>+++</code> and which closes that fence per C-20 carries a frontmatter (metadata header). The header MUST be block 0 of the C-02 split — one block, whatever blank lines it holds — and MUST render as a two-column properties table (C-20.3), never as prose: CommonMark would otherwise read the opening <code>---</code> as a thematic break and the closing one as a setext underline. Values are plain text — no Markdown, math, smart typography or <code></code> rewriting applies inside the header. View · <i>Show Frontmatter</i> (default on, persisted as <code>mdv6_show_frontmatter</code>, C-04) MUST hide the table when off; hiding is display-only — the header stays block 0, so block indices, bookmarks, scroll anchors, line citations and find address the same blocks either way (I-018). A candidate that fails C-20’s recognition rules MUST render exactly as it would with no frontmatter support (E-32). Block 0 as a header (F-158): hidden, it occupies no height, no block padding and no stack spacing, so the first body block sits where it would in a document without a header; it is never the D / 0 anchor (R-27, R-28) — the next block is taken instead; a line citation into it (C-19) scrolls to the top and flashes nothing. Shown, the table is a block like any other: it takes the C-18.7 hovered-block stripe, its values are text-selectable as prose (R-22), it is followed by the theme’s <code>paragraphBottomSpacing</code> as a GFM table is, and a citation into it flashes it. Either way, a bookmark or placeholder whose anchor is block 0 (possible when shown, or from a bookmark made before hiding) is titled Frontmatter, never by the header’s first line (R-27). A document whose only block is a hidden header has no anchor: D and 0 beep and add nothing (F-174). [Help <code>\$Frontmatter</code>; <code>Frontmatter.swift</code>, <code>FrontmatterView.swift</code>; commits 9d80903, 617e49e, d3e0b0e]
R-46	A <code>```mermaid</code> fence whose first directive line (C-06.4) names a type the native library does not speak — <code>gantt</code>, <code>pie</code>, <code>timeline</code>, <code>journey</code>, <code>quadrantChart</code>, <code>requirementDiagram</code>, <code>mindmap</code>, <code>gitGraph</code>, <code>flowchart-elk</code>, and any other — MUST render as a diagram through the bundled <code>mermaid.js</code> in a <code>WKWebView</code> per C-06.5, instead of the R-10 fallback. The web path MUST load nothing but the generated page and the bundled script, MUST make no network request, MUST pin <code>mermaid.js</code> by version and SHA-256 (C-13), MUST use mermaid’s dark theme on a dark application theme and default otherwise, MUST paint its background in the theme’s code-block background, and MUST fall back per R-10 when <code>mermaid.js</code> rejects the source, produces no SVG, or reports no positive height. While the first measurement is pending the block shows a small spinner at its 60 pt minimum height (K-19); for accessibility the block is one element labelled <i>Mermaid diagram</i>, hinted <i>Use Show Mermaid Source to view the diagram code</i>. The K-14 Mermaid ceiling applies before either path. [Help <code>\$Diagrams</code> and <code>math</code>; <code>MermaidWebRenderer.swift</code>; commits a02b2f6 ... 7b36bd2; [D-04] (#D-04), [D-46] (#D-46)]
R-50	A fenced block whose fence word (C-05: the info string’s first word, lower-cased) is diff or patch MUST be coloured line by line per C-05.1 instead of by tree-sitter: added lines in the palette’s diff-add colour on its diff-add background, removed lines likewise in the diff-remove pair, hunk headers (<code>@@ ... @@</code>) italic in the theme’s secondary text colour, file and commit metadata lines semibold, <code>\ No newline at end of file</code> notes italic in the palette’s comment colour; the <code>+/-</code> column stays visible. The block keeps the R-08 chrome (label diff/patch, wrap, copy). [Help <code>\$Rendered Markdown</code>; <code>DiffHighlighter.swift</code>; commit f2999eb]

ID	Statement
R-51	A raw HTML <code></code> tag in a prose block — which cmark-gfm would otherwise show as literal text — MUST render as the image it names, per C-22 : <code>src</code> resolved and gated exactly as a Markdown image’s destination (R-16 , C-16 , K-14), drawn at the size its <code>width/height</code> attributes ask for (as caps, never enlarging), with <code>alt</code> as its accessibility label. Tags inside fenced blocks and inline code spans MUST stay literal, as MUST every other HTML tag. [Help <code>\$Rendered Markdown; RawHTMLImages.swift</code> ; commits 3006049, 5c4e1b6, 68aa008]

2.3 Navigation and copying

ID	Statement
R-18	The application MUST maintain per-window back/forward stacks whose entries are (history entry , top block index). Loading a different file pushes the outgoing document’s entry and clears the forward stack except when the load is initiated by a bookmark (R-27), the placeholder (R-28), or a cold-start file argument (R-40); those three routes MUST NOT push, whether their target is in the current file or another file. <code>←</code> pops a live snapshot, pushes the current view onto the forward stack, loads the file as a selecting route (R-01 : history order and index untouched), and scrolls to the saved block; <code>→</code> is the mirror. When a history row is removed (swipe-delete, cap eviction), every snapshot holding that entry MUST be dropped from both stacks or skipped when popped, so <code>←</code> never displays a document that has no history row. A same-document jump from a slug #fragment link (R-19) or a TOC row (R-21) MUST push a snapshot and clear the forward stack; a line citation (R-19 , C-19) is a position move and MUST NOT push or clear either stack (C-19.4); find stepping (R-24) MUST NOT push. Re-opening the current path pushes nothing. [Help <code>\$Moving around; NavSnapshot, pushSameDocSnapshot</code>]

ID	Statement
R-19	A clicked link is handled in two steps. Resolution: a destination with no scheme is resolved by path arithmetic against the current document’s directory; a file: URL’s path is used as it is (the Markdown renderer resolves relative destinations against the document’s base URL before the click reaches the application, so a relative link normally arrives already in this form — F-117); any other scheme is not a path. Classification: the application MUST navigate in-app when the resolved path is an existing local file whose extension is, case-insensitively, md, markdown, or mdown; every other destination — a missing file, another extension, a file: URL to a non-Markdown file, a custom scheme — MUST be handed to the system opener, because a click is an explicit user action. A fragment MUST be percent-decoded exactly once as UTF-8 before comparison; invalid percent encoding has no matching slug. Fragment kinds. A fragment is one of two kinds, tested in this order: a line citation — the C-19.1 grammar — which selects a <i>line range</i>; otherwise a slug fragment, compared per C-11. A fragment that is neither (empty, or an unmatched slug) behaves as before. The line form wins when it parses, so a heading slug that happens to look like 110 is unreachable by fragment — an accepted consequence, since GitHub emits the same form and resolves it the same way. A line citation MUST scroll to the block resolved by C-19 and flash it (C-19.3); a same-document line citation MUST NOT push a snapshot or select a TOC row (it is not a heading choice, unlike D-41’s rule for slug fragments), while a cross-file line citation loads the target as an adding route and suppresses R-06 restoration exactly as a slug fragment does. A same-document slug fragment MUST scroll to the first matching C-11 slug or do nothing when none matches (E-06). A local path plus slug fragment MUST load the target as an adding route, suppress R-06 scroll restoration, then scroll to its first matching slug; if no slug matches, the loaded file MUST remain at the top. Only C-02 single-line ATX #-### headings are slug targets (E-22, D-17). [ContentView.handleClick]
R-20	The history sidebar MUST list every file opened by an adding route (R-01), most recently added first — a selecting route leaves the order unchanged — capped at 100 entries (the oldest is evicted), persisted across launches; a row MUST support swipe-to-delete. The sidebar MUST be collapsible (S, View menu, hover chevron) with the collapsed state persisted, and resizable by dragging its divider between 180 and 400 pt. [HistoryManager; Help \$\$Sidebars]
R-21	The inspector MUST show a table of contents of the document’s single-line ATX #, ##, ### headings (C-02), each row jumping to its block, with a search field that filters rows; and a collapsible bookmarks pane with a draggable height. The inspector is shown and hidden by View · Show/Hide Inspector (0) and the toolbar’s sidebar.right button (§5.1); its visibility and width (180–520 pt, dragged at its left edge) MUST persist. Heading text in the TOC MUST show math as Unicode (C-07.3), not as LaTeX source. [Help \$\$Sidebars; commit bcd2150]
R-22	Prose blocks MUST support standard macOS text selection (drag to select; C copies the rendered text through the system pasteboard). A TOC heading block — a block listed in tocHeadings (C-02 rule 7: single-line ATX #-###); an h4–h6 or setext heading is prose for every rule in this row (E-22) — MUST NOT be text-selectable; the pointer over one MUST be the pointing hand; a click on one (a tap without drag — modifier keys are not distinguished) MUST copy that heading’s section (C-12) as Markdown source to the pasteboard and flash the section for 0.6 s; a repeated click copies again and restarts the flash. There is no block-level selection model (it was removed in commit c50817a). [commit c50817a; ContentView.copySection, BlockTextSelection]

ID	Statement
R-23	E MUST open the current file in the chosen external editor; <i>File</i> → <i>Edit</i> → <i>Choose Editor...</i> picks one and <i>Forget Editor</i> clears it; with no editor set, E MUST prompt to choose. [Help <code>\$Editor integration</code>]
R-48	Navigate · <i>Next File</i> (<code>] </code>) and <i>Previous File</i> (<code>[</code>) MUST display the history row immediately below or above the displayed one, as a selecting route (R-01: history order and index untouched, so stepping never hoists a file to the top mid-walk), pushing a back snapshot exactly as a sidebar-row click does (R-18). They MUST stop at the ends of the list (no wrap) and are disabled at the corresponding end and while the window is EMPTY . and MUST perform Next and Previous File respectively in the target window; only a Tab key held with Control is consumed — an unmodified or -only Tab still reaches the focused control (find bar, search fields). [Help <code>\$Moving around; ContentView.stepFile, FileStepMonitor; commit 07658bb</code>]
R-49	The document pane MUST scroll from the keyboard without the reader first clicking into it, per K-18: ↓/↑ by a line step; Page Down/Page Up and Space/ Space by a page; Home/End to the top and bottom of the document. The keys act on the window they are typed into, only when no <code>,</code> or <code>^</code> modifier is held, and only while the window's first responder is nothing, the window itself, or a non-editable text view (a selectable prose block); any other first responder — a text field, the history list, a keyboard-focused button — keeps its keys. A handled key MUST NOT beep, push a back snapshot, or change the TOC selection. Scrolling is clamped at both ends by the scroll view's own bounds, including its content insets under the full-size title bar; <i>End</i> re-aims at the bottom on later frames until the lazily laid-out document stops growing (K-18). [Help <code>\$Moving around; ScrollKeyMonitor.swift; commit be885e5</code>]

2.4 Find, search, bookmarks

ID	Statement
R-24	F MUST open an in-document find bar. An empty query MUST produce no matches, show “No matches”, and disable stepping; a non-empty query, including whitespace-only input, MUST be matched verbatim as a case-insensitive substring over each block’s source (no trimming, no diacritic folding — unlike C-03). A reload (R-05) while the bar is open MUST recompute <i>m</i> and return to the first occurrence. <i>m</i> counts occurrences in document order, and the bar MUST show “<i>n</i> of <i>m</i>” or “No matches”. G / G MUST step per occurrence, wrap at the ends, and scroll the occurrence’s block into view; Esc MUST close. A matching block MUST be tinted as a whole when it is a code fence (``~/~~), a \$\$ math fence (C-02 rule 3), a GFM table (first line contains ; the second line, trimmed, is non-empty, contains at least one -, and consists only of -, :, , space), or contains ![anywhere. Every other matching block MUST be inline-highlighted: the block is re-rendered as inline text (leading #, >, and ordered-list markers stripped, -/*/+ bullets shown as •, inline Markdown interpreted, math shown as \$...\$ source) and every occurrence of the query in that text is marked, so an occurrence inside markup can be counted but unmarked, and vice versa (E-17). All occurrences in the current match’s block share the stronger tint; the current occurrence is not otherwise distinguished from siblings in that block (D-21). An inline-highlighted block MUST keep its theme typography — face, size, weight, colour and leading of its heading level or body text, the h1/h2 rule, and the inline code, strong, emphasis, strikethrough and link styles — per C-09.1, so a block does not change size or face when a match lands in it. A frontmatter header (R-44) with a match always takes the inline-highlight path — whatever the tint tests above would say (a header may contain ![or a \ -and-rule pair) — and is shown verbatim — block 0’s source lines exactly as written, fence lines (---, +++, ...) included, so what is displayed is what <i>m</i> counts — with no inline-Markdown interpretation, in the body style of C-09.1, even when <i>Show Frontmatter</i> is off: a match inside a hidden header shows the header’s source with the match marked for as long as the find bar is open, so find never lands on a block that draws nothing. When the sidebar has focus, F MUST route to global search (E-18). <i>The sidebar has focus</i> when the window’s first responder is a view inside the history sidebar — the history list, its revealed search field, or a search-hit row; a freshly opened window has no first responder, so its first F opens the find bar. [Help \$Find; ContentView find, shouldInlineHighlight, highlightedAttributedString]
R-25	F MUST focus a search field that queries the full-text index of every file in history (C-03): tokens are prefix-matched and ANDed; results (at most 80) MUST show the filename and a snippet with matched terms highlighted; choosing a result MUST open the file. [Help \$Find; Database.search]
R-26	The application MUST index a file’s content into the full-text index when it is added to history by an adding route (R-01: opened, or seeded as a directory sibling by R-02 — a selecting route does not re-index) and re-index history on launch, skipping any file whose modification time (whole seconds, K-06) is unchanged since its last indexing. Removing a file from history — swipe-delete or eviction by the 100-entry cap — MUST remove its row from the index (and its C-08 scroll position; bookmarks are kept), and launch MUST drop any index row whose path is not in history, so the search population is exactly the current history. File · <i>Close All</i> (R-47) removes every row at once by the same rule. [Database.indexFile/removeFile; HistoryManager.add/remove]

ID	Statement
R-27	D MUST add a bookmark at the block under the pointer if one is hovered, else at the topmost block whose frame intersects the viewport — except that a hidden frontmatter header is never the anchor and the next block is taken instead (R-44); titled by the nearest TOC heading (C-02 rule 7) at or within the previous 40 blocks, using its display text (inline Markdown stripped per C-12, math per C-07.3); else — for block 0 when it is a frontmatter header, the title Frontmatter (R-44) — the block’s first line with inline Markdown stripped, truncated to 60 extended grapheme clusters (K-06); else (line <i>n</i>) with <i>n</i> the 1-based block index when the stripped line is empty; (empty) only when the document has no blocks. The anchor is block index and fingerprint (C-08). Bookmarking the same block twice creates two rows. Bookmarks MUST persist in order; the first five MUST be bound to 1... 5; rows MUST be reorderable by drag and removable. Opening a bookmark MUST load its file if needed and scroll to the resolved anchor without pushing a back-snapshot (R-18). [Help \$Bookmarks; BookmarksManager]
R-28	0 MUST set a transient in-memory placeholder — anchored by the R-27 rule (hovered block, else topmost visible block, never a hidden frontmatter header; index and fingerprint) and recording the file path — and 0 MUST return to it, loading that file first (adding route, R-01) if another is displayed; 0 with no placeholder, or whose file no longer exists (E-27), MUST beep (§5.1); the placeholder MUST NOT survive relaunch and does not push a back-snapshot (R-18); <i>Clear Placeholder</i> in the row’s context menu (C-18.9) MUST remove it, after which 0 beeps. While set, the placeholder MUST be visible as the first row of the bookmarks pane (R-21), above a divider, titled by the R-27 title rule for its block (Frontmatter for a block-0 header, R-44) and carrying the 0 badge; clicking the row is 0; setting it MUST expand a collapsed bookmarks pane and mark the row <i>current</i> (C-18.9). [Help \$Bookmarks; setPlaceholder, jumpToPlaceholder, placeholderRow]

2.5 Appearance and preferences

ID	Statement
R-29	The reader MUST be able to choose one of the nine named themes or <i>System</i> from the toolbar; <i>System</i> MUST resolve to high-contrast in Light appearance and twilight in Dark and MUST switch live when macOS appearance changes. Each theme MUST restyle the article pane, code palette, and the diagram <i>Document</i> style; the choice MUST persist (mdv6_theme_id). [ThemeManager; TYPOGRAPHY.md]
R-30	Let <i>s</i> be the stored zoom factor and <i>d</i> be +0.10 for = or −0.10 for -. Each step MUST set $s' = \text{clamp}(\text{roundHalfAway}(10s)/10 + d, 0.60, 2.50)$; roundHalfAway rounds a half-integer away from zero. View → Actual Size MUST set $s' = 1.0$. The factor MUST persist (mdv6_font_scale) and MUST scale body text, document math (R-13), inline code, and fenced code blocks (C-05: fence text at $0.85 \times \text{base} \times s'$). The zoom HUD MUST appear after the change for 0.9 s and show $\lfloor 100s' + 0.5 \rfloor$ %. [ThemeManager fontScale; CodeRenderer.render]
R-31	? (Help → mdv6 Help) MUST open the bundled Help.md , copied (overwritten) to ~/Library/Application Support/mdv6/Help.md on every ? so the copy matches the running build and has a stable path for history and bookmarks. Because each copy changes the file’s mtime, the C-08 scroll position for Help is never restored across ? invocations, and a ? while Help is displayed triggers a reload (R-05). [HelpManager]
R-32	Every preference in C-04 MUST persist via UserDefaults under the listed key and MUST be honoured on the next launch.

ID	Statement
R-42	The window chrome MUST follow C-18: the window title is the displayed file's name; the toolbar is the five icon buttons of §5.1; the history sidebar and the inspector carry the section headers, row anatomy, hover/selection states, hotkey badges and reveal-on-click search fields of §5.5; a document rendered block-by-block keeps the vertical rhythm of TYPOGRAPHY.md at every block boundary (I-014); a single-line own-paragraph \$\$...\$\$ is centred like a multi-line one (C-07.1). Colours and type come from the active theme (C-09, TYPOGRAPHY.md); the structure does not vary by theme. [reference/*.png; ContentView.sidebar/tocPane/bookmarksHeader/bookmarkRow/toolbarContent in the original]

2.6 Launcher, packaging, diagnostics

ID	Statement
R-33	bin/mdv6 MUST implement the surface in §5.2: locate the app bundle per the documented search order, open files/directories by absolute path, read stdin into a temporary .md for -, print the bundle version for --version, and exit 1 with mdv6: no such file: <path> on stderr for a missing argument. [bin/mdv6]
R-34	make (default) MUST build a runnable build/mdv6.app from a clean checkout with only the Swift toolchain — and, on the first build only, network access to fetch the pinned mermaid.js, whose digest it verifies (C-13) — copying every resource the app needs (C-13); make install MUST place it in /Applications, register it with LaunchServices, and symlink the CLI. Every make dist invocation MUST refuse before building unless HEAD carries an exact vX.Y.Z tag; a command-line VERSION value MUST NOT bypass or replace that tag. [Makefile; build.sh]
R-35	The application MUST NOT print document content, file contents, or query strings to any log at any verbosity — including the logs of the WebKit processes that host its Mermaid pages (C-06.5). The only diagnostics it emits are NSLog lines prefixed [mdv6] — persistence-store failures (E-12, may name the file path), bundled-font registration failures, and an external-editor launch failure (R-23, whose error text may include the file path) — and the font-registration lines SwiftMath prints once per font on first use.
R-36	For every input admitted by K-14, the application MUST NOT terminate because of document content. A repair-layer or third-party parse/layout failure MUST degrade to the fallback of R-10/R-14/C-14, and every path that reaches a third-party parser MUST be preceded by the applicable validation and sanitisation. Inputs rejected by K-14 MUST follow E-28 without entering a third-party parser. [NOTES \$Mermaid: ELK layout asserts]
R-37	The repository MUST carry an automated test suite runnable with swift test from a clean checkout, covering at least the pure contracts (C-02 block split, C-03 query construction, C-07.1 delimiters and C-07.3 plain text, C-08 fingerprint/resolve, C-10 smart typography, C-11 slugs, C-12 sections) and the Mermaid/LaTeX sanitisers (C-06.1, C-07.2), and CI MUST run it on every push to main and on every pull request. [Tests/mdv6Tests, Tests/mdv6RenderTests; .github/workflows/build.yml; D-01, §9.0]

ID	Statement
R-38	Fenced code blocks tagged <code>swift</code> and <code>sql</code> MUST be syntax-highlighted with tree-sitter like the languages of K-05: the <code>tree-sitter-swift</code> and <code>tree-sitter-sql</code> grammars (parser, scanner, and a <code>highlights.scm</code>) vendored under <code>mdv6/Grammars/</code> , pinned in its README, compiled into <code>CGrammars</code> , and resolved from the fence hints in C-05 (<code>swift</code> ; <code>sql</code> , <code>sqlite</code> , <code>postgresql/postgres</code> , <code>mysql</code> , <code>plsql</code> , <code>tsql</code>). Highlighting quality MUST match the existing languages: keywords, strings, comments, numbers, types, and function names each map to a palette capture. [mdv6/Grammars/swift, mdv6/Grammars/sql; D-15]
R-43	Fenced code blocks tagged <code>cpp</code> (or <code>c++</code> , <code>cplusplus</code> , <code>cc</code> , <code>cxx</code> , <code>cp</code> , <code>hpp</code> , <code>hxx</code> , <code>hh</code> , <code>metal</code> , <code>msl</code>), <code>json</code> , <code>lua</code> , <code>opencl</code> (or <code>cl</code> , <code>opencl-c</code>), <code>perl</code> (or <code>pl</code> , <code>perl5</code>) and <code>markdown</code> (or <code>md</code> , <code>gfm</code>) MUST be syntax-highlighted with tree-sitter like the languages of K-05: the <code>tree-sitter-cpp</code> , <code>tree-sitter-json</code> , <code>tree-sitter-lua</code> , <code>tree-sitter-opencl</code> , <code>tree-sitter-perl</code> and <code>tree-sitter-markdown</code> grammars (parser, scanner where the grammar has one, and a <code>highlights.scm</code>) vendored under <code>mdv6/Grammars/</code> , pinned in its README, compiled into <code>CGrammars</code> , and resolved from the fence hints in C-05. Highlighting quality MUST match R-38 — keywords, strings, comments, numbers, types and function names each map to a palette capture — with the C-05 caveat that a capture name no <code>CodePalette</code> entry covers renders in the plain colour; the vendored queries are upstream's except where <code>mdv6/Grammars/README.md</code> lists a derivation. [mdv6/Grammars/{cpp,json,lua,opencl,perl,markdown}; D-45]
R-39	The repository MUST contain the render harness and corpus specified by C-17: <code>tools/render-harness/</code> MUST be a SwiftPM executable that links the application's pipeline code rather than copying it; <code>test-docs/mermaid/*.mmd</code> MUST contain one licence-cleared raw Mermaid diagram per file; and <code>test-docs/render-cases.json</code> MUST enumerate the deterministic render checks. The C-17 commands MUST make T-13, T-17, and T-19 reproducible from a clean checkout. [tools/render-harness; test-docs/mermaid, test-docs/render-cases.json; D-01]
R-40	On launch with no file argument, the main window MUST attempt exactly the first history entry (the most recently added path, R-20) and restore its scroll position (R-06). If that entry is unreadable, it MUST remain in history and the window MUST enter <code>EMPTY</code> ; later rows MUST NOT be tried automatically. A cold-start file argument (<code>LaunchServices</code> , <code>bin/mdv6 FILE</code>) MUST pre-empt automatic history restoration, display that file, and MUST NOT create a back-stack snapshot for the unseen history head (R-18). The window MUST otherwise enter <code>EMPTY</code> only when history is empty. Automatic window-state restoration MUST be disabled (E-30): a launch creates exactly one window regardless of how many were open at the previous quit. [ContentView onAppear; application(_:open:)]
R-41	Before reading, parsing, laying out, downloading, or decoding untrusted document content, the application MUST enforce the applicable K-14 ceiling. Content over a ceiling MUST follow E-28; admitted content MUST remain subject to R-36. Remote fetches MUST additionally obey C-16. [ContentLimits; ImageLoading]

2.7 Printing

ID	Statement
R-45	File · <i>Print...</i> (P) MUST print the displayed document through the standard macOS print panel — whose PDF menu is the product’s Save-as-PDF — by re-rendering every C-02 block through the same pipeline the screen uses (C-21). Printed text MUST be vector glyphs; a standalone display formula and every inline formula that SwiftMath accepts MUST print as vector glyphs (a rejected one prints its E-10 fallback, which may be a bitmap) — no image object is drawn inside a formula’s rectangle except, for an inline formula, one fully transparent 1×1-pixel spacer that reserves its place in the line (F-184) — except in a block where C-21.4 cannot place the formulas, whose inline formulas then print as bitmaps at 12 px/pt (E-33; T-53 requires no such block in <code>test-docs/math.md</code>); a Mermaid diagram MUST print as a diagram, vector where C-21.3 obtains a PDF of it, and a fence that fails to render MUST print as its source in a plain code block, never as an empty box. Page breaks MUST fall between blocks, except inside a block taller than the printable page height (C-21.5). The printed page MUST use the high-contrast theme whatever the screen theme, a type scale derived from the paper’s column (C-21.1) and not from the screen zoom, soft-wrapped code, no chrome (no hover toolbars, find tint, stripe, flash or selection), and no remote fetch (remote images print as the blocked placeholder). Frontmatter prints as its table when <i>Show Frontmatter</i> is on and not at all when it is off. P with no document or one with no blocks (R-04: a zero-byte or whitespace-only file), or while a print session is already open, MUST beep and do nothing else. [Help <code>\$Printing</code>; <code>PrintController.swift</code>; commits 27f7449, 800e941 ... 5c4e1b6]

3. Behavior and state model

3.1 Document lifecycle

A window holds at most one **current document**. Its states and transitions:

State	Meaning	Enters via	Leaves via
EMPTY	No file loaded; the drop target / Open... prompt is shown. History MAY be empty or may retain an unreadable initial head (R-40). A launch creates exactly one window (E-30).	launch with empty history; failed load of the initial history head with no prior document (R-40, E-03); swipe-delete or W of the last history row while it is displayed (R-20, R-47); <i>Close All</i> confirmed in any window — every window enters EMPTY (R-47, E-36)	any open route (R-01) → LOADING ; swipe-delete of a retained history row (the unreadable head of R-40, or any row while no document is displayed) → stays EMPTY with the row removed

State	Meaning	Enters via	Leaves via
LOADING	Outgoing document's scroll position persisted when one exists and its row is not being removed (R-06); file read and decoded (R-04), split into blocks (C-02); adding route only: history row added or moved to the top (R-20) and file indexed (R-26); selecting route: history and index untouched; scroll anchor looked up unless R-19 suppresses it.	adding or selecting route (R-01); launch with non-empty history (R-40, selecting); swipe-delete or W of the displayed row with other rows remaining (R-20, R-47, selecting); Next / Previous File (R-48, selecting)	success → VIEWING (an empty file is a success, R-04); unreadable file → the previous state (VIEWING of the prior document, or EMPTY if there was none), with no history or selection change (E-03). An unreadable initial history head therefore leaves the row in history and enters EMPTY (R-40).
VIEWING	Blocks rendered lazily; watcher armed on the path (R-05); find/TOC/bookmarks operate on the cached split. In-flight renders (diagram layout, math) are cancelled when the document changes.	LOADING	open of another file → LOADING; file changed on disk with a readable read → RELOADING; a read that fails or returns a transient empty (E-21) → stays VIEWING; file deleted → stays VIEWING (E-21); swipe-delete or W of the displayed history row → LOADING of the new first history entry (selecting route; the deleted entry's position is not persisted, its row being gone, and no snapshot is pushed; the deleted entry's snapshots dropped, R-18), or EMPTY when no rows remain (R-20, R-47); <i>Close All</i> confirmed → EMPTY (R-47); window close (W or the close button) → CLOSED; print (R-45), keyboard scrolling (R-49) and the <i>Show Frontmatter</i> toggle (R-44) → stays VIEWING

State	Meaning	Enters via	Leaves via
RELOADING	New content replaces <code>rawMarkdown</code> in place; scroll position kept; text selection not preserved. Transient empty/undecodable reads are ignored (R-05, E-21).	watcher event, coalesced 50 ms; a further watcher event while RELOADING is processed exactly as one during VIEWING — a readable read replaces the in-flight reload and completes into VIEWING, an unreadable one is ignored and the state is unchanged	→ VIEWING
CLOSED	Scroll position persisted (R-06); watcher cancelled.	window close, quit	terminal

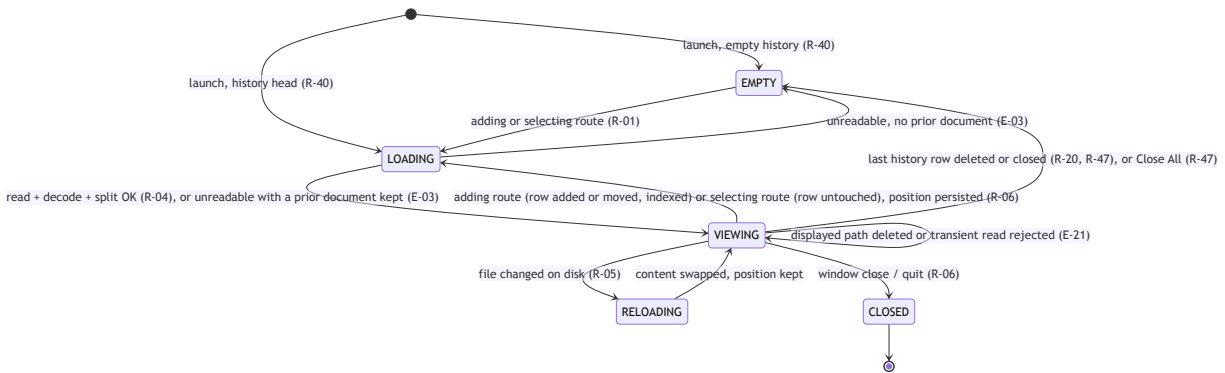


Figure 3.1 — document lifecycle per R-01, R-04..R-06, R-20, R-40, R-47, E-03, E-21. The table is normative; the diagram is illustrative.

The table above is **total**: every state-and-event pair the application can be in is named, and a pair it does not name is unreachable from that state (a launch while VIEWING, a watcher event while EMPTY, a load outcome while RELOADING). No row leaves a reachable pair unspecified. (F-140) The v0.14 commands are placed by the same rule: W and Next / Previous File are disabled in EMPTY (R-47, R-48), P in EMPTY beeps (R-45), and *Close All* from EMPTY with rows retained (R-40) removes them and stays EMPTY. After the last window is CLOSED the application keeps running with no document window; the next open event, O, bookmark, 0, ? or Dock click creates a window that starts this lifecycle afresh (R-01, E-38).

3.2 Render pipeline for one block

Every visible block goes through the same path on each render (the split itself happens once per load, R-04):

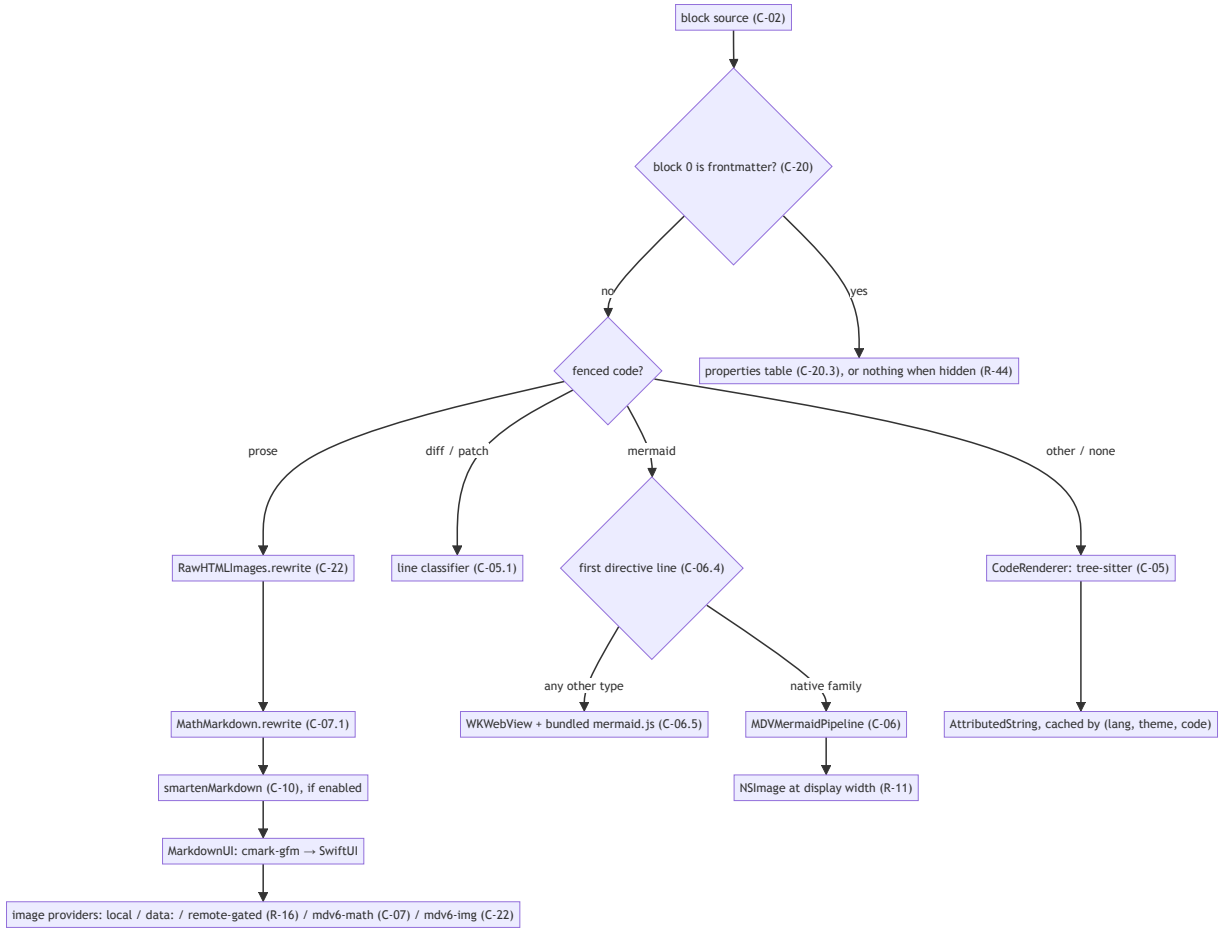


Figure 3.2 — per-block render path per [R-07..R-17](#), [R-44](#), [R-46](#), [R-50](#), [R-51](#). While the find bar has a match in a block, that block takes the [R-24](#) inline-highlight path instead, frontmatter included. Each edge corresponds to a §4 contract; the diagram is illustrative.

Order matters in two places and is normative: ** rewriting precedes math rewriting** (C-22), so a \$ inside an **src** or **alt** never opens a math span; and **math rewriting precedes smart typography** (R-17), so that --, ..., and quotes inside \$...\$ are never curled or dashed. Printing (C-21) runs the same order.

3.3 Durable artifacts

Artifact	Location	Written when	Read when
History list	<code>UserDefaults["mdv6_history"]</code> (JSON per C-15, ≤ 100 entries)	adding-route open (R-01), swipe-delete, cap eviction	launch
Full-text index	<code>mdv6.db</code> tables <code>articles</code> , <code>articles_fts</code> (C-03)	every adding-route open (mtime-gated), launch re-index; rows removed on swipe-delete and eviction (R-26)	F search
Bookmarks	<code>mdv6.db</code> table <code>bookmarks</code> (C-08)	D, reorder, remove	launch, Bookmarks menu, inspector
Scroll positions	<code>mdv6.db</code> table <code>scroll_positions</code> (C-08)	window close / quit / file switch (R-06); row removed with the history row (R-26)	file load

Artifact	Location	Written when	Read when
Preferences	UserDefaults keys in C-04	on change	launch
Help file	~/Library/Application Support/mdv6/Help.mdbundle, R-31	every ? (overwritten from the	?
Render caches	in-memory only: code AttributedString (256 entries), math images (2048), Mermaid layouts (96) and rasters (192, ≤ 192 MB)	render	render

mdv6.db MUST be opened with `SQLITE_OPEN_FULLMUTEX`, `journal_mode = WAL`, `synchronous = NORMAL` ([I-006](#)). Its `meta` table holds `schema_version` (currently 4); `migrate()` MUST apply forward migrations by comparing it, each migration's statements and the version bump inside **one** transaction (`BEGIN IMMEDIATE ... COMMIT`), so a crash mid-migration leaves the previous schema and version intact. A migration whose statement fails is rolled back and logged ([E-12](#)); the next launch retries it.

4. Interfaces / contracts

C-01 Application bundle and document types

```
mdv6.app/
  Contents/Info.plist      CFBundleIdentifier com.mdv6.app, LSMinimumSystemVersion 13.0,
                           CFBundleShortVersionString 1.0.0
                           CFBundleDocumentTypes: extensions [md, markdown, mdown];
                           LSItemContentTypes [net.daringfireball.markdown, public.plain-text]
  Contents/MacOS/mdv6     SwiftPM executable
  Contents/Resources/     AppIcon.icns · *.otf (Alegreya, Besley, OpenDyslexic) ·
                           *-highlights.scm (9) · mathFonts.bundle/ (Latin Modern Math + plist)
                           · mdv6 (CLI script, for "Install Command Line Tool...") · Help.md
                           · mermaid.min.js (11.4.1, SHA-256-verified at build, C-13; R-46) · mermaid
Entitlements: app-sandbox = false; files.user-selected.read-only = true
```

C-02 Document split: ParsedDocument

```
struct ParsedDocument {           // computed once per load (R-04); equality on `raw`
  let raw: String
  let blocks: [String]           // see rules
  let blockLines: [Range<Int>]   // half-open 1-based source lines per block; definition: rule 8
  let lineCount: Int             // lines in `raw` after normalization (C-02 rule 6)
  let tocHeadings: [TOCHeading] // level 1...3, from each block's first line (rule 7)
  let frontmatter: [FrontmatterRow]? // rule 9: non-nil exactly when blocks[0] is a C-20 header
}
struct TOCHeading { level: Int; text: String /*display*/; slugText: String /*for #fragment*/; blockIndex: Int }
```

Split rules (normative):

1. Input is split on `\n`. A **blank line** (only whitespace) ends the current block.
2. A line whose first non-space characters are ````` or `~~~` opens a **fence**; blank lines inside a fence do not split; the fence closes at the next line starting (after spaces) with the same three-character marker — *as built, the closing run is not required to be at least as long as the opener* (a deviation from CommonMark; see [E-23](#)). An unclosed fence runs to the end of the input.

3. A line whose first non-space characters are `$$`, with no second `$$` on the same line, opens a **math fence**; it closes at the next line *containing* `$$`, or at the end of the input.
4. Indented code blocks (four spaces) are **not** recognised by the splitter: a blank line inside one splits it into two blocks (E-23).
5. Leading/trailing newlines of a block are trimmed; empty blocks are dropped.
6. Line endings are normalised before splitting: `\r\n` and lone `\r` MUST be treated as `\n`, so a CRLF document yields the same blocks and TOC as its LF equivalent.
7. `tocHeadings` contains each block whose trimmed text starts with `#`, `##`, or `###` and is not a fence, using its first line only; a block MAY span more than one line and only its first line carries the marker (rule 1 makes the one-line form the ordinary case). `text` and `slugText` are computed from the heading **body** — that first line with its leading `#` run and the single space after it removed — so the marker never reaches the TOC, a bookmark title, or a slug; `text` is the body with inline Markdown stripped (C-12 rules) and math converted per C-07.3, while `slugText` is the same without the math conversion. (F-142, F-143)
8. `blockLines[i]` is **half-open**: `lowerBound` is block `i`'s first source line (1-based) and `upperBound` is one past its last, so a block on lines 5–7 is `5.. < 8` and line `n` lies in block `i` exactly when `blockLines[i].lowerBound ≤ n < blockLines[i].upperBound`. `lineCount` is the number of lines the normalized input splits into (rule 6; a trailing `\n` does not create a final empty line). The map derives from the same split as `blocks`, so `blocks[i]` is `raw`'s lines `blockLines[i]` with rule 5's trimming applied. Both are computed once per load and stored on the document, so a line→block lookup (C-19) is $O(\log |\text{blocks}|)$ and answers at most once per distinct `raw` (I-004). Resolution is **total**: a line no block contains — one past `lineCount`, inside the run of blank lines a rule-5 drop or a C-07.1 insertion removed, or before the first block — resolves to the **last** block whose `lowerBound ≤ n`, or to the **first** block when no block satisfies that (the line precedes every block); a document with no blocks resolves to nothing.
9. **Frontmatter first**. After rule 6's normalisation and before rule 1, the input is tested for a C-20 header. When one is found, the header — its opening fence line through its closing fence line, no trailing newline — is `blocks[0]`, whatever blank lines it contains, and rules 1–5 split only the text after the closing fence's line break; `blockLines[0]` spans lines `1.. < (c + 1)` where `c` is the closing fence's line, and every later block's lines are counted in the whole document, so rule 8 and C-19 are unchanged. `frontmatter` holds the header's C-20.2 rows and is nil when there is no header. Because a header opens with `---/+++`, it is never a rule-7 TOC heading. A header with no blank lines of its own **and followed by a blank line** splits into the same `blocks[0]` and the same later blocks either way, so bookmarks and scroll anchors (C-08) written before v0.14 still resolve; when body text follows the closing fence with no blank line, the pre-v0.14 split had one block where rule 9 makes two, every later index shifts by one, and such an anchor resolves by fingerprint where it can and otherwise lands one block early (C-08's clamped-index fallback) — an accepted one-time migration cost (F-166).

C-03 Full-text index

```
CREATE TABLE articles (id INTEGER PRIMARY KEY, path TEXT NOT NULL UNIQUE, filename TEXT NOT NULL,
    content TEXT NOT NULL DEFAULT '', indexed_at INTEGER NOT NULL,
    file_mtime INTEGER NOT NULL DEFAULT 0, file_size INTEGER NOT NULL DEFAULT 0);
CREATE VIRTUAL TABLE articles_fts USING fts5(filename, content, path UNINDEXED,
    content='articles', content_rowid='id', tokenize='unicode61 remove_diacritics 2');
-- triggers keep articles_fts in step with INSERT/UPDATE/DELETE on articles
```

Query construction: split the input on whitespace; drop the characters " () : * ^ from each token; discard tokens that become empty; wrap every survivor as "token"; join with spaces (FTS5 implicit AND). Dropping is **deletion, not escaping**: the six characters are removed wherever they occur, so `a"b` and `ab` produce the same term and a query cannot distinguish the two (F-147, accepted — the six characters are FTS5 syntax and every survivor is a quoted literal prefix). A query with no surviving tokens performs no search and yields no results. Results MUST use `ORDER BY rank ASC, path COLLATE NOCASE ASC, path ASC LIMIT 80`, with `snippet(articles_fts, 1, char(2), char(3), '...', 14)` — U+0002/U+0003 bracket matched terms and the UI renders them highlighted. The two path keys make equal-rank ordering deterministic, in-

cluding the 80-row boundary.

C-04 Preferences (UserDefaults)

Key	Type	Default	Meaning
mdv6_theme_id	String	high-contrast	Theme id or system (R-29)
mdv6_font_scale	Double	1.0	Zoom factor (R-30), clamped on read
mdv6_smart_typography	Boolean	true	View → Smart Typography (R-17)
mdv6_load_remote_images	Boolean	false	View → Load Remote Images (R-16)
mdv6_sidebar_collapsed	Boolean	false	History sidebar hidden (R-20)
mdv6_inspectorVisible	Boolean	false	TOC/bookmarks inspector shown (R-21)
mdv6_inspectorWidth	Int	240	Inspector width, clamped to [180, 520]
mdv6_bookmarksExpanded	Boolean	false	Bookmarks pane open
mdv6_bookmarksHeight	Int	240	Bookmarks pane height, clamped at use (K-04)
mdv6_editor_bundlePath	String	""	External editor bundle path (R-23)
mdv6_history	Data (JSON)	[]	History entries (R-20, C-15)
mdv6.mermaid.style	String	document	Diagram style (R-09; native path only, R-46)
mdv6_show_frontmatter	Boolean	true	View → Show Frontmatter (R-44)

A stored value of the wrong type, outside its range, or not in its enumeration falls back to the default at use: an unknown `mdv6_theme_id` resolves to **high-contrast** (the picker shows no selection until the reader picks one); `mdv6_font_scale` is clamped on read and each step applies the R-30 round-half-away formula; widths and height are clamped per K-04; an unknown `mdv6.mermaid.style` reads as **document**.

C-05 Code highlighting: CodeRenderer

```
func render(code: String, languageHint: String?, theme: MDVTheme) -> AttributedString // synchronous,
```

- Language resolution: lower-case the info string, keep its first word; direct names `c go rust bash javascript yaml toml python ruby (+ swift sql per R-38)` and `cpp json lua openssl perl markdown (R-43)`; aliases `js jsx javascriptreact node → javascript`, `sh zsh shell → bash`, `python3 → python`, `rb → ruby`, `yml → yaml`, `rs → rust`, `golang → go`, `h objective-c objc → c`, `sqlite postgresql postgres mysql plsql tsql → sql (R-38)`, `c++ cplusplus cc cxx cp hpp hxx hh metal msl → cpp` (a metal fence is the C++ grammar — D-45), `cl openssl-c → openssl`, `perl5 → perl`, `md gfm → markdown`; anything else → plain.
- Prompt-aware fence words (R-08 *Copy Without Prompts*): the **raw** first word of the info string, lower-cased, is one of `bash sh zsh fish shell console` — a separate test from language resolution (`fish` and `console` highlight as plain yet are prompt-aware; `shell-session` is not).
- Highlighting: parse with a fresh `Parser` per call, run the grammar's `highlights.scm`, colour each capture from the theme's `CodePalette` by capture-name components; `comment` captures are italic. If the query fails to compile, that language falls back to plain for the rest of the session. Fence text is set in the system monospace face at $0.85 \times \text{baseFontSize} \times \text{the zoom factor (R-30)}$.
- Markdown is the one language highlighted by two grammars (R-43): the block grammar's captures are applied first, then every (`inline`) span it leaves whole is re-parsed with `tree-sitter-markdown-inline`

and its captures are applied on top, so an inline capture wins on its own range. The embedded query failing to compile leaves the block captures standing — it is not a `plainForSession` case.

- Result cache: key (`language`, `diff flag`, `theme id`, `zoom factor`, `hash(code)`), at most 256 entries (flushed whole when full).
- **Diff fences (R-50)**. A fence word `diff` or `patch` selects the C-05.1 line classifier and bypasses tree-sitter; it is tested on the same lower-cased first word as language resolution, and neither word is a K-05 language.

C-05.1 Diff line classification. Split the fence body on `\n` (empty lines kept). Keep two counters, *o* and *n* (old and new lines left in the current hunk), both 0 at the start. For each line, in order:

1. If *o* > 0 or *n* > 0 (inside a hunk body): a line starting + is **added** (*n* −= 1); − is **removed** (*o* −= 1); `\` is a **note**; anything else — a space, or an empty line whose leading space was stripped — is **context** (*o* −= 1, *n* −= 1).
2. Else, if the line is a hunk header — split on spaces into at most four fields, the first is `@@`, the second is `-1[,s]` and the third `+1[,s]` with every part an integer — it is a **hunk header**, and *o*, *n* are set to the two counts *s* (an omitted *s* is 1).
3. Else a line starting `---` or `+++` is **meta**; otherwise a line starting + is **added**, − **removed**, `\` a **note**, a space or an empty line **context**, and anything else **meta** (so `diff --git`, `index ...`, and commit headers are meta, and a snippet without `@@` headers still colours its `+/-` lines).

The counters are what tell a removed `-- comment` line (`--- comment` inside a hunk) from a `--- a/file` header. Styling: every line starts in the palette’s plain colour and the system monospace face (C-05 size); **added** → foreground `diffAdd` on background `diffAddBg`; **removed** → `diffRemove` on `diffRemoveBg` (the background covers the line’s characters, not the line’s full width); **hunk header** → the theme’s `secondaryText`, italic; **meta** → semibold; **note** → the palette’s `comment` colour, italic; **context** unchanged. The four diff colours are palette fields, by theme (sRGB hex):

Theme	<code>diffAdd</code>	<code>diffAddBg</code>	<code>diffRemove</code>	<code>diffRemoveBg</code>
high-contrast (and every light theme without its own palette)	<code>#417F37</code>	<code>#DAFBE1</code>	<code>#CF222E</code>	<code>#FFEBE9</code>
any dark theme without its own palette (oneDark)	<code>#98C379</code>	<code>#1E3A2B</code>	<code>#E06C75</code>	<code>#3E1C20</code>
sevilla	<code>#4F7138</code>	<code>#E3E5C9</code>	<code>#8C2A1A</code>	<code>#F0DCD0</code>
charcoal	<code>#7EE787</code>	<code>#0E2B1A</code>	<code>#FF7B72</code>	<code>#3D1416</code>
solarium-dark	<code>#519900</code>	<code>#EAE9CD</code>	<code>#DC322F</code>	<code>#F0D8D2</code>
solarium-mint	<code>#65D900</code>	<code>#0F2E1A</code>	<code>#DC322F</code>	<code>#3A1817</code>
phosphor	<code>#CFCFCF</code>	<code>#1A1A1A</code>	<code>#888888</code>	<code>#141414</code>
twilight	<code>#A6E3B0</code>	<code>#122319</code>	<code>#F8B3B0</code>	<code>#271419</code>
standard-erindark	<code>#4F7138</code>	<code>#E3E5C9</code>	<code>#8C2A1A</code>	<code>#F0DCD0</code>
standard-erindark	<code>#C9C1A1</code>	<code>#222B1A</code>	<code>#847C6A</code>	<code>#2A211A</code>

`reference/ORIGINAL-DIFF.png` shows the structure (§9.7). Phosphor and Standard Erin Dark distinguish additions from removals by brightness, not hue, as their palettes do everywhere else (`TYPOGRAPHY.md`).

C-06 Mermaid pipeline: MDVMermaidPipeline

```
static func prepare(source: String, theme: DiagramTheme) throws -> MDVMermaidPrepared // parse + repair
static func rasterize(_ p: MDVMermaidPrepared, width: CGFloat, scale: CGFloat) -> NSImage? // CoreText
static func displaySize(for p: MDVMermaidPrepared, width: CGFloat) -> CGSize // whole points; shared b
```

C-06.1 Source sanitisation (before parsing), in this order:

#	Rule	Reason
0	Drop the leading preamble exactly as C-06.4 skips it : blank lines, %% comments, and every line of a %%{ ... }%% directive, single- or multi-line, that precedes the first directive line (a front-matter block is rule 1's).	the library drops only lines beginning %, so a multi-line directive's inner lines reach the parser and fail (F-152)
1	Drop a leading YAML front-matter block (--- ... ---).	parser rejects it (<code>invalidHeader</code>)
2	xychart: line "name" [...] / bar "name" [...] → line [...] / bar [...].	parser knows only the unnamed form
3	On style/classDef/linkStyle lines: expand #rgb/#rgba to 6/8 digits; map these CSS colour names (case-insensitive, only where they follow fill:, stroke: or color:) to hex: white black red green blue yellow orange purple gray grey lightgray lightgrey darkgray silver pink lightblue lightgreen lightyellow gold teal navy maroon olive cyan magenta brown beige ivory lavender coral salmon tomato crimson indigo violet khaki tan wheat mintcream honeydew aliceblue whitesmoke gainsboro snow, plus transparent and none → #00000000. Any other name is passed through (and renders black).	3-digit hex and names render black
4	stateDiagram: fold every ID: text description line for an ID into one state "a b" as ID alias inserted after the header.	parser keeps only the first registration
5	id[/text/] and id[\text\] (parallelograms) → id[text].	not in the parser's shape table
6	Strip inline formatting tags <i> <u> <s> <small> <sup> <sub> <code> <tt> <mark> (open and close), keeping their content; leave .	rendered literally

Rule 3's colour values. The names above map to the SVG 1.1 colour keywords, case-insensitively: white #ffffff · black #000000 · red #ff0000 · green #008000 · blue #0000ff · yellow #ffff00 · orange #ffa500 · purple #800080 · gray #808080 · grey #808080 · lightgray #d3d3d3 · lightgrey #d3d3d3

· darkgray #a9a9a9 · silver #c0c0c0 · pink #ffc0cb · lightblue #add8e6 · lightgreen #90ee90 · lightyellow #ffffe0 · gold #ffd700 · teal #008080 · navy #000080 · maroon #800000 · olive #808000 · cyan #00ffff · magenta #ff00ff · brown #a52a2a · beige #f5f5dc · ivory #ffffff · lavender #e6e6fa · coral #ff7f50 · salmon #fa8072 · tomato #ff6347 · crimson #dc143c · indigo #4b0082 · violet #ee82ee · khaki #f0e68c · tan #d2b48c · wheat #f5deb3 · mintcream #f5fffa · honeydew #f0ffff · aliceblue #f0f8ff · whitesmoke #f5f5f5 · gainsboro #dcdcdc · snow #ffaafa; transparent and none both map to #00000000. The list is exhaustive: any other name is passed through (and renders black). (F-139)

C-06.2 Post-parse and post-layout repairs:

Diagram	Repair
flowchart, stateDiagram	Subgraph ownership: a node listed in several subgraphs belongs to the last one (Mermaid.js semantics); it is removed from the others. (Prevents the ELK assert , E-01.)
stateDiagram	classDef , class A,B name , and style lines read from the source are applied to the model (classDefs , classAssignments , nodeStyles).
flowchart, stateDiagram	Whole-label \$\$\$ nodes: label replaced by a blank placeholder measured to the math image's size; image composited after rasterising, centred, at a pixel-snapped origin (R-15, I-009).
sequenceDiagram	
 → newline in notes; → space in actor labels; message labels with
 are blanked and drawn by mdv6, lines stacked upward from the arrow (13 pt pitch, 11 pt font, muted colour).
sequenceDiagram	Actor gaps widened until every message label fits between its endpoints (+ 24 pt; self-messages + 36 pt); all x coordinates remapped piecewise-linearly through old→new actor centres.
sequenceDiagram	Multi-line message rows: the message and everything below shifted down $(n - 1) \times 13 + 4$ pt; spanning blocks, lifelines, and the diagram height grow.
sequenceDiagram	A block whose last item is a note is extended to enclose it (+ 8 pt). autonumber draws a filled disc ($r = 8$ pt) with the 1-based index at each arrow's tail.

C-06.3 Document theme. The *Document* style derives a **DiagramTheme** from the active **MDVTheme**: background = code-block background, foreground = text colour, node surface = page colour mixed 25 % toward the code background on light themes (lifted 16 % toward foreground on dark), lines/borders/muted = fixed mixes of background and foreground.

C-06.4 Dispatch (R-10, R-46). The **first directive line** of a Mermaid source is its first line that, after trimming spaces and tabs, is non-empty and is not (a) inside a leading --- ... --- front-matter block (a line exactly --- opens it, the next line exactly --- closes it), (b) a **%{ ... }%** directive — a line starting **%{** opens one, and it ends on the first line (that one included) containing **}%** — or (c) a **%** comment; it is compared lower-cased. The diagram goes to the **native** pipeline exactly when that line is **flowchart**, **graph**, **sequencediagram**, **classdiagram** or **erdiagram**, or starts with one of them followed by a whitespace character, or starts with **statediagram** (covering **stateDiagram-v2**) or **xychart** (covering **xychart-beta**).

Every other line — including `flowchart-elk`, `graphfoo`, and an empty source with no directive line — goes to the **web** path. Dispatch reads the unsanitised source; C-06.1 runs only on the native path, and the web path hands `mermaid.js` the source as written (`mermaid.js` reads its own front matter and directives).

C-06.5 Web renderer (R-46). One `WKWebView` per web-dispatched block, loading with no base URL a generated page that contains, in order: a `<meta charset="UTF-8">`; a style sheet setting `html, body` to the theme's `secondaryBackground` as a CSS `rgb(...)`, zero margins, the diagram container padded 12px 18px, and `svg { max-width: 100%; height: auto; display: block }`; the source HTML-escaped (`&`, `<`, `>`) inside `<div class="mermaid">`; the bundled `mermaid.min.js` inline; and a script that calls `mermaid.initialize({ startOnLoad: false, theme: '<'dark' when the theme is dark, else 'default'>`, `securityLevel: 'strict'`, `logLevel: 'fatal' }`) then `mermaid.run()`. On success the script reports `{ ok: true, height: h }` to a `mermaidHeight` message handler, where h is the rendered SVG's bounding height plus 24 (the vertical padding) — after two animation frames, and again on every `ResizeObserver` change of the SVG (some types size themselves after the promise resolves); on a rejected promise or no SVG it reports `{ ok: false, error }`. The Swift side:

- sets the block height to $\max(h, 60)$ pt; ignores a report within 0.5 pt of the last one; shows a small spinner over the block until the first successful report and never again after it (a theme change keeps the old render on screen until the new one reports);
- reloads the page only when (source, theme id, `isDark`) changes — never on its own height write, which would loop;
- treats `ok: false` or a height ≤ 0 as failure and shows the R-10 fallback;
- MUST refuse every navigation and subresource load other than the initial page (a navigation-policy and content-rule-list denial), so a diagram's links, images or `click` callbacks reach nothing (D-46);
- MUST remove the `mermaidHeight` handler when the view is dismantled;
- MUST NOT use private API to make the web view transparent — the page paints the background itself;
- **host integration (F-151):** MUST NOT show WebKit's context menu — a right-click on the diagram shows the R-09 block menu (*Copy Code*, *Show Mermaid Source*); MUST pass every scroll-wheel and trackpad-scroll event to the article's scroll view (the page itself never scrolls: its height is the reported height); MUST NOT accept first responder, so a click on the diagram leaves R-49's keyboard scrolling working; and MUST NOT allow text selection inside the diagram (as with a native diagram, which is an image);
- **width (F-156):** the SVG is laid out at the block's width minus the 18 px side padding and scales down, never up (`max-width: 100%; height: auto`); there is no natural-width cap, no raster, and no zoomed-height cap — R-11 and K-07 bind only the native path;
- **quiet and closed (F-164):** `mermaid` is initialised with `logLevel: 'fatal'` in addition to the options above, the web view is not inspectable, and no console message is forwarded to the application's log (R-35).

`reference/ORIGINAL-GANTT.png` shows a web-path diagram's structure (§9.7). The web path has no pinch-zoom, style menu or PNG export (R-09); its context menu is the block menu above. Printing uses the same page without the padding (C-21.3).

C-07 LaTeX math

C-07.1 Rewriting. `MathMarkdown.rewrite(block, fontSize, headingSizeEms, color)` replaces each math span in a prose block with an image reference

```
![] (mdv6-math://inline/<base64url(latex)>?s=<size pt, 1 decimal>&c=<RRGGBBAA>) // $...$ only
![] (mdv6-math://display/<base64url(latex)>?s=...&c=...) // every $...$$, own
```

The host selects the typesetting mode (K-08: `inline` $\rightarrow$ `.text`, `display` $\rightarrow$ `.display`) and is decided by the delimiter alone. Placement is a separate rule: a `$...$$` whose opening is at line start and closing at line end — **whether the span occupies one source line (`$...$`) or several (a C-02 rule 3 fence)** — is emitted as its **own paragraph** (blank lines inserted, indentation preserved) so MarkdownUI's block-image path renders it centred via `MathDisplayView`; every other span — `$...$`, and a `$...$$` mid-line — is

an inline image via `MathInlineImageProvider`, so a mid-line `$$\sum_{i=1}^n$$` shows display-style limits inside the sentence. `base64url` is RFC 4648 §5 without = padding; decoders MUST re-pad to a multiple of four. Delimiter rules (Pandoc `tex_math_dollars`): an opening `$` is followed by non-whitespace; a closing `$` is preceded by non-whitespace and not followed by a digit; a span contains no bare `$` and never crosses a backtick; `\$` is literal; fenced blocks and inline code are never rewritten; an empty `$$` pair is literal. **Scan (F-146)**. A block is scanned left to right and, at each position, `$$` is tested before `$`, so a `$$$` span wins over the `$...$` reading of the same characters. The scan skips a backslash and the character after it verbatim, so an escaped `$` can never open a span. A candidate that reaches a backtick before its closer, or that reaches the end of the block with no closer, is **not** a span: the opener is left literal and the scan resumes *after* the opener’s run. A `$$$` whose body is entirely whitespace is literal. The scan runs over the whole block, not per source line, so a display span may cross line breaks (which is what the C-02 rule 3 fence form relies on).

C-07.2 Typesetting. `MathImageCache.rendered(for: MathSpec)` typesets with `MathImage(latex, fontSize, textColor, labelMode: display ? .display : .text)` after (a) registering the extra symbols and (b) applying the rewrites below, and bakes the result to a bitmap at the screen scale (I-008). Cache: 2048 entries keyed by the URL.

(a) Registered symbols (Latin Modern Math has the glyphs)	(b) Command rewrites (regex, in order)
relations: <code>gtrsim</code> <code>lesssim</code> <code>gtrapprox</code> <code>lessapprox</code> <code>leqslant</code> <code>geqslant</code> <code>lll</code> <code>ggg</code> <code>nless</code> <code>ngtr</code> <code>nleq</code> <code>ngeq</code> <code>doteq</code> <code>triangleq</code> <code>therefore</code> <code>because</code> <code>implies</code> <code>impliedby</code> <code>models</code> <code>vDash</code> <code>Vdash</code> <code>nparallel</code> <code>nmid</code> <code>subsetneq</code> <code>supsetneq</code> <code>nsubseteq</code> <code>nsupseteq</code> <code>squasubseteq</code> <code>squasupseteq</code> <code>precsim</code> <code>succsim</code> ; arrows: <code>hookrightarrow</code> <code>hookleftarrow</code> <code>righttharpoonup</code> <code>lefttharpoonup</code> <code>rightleftharpoons</code> <code>leftrightharpoons</code> <code>nearrow</code> <code>searrow</code> <code>swarrow</code> <code>narrow</code> <code>longmapsto</code> <code>twoheadrightarrow</code> <code>rightsquigarrow</code> <code>leadsto</code> <code>rightrightarrowbar</code> <code>leftleftarrows</code> ; ordinary: <code>dots</code> <code>dotsc</code> <code>dotsb</code> <code>varnothing</code> <code>hslash</code> <code>mho</code> <code>Box</code> <code>square</code> <code>blacksquare</code> <code>bigstar</code> <code>checkmark</code> <code>ddagger</code> <code>S</code> <code>P</code> <code>pounds</code> <code>copyright</code> <code>degree</code> <code>beth</code> <code>gimel</code> <code>wp</code> <code>nexists</code> <code>complement</code> <code>#</code> <code>_</code> ; big operators: <code>iint</code> <code>iiint</code> <code>oiint</code> <code>bigsup</code> <code>bigcup</code> <code>bigodot</code> <code>bigotimes</code> <code>biguplus</code> ; binary: <code>intercal</code> <code>leftthreetimes</code> <code>rightthreetimes</code> <code>divideontimes</code>	<code>\operatorname{X}</code> / <code>\operatorname*{X}</code> → <code>\mathrm{X}</code> ; <code>\dfrac</code> / <code>\tfrac</code> → <code>\frac</code> ; <code>\boldsymbol</code> → <code>\bm</code> ; <code>\bmod</code> → <code>\mathrm{mod}</code> ; <code>\pmod{n}</code> → <code>\mathrm{mod}{n}</code> ; <code>\not=</code> → <code>\neq</code> ; <code>\lVert</code> / <code>\rVert</code> → <code>\Vert</code> ; <code>\lvert</code> / <code>\rvert</code> → <code>\vert</code> ; <code>\big</code> <code>\Big</code> <code>\bigg</code> <code>\Bigg</code> (with optional <code>l r m</code>) before a delimiter → removed; <code>\coloneqq</code> → <code>:=</code> ; <code>align*</code> / <code>equation*</code> / <code>gather*</code> / <code>multline*</code> → <code>unstarred</code> ; <code>align</code> → <code>aligned</code> ; <code>multline</code> → <code>gather</code> ; <code>\begin{equation}</code> / <code>\end{equation}</code> → removed

`\boxed{...}` is implemented in the vendored `SwiftMath` (`MTBoxed` atom, `MTBoxDisplay`: frame of fraction-rule thickness with 0.35 em padding). Unsupported and shown as source: `\underbrace`, `\overbrace`, `\stackrel`, `\substack`, `\&`. The open/close vertical-bar variants `\lVert` / `\rVert` and `\lvert` / `\rvert` name the same glyphs as `\Vert` (U + 2016) and `\vert` (|), which `SwiftMath` does accept, so (b) rewrites them instead of falling back.

C-07.3 Plain-text form (`MathMarkdown.plainText`), used by the TOC, bookmark titles, and mixed Mermaid labels: same delimiter rules; `\frac{a}{b}` → `a/b`, `\sqrt{x}` → `√x`, wrappers (`\text` `\mathrm` `\mathbf` `\mathit` `\mathcal` `\mathbb` `\operatorname` `\boldsymbol` `\bm` `\hat` `\vec` `\bar` `\tilde`) → their content; `~/_` followed by a character or `{...}` → Unicode super/subscript when every character has one (digits, `+ - n i / + - i j n k x`), else kept verbatim; Greek letters, common relations/operators/arrows/sets → Unicode; unknown commands → their name; braces removed; whitespace collapsed.

C-08 Anchors: bookmarks and scroll positions

```
CREATE TABLE bookmarks (id INTEGER PRIMARY KEY, path TEXT NOT NULL, title TEXT NOT NULL,
    sort_order INTEGER NOT NULL, created_at INTEGER NOT NULL,
    block_index INTEGER NOT NULL DEFAULT 0, block_fingerprint TEXT NOT NULL DEFAULT '');
CREATE TABLE scroll_positions (path TEXT PRIMARY KEY, block_index INTEGER NOT NULL,
    block_fingerprint TEXT NOT NULL, updated_at INTEGER NOT NULL, file_mtime INTEGER NOT NULL DEFAULT 0);
```

fingerprint(block) MUST split block at every Unicode whitespace scalar, discard empty pieces, join the pieces with U+0020 SPACE, apply locale-independent Unicode lowercase without compatibility or canonical normalization, and retain the first 80 extended grapheme clusters. resolve(blocks, storedIndex, fingerprint) = the first block whose fingerprint equals the stored one; else storedIndex clamped to [0, |blocks| − 1]; else 0 for an empty document. A scroll position is restored only when the stored anchor resolves **and** the file’s modification time is within 1 s of the stored file_mtime **and** the index is in bounds (E-08).

C-09 Theme contract (MDVTheme), the fields behaviour depends on

```
struct MDVTheme {
    let id: String; let isDark: Bool
    let text, secondaryText, tertiaryText, heading, strong, link, accent, background, secondaryBackground
    var bodyFontFamily: FontFamily; var baseFontSize: CGFloat // default 16
    var h1SizeEm = 1.75, h2SizeEm = 1.4, h3SizeEm = 1.15; h4SizeEm 1.0, h5SizeEm 0.875, h6SizeEm 0.85 (
    var headingSizeEm: [CGFloat] // [h1..h6], used by markdownTheme and by math in headings (R-13)
    var articleMaxWidth: CGFloat?; var articleHorizontalPadding: CGFloat
    var smartTypographyAllowed: Bool // false for phosphor, standard-erin-light, standard-erin-dark
    var codePalette: CodePalette? // default: oneDark (dark) / githubLight (light)
    var showH1Rule, showH2Rule: Bool // heading rules (C-18.7, C-09.1); defaults and per-theme values
    var headingFontWeight, strongFontWeight: Font.Weight // C-09.1; TYPOGRAPHY.md
    var paragraphLineSpacingEm: CGFloat // body leading, default 0.30 (K-10); C-09.1
}
static let all = [highContrast, sevilla, charcoal, solariumDaylight, solariumMoonlight, phosphor, twili
```

Code blocks always use the system monospace face regardless of bodyFontFamily (TYPOGRAPHY.md).

C-09.1 Find-highlight typography (R-24). A block on the inline-highlight path is drawn as one attributed Text styled from its ATX heading level ℓ — the count of leading # (1–6) when followed by a space after trimming, else body text:

ℓ	size factor e	colour	weight	line spacing	below the text
1	h1SizeEm	heading	headingFontWeight	0.25em	when showH1Rule: padding 0.3× the body size, then the divider; else nothing
2	h2SizeEm	heading	headingFontWeight	0.25em	when showH2Rule: padding 0.3× the body size, then the divider; else nothing
3	h3SizeEm	heading	headingFontWeight	0.25em	—
4 / 5 / 6	1.0 / 0.875 / 0.85	heading (h6: tertiaryText)	headingFontWeight	0.25em	—
body	1.0	text	regular	paragraphLineSpacingEm	

Size = round(baseFontSize × s × e) with s the zoom factor (R-30); the body size is baseFontSize × s , and

the padding above a rule is the article heading’s own (MarkdownUI resolves its 0.3em against the body size, and pads only above a rule), so a heading’s height is the same on both paths (F-182, T-60); in the theme’s body face (`bodyFontFamily`, a custom face at a fixed size). Inline runs: `code` → system monospace at $\text{round}(0.90 \times \text{size})$ on `secondaryBackground`; `strong` → `strongFontWeight` in the `strong` colour; `emphasis` → italic; `strikethrough` → single strike; a link → the `link` colour. The renderer’s own bold/italic/monospace presentation intents are cleared after they are mapped, so they are not applied twice. Match marks are drawn over these styles as R-24 states.

C-10 Smart typography (`smartenMarkdown`)

Applied to one block; the block is returned unchanged if it is a fence, is a GFM table by R-24’s `test` (first line contains `|`; the second line, trimmed, is non-empty, contains at least one `-`, and consists only of `-`, `:`, `|`, space), or is a thematic-break line. Otherwise, outside inline code spans (a run of n backticks closes only on a run of exactly n), link/image URL parts (`()` ... matching `)`), and `<...>` spans: `"` and `'` → directional quotes chosen from the preceding character — an **opening** quote when the preceding character is absent, whitespace, or one of `(` `[` `{` `<` `"` `'` `-` `-` `-` `/`, and a **closing** quote otherwise (F-144; the predicate is normative, not illustrative); `---` → `-`; a run of **four or more** dashes is left unchanged; `--` between letters/digits → `-`; `--` → `-`; other `--` runs unchanged (CLI flags survive); `...` → `...` (a longer dot run keeps its extra dots).

C-11 Heading slug

`slug(s)` = lower-case `s`; keep letters and digits; keep `-` and `_` when something precedes them; drop every other non-whitespace character; **every** run of whitespace becomes one `-` when something precedes it — including a run that follows a `-` or a dropped character, so that `a - b` → `a---b` and `C++ & Rust` → `c---rust` as on GitHub; strip trailing `-/_`. Applied to both the link fragment and `TOCHeading.slugText`; equality selects the target, and when several headings share a slug the **first in document order** wins. GitHub’s numeric disambiguation suffixes (`-1`, `-2`) are not generated (D-17).

C-12 Section and inline-stripped text

`section(headingAt i) = blocks [i, j)` where j is the index of the next **TOC heading** (C-02 rule 7 — an `h4`–`h6` or `setext` heading never ends a section) with level $\leq$ the level of i , or the block count. Copy output = those blocks joined with `\n\n`. `stripInlineMarkdown` removes trailing `#s`, `**`, `__`, backticks, unescaped `*`, **both** underscores of an `__` emphasis pair whose opening `_` is not preceded by a letter or digit (word-internal underscores such as `snake_case` are kept), and reduces `[text](url)` to `text`. The removals are applied in **this order**, which is normative because the order is observable (F-141): (0) every C-22.1 `<img ...>` tag is removed whole — the image renders in the heading, but a TOC row, bookmark title or slug never shows the tag’s source (F-171; a tag inside a code span is kept like any code); (1) trailing `#s`, and the surrounding whitespace is trimmed; (2) `[text](url)` → `text`; (3) the `__` pairs; (4) a left-to-right scan that drops `**` pairs, every backtick, every `_` of a `__` pair, and every unescaped `*`, keeping the escaped character alone when it follows a backslash (`*`, `_`, backtick, `#`, `[`, `]`, `\`); (5) the surrounding whitespace is trimmed again.

C-13 Build outputs (`build.sh`)

```
swift build -c {debug|release}
build/mdv6.app/Contents/{MacOS/mdv6, Info.plist, Resources/{AppIcon.icns, *.otf, *-highlights.scm,
                                     mathFonts.bundle/, mdv6, Help.md, mermaid.min.js, mermaid.LICENSE.txt}}
codesign --force --sign - --entitlements mdv6/mdv6.entitlements build/mdv6.app      # ad hoc
```

mermaid.js (R-46). `Vendor/mermaid/mermaid.min.js` is not committed (`.gitignore`). When it is absent, `build.sh` (through `tools/fetch-mermaid.sh` ensure) **MUST** download `https://cdn.jsdelivr.net/npm/mermaid@1` to a temporary name and move it into place; whether downloaded or already present, it **MUST** verify the file’s SHA-256 equals `a43bc1afd446f9c4cc66ac5dd45d02e8d65e26fc5344ec0ef787f88d6ddb6f9e` and **MUST** exit 1 on a mismatch — naming the expected and actual digests and telling the engineer to delete the file or update the pinned digest with the version — before copying it into `Resources/`. A stale,

tampered or regressed copy therefore fails the build instead of shipping. The script is **not** a SwiftPM resource and lives outside every target’s source directory: a declared resource that is absent fails **swift build**, and an excluded one that is absent warns, either of which would break [R-37](#)’s clean-checkout **swift test** (F-183). The application finds it flat in **Resources/**; the tests and the harness find it at its repository path, and a web-path test is skipped with a recorded reason when it has not been fetched. **Licence (F-165)**: the bundle MUST also carry **mermaid.LICENSE.txt** — mermaid’s MIT licence and the licence notices of the third-party code its minified build embeds, as published with 11.4.1 — committed at **Vendor/mermaid/LICENSE.txt** (it is small and changes only with the pin; v0.14.2 assembled it from <https://cdn.jsdelivr.net/npm/mermaid@11.4.1/LICENSE>, the bundle’s own *Bundled license information* block — DOMPurify 3.2.1, js-yaml 4.1.0, lodash-es, cytoscape’s embedded notices — and <https://cdn.jsdelivr.net/npm/dompurify@3.2.1/LICENSE>) and copied into **Resources/**; a pin bump MUST update it in the same commit.

The vendored SwiftMath resolves **mathFonts.bundle** from **Bundle.main** first and from **Vendor/SwiftMath/mathFonts.bundle** (by **#filePath**) when running unbundled (**swift run**).

C-15 History persistence (mdv6_history)

```
[ { "id": "<UUID>", "path": "/abs/path/to/file.md", "addedAt": <seconds since 2001-01-01 as Double> }, ... ]
```

Swift Codable encoding of [HistoryEntry] (id: UUID, path: String, addedAt: Date, keys as shown, default JSONEncoder date strategy). Order is most recent first. **filename** is derived (last path component), not stored. A value that fails to decode MUST yield an empty history, never a crash; unknown keys MUST be ignored.

C-16 Remote-image network contract

When **mdv6_load_remote_images** is true, a remote image MUST be fetched with an ephemeral **URLSession** that has no persistent cache, cookie storage, credential storage, or shared authentication state. The request MUST be an unauthenticated **GET** to the document-provided **http** or **https** URL; it MUST send no **Cookie**, **Authorization**, or **Referer** header. The only document-derived request data MAY be the original URL and the sequence of redirect URLs. Redirects MUST be followed at most five times and only while every target remains **http** or **https**; any other redirect MUST fail. Connection timeout is 15 s and total resource timeout is 30 s. The body MUST be streamed and cancelled as soon as it exceeds 32 MiB. A successful response MUST have a 2xx status, an **image/*** media type, and decode through **ImageIO** within [K-14](#). Any violation MUST produce the [E-11](#) failure placeholder. No response body or decoded remote image may be written to disk.

C-17 Render harness and corpus

The executable name is **render-harness**; every invocation runs as **swift run --package-path tools/render-harness render-harness ...** from the repository root. The harness links the application’s pipeline code ([R-39](#)) and renders through the same block views the window uses.

Invocation	Behaviour	Exit
render-harness INPUT --output FILE [--width PT] [--scale S] [--theme ID]	Render one .md document or one raw .mmd diagram to PNG. Defaults: width 860 pt, scale 2, theme high-contrast ; --theme takes a C-09 theme id (T-45). The parent directory of FILE MUST exist; the command MUST NOT create it.	0 success; 1 render/fallback failure; 2 usage, unreadable input, unknown theme id, or unwritable output

Invocation	Behaviour	Exit
<code>render-harness --scan ROOT</code> <code>--output-dir DIR</code>	Recursively discover non-hidden <code>.mmd</code> files and Mermaid fences in <code>.md</code> files, ordered by the UTF-8 bytes of relative path and then fence index. Render each case to a collision-free relative PNG path under <code>DIR</code> ; raw <code>.mmd</code> is one case. A case that C-06.4 dispatches to the web path is not rendered — the harness is offline and hosts no WebKit — and is reported with status <code>web</code> , an expected result; every fallback of a natively dispatched case is a failure.	0 when every case has its expected result; 1 when any case fails; 2 for usage or I/O failure
<code>render-harness INPUT --print-pdf</code> <code>FILE [--paper letter\ a4]</code>	Run the C-21 print pipeline over one <code>.md</code> document, or one raw <code>.mmd</code> diagram wrapped as a single <code>```mermaid</code> fence (default paper Letter), and write the paginated result to <code>FILE</code> as PDF instead of presenting a panel — the scripted half of T-53. Web-dispatched diagrams print as their C-21.3 fallback (source) here, for the same reason. On stdout, one JSON object per block: <code>index</code> , <code>kind</code> (<code>prose</code> , <code>code</code> , <code>formula</code> , <code>diagram</code> , <code>frontmatter</code>), <code>page</code> (1-based), <code>rect</code> (points, page coordinates) of where it was drawn — two objects when C-21.5 sliced it — and <code>images</code> , the image objects drawn inside that rect as { <code>"pixels": [w, h]</code> , <code>"rect": [x, y, w, h]</code> }, plus <code>formulas</code> , the rects of its math spans.	0 success; 1 render failure; 2 usage, unreadable input, or unwritable output
<code>render-harness --check MANIFEST</code> <code>[--case ID]</code>	Execute every manifest case, or exactly <code>ID</code> , compare dimensions and pixels with its golden when present, and evaluate its named metric. Cases MUST run in manifest order.	0 when all selected cases pass; 1 for any mismatch or render failure; 2 for invalid manifest, unknown ID, usage, or I/O failure

`test-docs/render-cases.json` MUST be [UTF-8](#) JSON with this shape; paths are repository-relative, ids are unique, and unknown keys are ignored:

```
{
  "version": 1,
  "cases": [
    {
      "id": "unique-string",
      "input": "test-docs/example.md",
```

```

    "kind": "markdown-or-mermaid",
    "width": 860,
    "scale": 2,
    "expect": "render | fallback | web",
    "golden": "test-docs/goldens/example.png",
    "metric": "pixel-or-ink-or-sequence-layout"
  }
]
}

```

For a pixel comparison, let N be the number of pixels in either equal-sized image and let D be the pixels for which at least one 8-bit RGBA channel differs by more than 8. The mismatch fraction is

$$q = \frac{|D|}{N}.$$

The comparison **MUST** pass when $q \leq 0.001$; $N = 0$, unequal dimensions, a missing golden, or a missing output is a failure. **ink** uses §7.1. **sequence-layout** checks the geometric assertions in T-19. Each case **MUST** emit one JSON object on stdout with **id**, **status** (**pass**, **fail**, **fallback**, or **web**), and **output**; human-readable diagnostics go to stderr. Case output **MUST** be byte-for-byte stable for identical inputs, dependencies, width, scale, and backing environment.

C-19 Line citations (#L... fragments)

C-19.1 Grammar. A **line citation** is the fragment of a link destination, after the single UTF-8 percent-decode of R-19, matching

`~L([1-9][0-9]*)?(?:-(?:L)?([1-9][0-9]*))?$` case-insensitive, anchored

so #L10, #L10-L12, #l10-l12, and #L10-12 all parse. Line numbers are **1-based** and refer to the target document's source lines after C-02 rule 6 normalisation — the same coordinate space as **blockLines/lineCount**. The leading digit excludes 0, so #L0 and #L00 do not match; a fragment that does not match this pattern in full (including #L, #L0, #L10C5, #L10, and any fragment with a trailing or interior token) is **not** a line citation and falls through to C-11 slug matching (R-19).

C-19.2 Range normalisation. Let the parse yield the start a and, when the fragment names two numbers, the second number b (absent otherwise). The **resolved start line** is

$$a' = \begin{cases} \min(a, b) & b \text{ present} \\ a & b \text{ absent} \end{cases}$$

so a reversed range (#L12-L10) is read as the range it names rather than as an error. The range is resolved by its start line only; the end is informational and is not used for scrolling or for the flash extent (C-19.3).

C-19.3 Resolution and effect. The target block is the last index i with `blockLines[i].lowerBound ≤ a'` — the block that **contains** line a' — falling back to the first block when no index satisfies the predicate; C-02 rule 8 owns the total resolution. On a successful resolution the application **MUST** (a) set the top visible block to the target and scroll it to the top of the viewport — the same scroll a TOC row performs (R-21), without the snapshot or the row selection (C-19.4) — and (b) **flash** that single block for the R-22 duration (0.6s) using the existing section-flash treatment — **theme.accent** at 0.18 opacity over the block, restarting on a repeat click. Because a C-02 block is the smallest addressable render unit (§3.2 renders one **Markdown** view per block), the flash covers the **whole block**, not the line run; the citation therefore highlights the *paragraph* containing the cited line. Line-accurate highlighting inside a block is explicitly **not** specified: the render path rewrites block text before layout (C-07.1 inserts blank lines around own-paragraph \$\$\$ spans) and cmark-gfm normalises the paragraph flow, so no line identity survives to the view tree. A resolution that finds no block (an empty document) scrolls to the top and flashes nothing.

C-19.4 Interaction with the rest. A line-citation jump is a *position* move, not a *choice*: it MUST NOT set the TOC selected row (E-29, D-41), MUST NOT push a back snapshot on a same-document jump (R-18), and MUST NOT be recorded as a bookmark anchor. It MUST cancel an in-progress flash and take it over (one flash at a time). It works in every document state that has blocks; in **EMPTY** there is no fragment to resolve against, so a link click cannot arise.

C-19.5 Scope. A line citation applies only where R-19's classification admits the destination in-app: local files whose extension is `md`, `markdown`, or `mdown`. A citation to any other extension — the `Foo.swift#L412-L418` form GitHub emits for source files — takes R-19's system-opener branch (E-05) and never reaches this contract; rendering a cited code file is out of scope and is not implied by these rows.

C-20 Frontmatter (R-44)

C-20.1 Recognition. Line 1 of the (C-02 rule 6 normalised) document is the **opening fence** and must be exactly `---` (YAML) or `+++` (TOML) — nothing before it, no leading or trailing space. Scanning from line 2, the header ends at the first line that is exactly a **closer**: `---` or `...` for YAML, `+++` for TOML; an indented `---` inside a folded value is not a closer. No closer anywhere → no header. **YAML content guard**: because `---` is also a thematic break, every YAML line before the closer must look like YAML — it is empty, starts with a space or tab (a continuation, which may hold anything), starts with `#` (a comment), is a sequence item (`-` alone or starting `-`), or has a **mapping colon**: a `:` followed by a space, a tab, or the end of the line (so `type:` and `summary: >- pass, http://example.com` and a prose sentence do not). One failing line → no header. **A candidate whose C-20.2 reduction yields zero rows is not a header either** (F-150): a `---`, blank line, `# Title`, blank line, `---` opening — two rules around a title, whose `# Title` the guard admits as a comment — therefore renders as ordinary Markdown, and so does `---` directly followed by `---`. TOML has no guard: `+++` means nothing to CommonMark, and multi-line TOML arrays close with a `]` at column 0. False negatives are cheap and false positives are not; a rejected candidate renders as ordinary Markdown, which is also what GitHub does with a header it cannot parse.

C-20.2 Rows (`frontmatterRows`, a display reduction, not a YAML or TOML parser — it never rejects). Drop the opening fence and the closer. Walk the remaining lines; skip empty-or-whitespace lines and lines starting `#`. Then:

- **TOML**: a line (trimmed) starting `[` is a row with that text as the key and an empty value; a line with no `=` is a keyless row; otherwise `key =` the trimmed text before the first `=`, `value =` the trimmed text after it. When the value's **bracket depth** — `[` minus `]`, ignoring brackets inside `"..."/'...'` and everything after an unquoted `#` — is positive, following lines join the value, one per line (dedented as below), until the depth returns to ≤ 0 .
- **YAML sequence item** at column 0: a keyless row holding the line.
- **YAML mapping line** (C-20.1's colon): `key =` text before the colon, trimmed; `scalar =` text after it, trimmed; the **continuation group** is every following line that is empty or starts with a space or tab, up to the next column-0 line, trailing empty lines dropped, dedented by the group's smallest indent over its non-empty lines. Value: no group → the scalar; scalar `>`, `>-`, `>+` → the group **folded** (consecutive non-empty lines joined by one space, each run of k empty lines becoming k line breaks); `|`, `|-`, `|+` or empty scalar → the group joined with line breaks (a nested mapping or sequence kept verbatim); any other scalar → the scalar followed by the group, line-broken.
- **Anything else**: a keyless row holding the trimmed line.

Keys and single-line values lose **one** matching pair of surrounding `"` or `'` (only when both ends match and the text has at least two characters). A keyless row's value spans the table's full width.

C-20.3 Table. A **VStack** of rows with no spacing: a keyed row is a key cell then a value cell; a keyless row is a value cell alone. Every cell has 6 pt vertical and 13 pt horizontal padding and a 1 pt border in the theme's `border` colour; fills alternate `background` (row 0, 2, ...) and `secondaryBackground` — the GFM table style (R-07), so a header and a table read as siblings. The key column is as wide as the **widest key at its natural, unwrapped, padded width**; keys are semibold, values regular and wrapping, both in the theme's body face at `round(baseFontSize × s)` with s the zoom (R-30), line spacing `round(0.25 × size)`, in the `text` colour. A cell's fill and border stretch to its row's height, so a wrapped value never leaves its key

with a short band. Values are plain Text: a ****** or **#** in a value is part of the value. A header whose rows are empty is not a header (C-20.1). `reference/ORIGINAL-FRONTMATTER.png` shows the structure (§9.7). Print scales the paddings by the C-21.1 factor (C-21.6).

C-21 Printing (R-45)

`reference/ORIGINAL-PRINT-MATH.png` shows the printed blocks' structure before pagination (§9.7).

C-21.1 Page and type. The print info is a copy of the shared one with 54 pt margins on all four sides, horizontal pagination *fit*, vertical pagination *automatic*, no centring, and AppKit's standard header and footer on (job title — the displayed file's name — with the date, and page numbers). The panel shows paper size, orientation and scaling. Let w_c be the paper width minus both margins (Letter: $612 - 108 = 504$ pt) and W the print theme's `articleMaxWidth` (**high-contrast**: 860 pt). The **print type scale** is

$$s_p = \min\left(1, \frac{w_c}{W}\right) \quad (\text{Letter: } s_p = 504/860 \approx 0.586, \text{ body } 16 \times s_p \approx 9.4 \text{ pt}),$$

$s_p = 1$ when the theme sets no `articleMaxWidth`. Body, headings (em-relative), inline code and fenced code take s_p in place of the screen zoom; the theme's absolute point margins — paragraph and heading spacing, code-block, table and thematic-break gaps, table cell padding, code-block chrome insets — are multiplied by s_p too, so the printed page keeps the window's proportions. W caps the **padded** article frame (K-13) while w_c is a content width, so the printed line holds about $w_c/(s_p(W - 2p - 2b)) = 860/768 \approx 1.12$ times the characters of the widest screen line — some 12 % more, not the same; the formula is kept as built because it is what the original prints and what its Help describes (D-47). The screen zoom never reaches print (I-017). The print theme is **high-contrast** whatever the screen theme; smart typography applies when the preference is on **and high-contrast** allows it.

C-21.2 Blocks and the container. Each C-02 block is rendered on its own, at width w_c , into a one-page vector PDF by SwiftUI's `ImageRenderer` (a hosting view in an offscreen window prints blank, because its content composites through the render server), from a view tree that mirrors the screen's block view with: the C-21.1 scales; code blocks soft-wrapped with no hover toolbar or context menu; `LocalImageProvider` with remote loading **off**; no find tint, stripe, flash or selection. Prose follows the §3.2 order (`<img>` → `math` → smart typography). Block pages are stacked top-down, on the theme's page background, with the gap between blocks i and $i + 1$ equal to s_p times the **screen's** inter-block gap for that pair at zoom 1 **under the print theme** (**high-contrast**) — the gap I-014 and C-18.10 pin, block-boundary heading and paragraph margins included; for a pair I-014 does not pin (one involving a fence, table, list, display formula, diagram or frontmatter table) the printed gap is s_p times whatever the screen build produces for it, which is implementation freedom and is not tested (F-177) — so printed headings keep the screen's rhythm, scaled (F-160; the original's flat $12 s_p$, which drops those margins, is not carried — D-47), in a flipped container view whose frame height is the stack's; each block PDF is drawn in `draw(_:)`, so text stays vector glyphs through to the printer or the saved PDF. A `CGPDFPage` does not retain its document, so each block's document **MUST** be retained as long as its page is drawn.

C-21.3 Mermaid. Let $w_d = \max(w_c - 2 \cdot 18 s_p, 1)$ (the diagram's drawn width inside the chrome's scaled 18 pt insets) and $w_\ell = \max(w_d/s_p, 1)$ (the width the screen's column would give it — diagrams lay out at screen width and print smaller, like the type, because their label sizes are absolute). Before the view tree is built, each Mermaid fence is rendered by its C-06.4 path: **native** — prepared through the layout cache (the native style preference `mdv6.mermaid.style` is honoured) and drawn into a PDF at w_ℓ through the same CoreGraphics/CoreText drawing the raster uses, then placed scaled by s_p (at most 1), centred; if the PDF cannot be made, a raster at $4 s_p$ px/pt; **web** — the C-06.5 page without padding in an offscreen borderless window that is **never ordered front** (so the print job flashes nothing on screen), laid out at w_ℓ , measured (the page's own report, with a polling fallback of three identical 50 ms reads or 3 s, because an unshown window may get no animation frames; K-17's 5 s ceiling on the wait), then turned into a PDF by WebKit and drawn at w_d ; if that fails, a snapshot at 4 px/pt from a window that is ordered front for the capture. Either way the diagram sits on the code-block background with $12 s_p$ pt vertical insets and a 6 pt corner

radius. A fence both routes fail on is re-tagged from `mermaid` to `text` and prints as a plain code block of its source.

C-21.4 Math and inline pictures. A block that is exactly one own-paragraph `$$$` (C-07.1) that SwiftMath accepts is drawn by drawing SwiftMath’s own **vector** image, centred, scaled down to w_c if wider, with $4s_p$ pt padding above and below; a rejected formula keeps the Markdown path so its message and source print. Every other formula is typeset up front with **MathSpec** at `baseFontSize` $\times s_p$ (heading-scaled per R-13) and a raster density of 12 px/pt (a separate cache entry from the screen’s, C-07.2); *The rest of this paragraph is informative — the original’s method; any method that meets R-45’s outcome conforms.* Each inline formula and each inline local or **data:** picture (C-22 or Markdown; remote ones are not fetched) becomes a **slot**: the block is laid out with an invisible placeholder image of the slot’s size in each slot — painted in the page background, with pixel dimensions $1 \times (k + 1)$ for slot k — the placeholders’ rectangles are read back from the block page’s content stream by that pixel signature, and the real content is drawn into each rectangle: the formula’s vector image, or the picture. When the placeholders cannot all be found, the block is laid out once more with the baked bitmaps and pictures in place. (*Normative again:*) any such method needs the renderer to accept already-resolved inline images, because **ImageRenderer** runs no `.task` and upstream MarkdownUI resolves inline images only there — the one patch of the vendored MarkdownUI (I-016).

C-21.5 Pagination. The container overrides `adjustPageHeightNew`: for AppKit’s proposed break, find the first block that straddles it; if that block is no taller than a page and does not start at the page’s top, the break moves up to the block’s top, so the block starts the next page; a block taller than a page is sliced at the proposed break. The result is clamped to `[limit, proposed]` — never below AppKit’s limit, never beyond the proposed bottom (AppKit asserts on an increase, and on a final partial page its limit can exceed the bottom) — with `heightAdjustLimit` 0.9, so blocks up to 90 % of a page move whole.

C-21.6 Frontmatter and session. Block 0 with a header prints as the C-20.3 table at size `baseFontSize` $\times s_p$ with paddings $\times s_p$ when *Show Frontmatter* is on, and prints nothing when it is off. The print panel is a sheet on the target window (app-modal without one); the operation shows a progress panel; while its session is open a second P beeps (R-45). The panel **MUST** be presented from a plain main-queue callout, not from inside the asynchronous pre-render task, because the SwiftUI-backed print panel of newer macOS interrogates the current concurrency executor and crashes when that task has ended (reproduced 5/5 in the original).

C-22 Raw `<img>` tags (R-51)

C-22.1 Rewrite. `RawHTMLImages.rewrite(block)` runs on a prose block before C-07.1 and returns the block unchanged when it does not contain `<img` or its first non-space characters are `` ` `` or `~~~`. Otherwise it scans left to right, tracking inline code exactly as C-07.1 does (a backtick run of length k opens a span that only a run of exactly k closes) and copying code verbatim. Outside code, `<img` (case-sensitive) up to the first following `>` that is **not inside** a `"..."` or `'...'` attribute value is a candidate tag (so `alt="a > b"` does not end it; F-171); with no `>` the `<` is copied and the scan moves on. From a candidate, the attributes `src`, `alt`, `width` and `height` are read case-insensitively as `name="..."`, `name='...'` or `name=value` (unquoted, ending at whitespace, `>` or a quote), whitespace allowed around `=`. `width/height` lose a case-insensitive `px` and must parse as a number > 0 , else they are absent. A tag with no or an empty `src` is copied as written; any other becomes

```
[!] (mdv6-img://<base64url(src)>?a=<base64url(alt)>&w=<width, 1 decimal>&h=<height, 1 decimal>)
```

with each query item present only when its attribute is (and `a` only for a non-empty `alt`); `base64url` as in C-07.1. Other tags, attributes and HTML are untouched and still show as text.

C-22.2 Display. `src` resolves like a Markdown image destination: a URL with a scheme is used as it is, otherwise it is a path resolved against the document’s directory (R-16). Given the image’s natural size (w_0, h_0) and ratio $r = w_0/h_0$, the display size is: both attributes $\rightarrow (\min(w, hr), \min(h, w/r))$; `width` only $\rightarrow w' = \min(w, w_0)$, $(w', w'/r)$; `height` only $\rightarrow h' = \min(h, h_0)$, $(h' r, h')$; neither $\rightarrow (w_0, h_0)$; all as caps, the image drawn aspect-fit within them and shrunk further by the column like any other image. A tag alone

in its paragraph uses the block image path; one inside text is an inline image whose point size is set to the display size (pixels untouched, so nothing is resampled). **Inline placement:** an inline image other than a math span (C-07.1) — `<img>` or Markdown — sits on the text baseline, its bottom edge on the baseline of its line; the line’s height grows to hold it, the other lines of the paragraph are unaffected, and no text wraps around it (it is a glyph in the line, not a float). Inline math is placed per R-12 and D-02, not by this rule (F-173). `reference/ORIGINAL-RAW-HTML-IMAGES.png` shows the block and inline cases (§9.7). Remote `src` obeys R-16 and C-16 — the clickable blocked placeholder on the block path and R-16’s non-clickable inline *Remote image blocked* image on the inline path when *Load Remote Images* is off — and every image obeys K-14. On the block path, a local image that cannot be loaded shows its `alt` text, or **Missing image:** `<src>` when `alt` is empty, in 12 pt secondary text; on the inline path it is left out of the line (the renderer draws no inline placeholder). The accessibility label is `alt`, or `src` when `alt` is empty. The same size function serves print (C-21.4), so a picture occupies the same proportion of the column on paper as on screen.

5. Interface specification

5.1 GUI: menus and shortcuts

Menu · item	Shortcut	Effect	Errors / disabled
mdv6 · Install Command Line Tool...	—	Symlink <code>/usr/local/bin/mdv6-helper</code> → <code>Contents/Resources/mdv6-command-line-tool</code> unprivileged attempt first, then an AppleScript <i>with administrator privileges</i> auth dialog	every outcome is an NSAlert (C-14): <i>mdv6-helper missing, Already installed,</i> <i>Install failed</i> (with the error message), <i>mdv6-command-line tool installed</i> ; cancelling the auth dialog leaves the symlink untouched with no alert
File · Open...	O	Open panel; loads into this window (R-01)	cancel: no-op
File · Open in New Window...	O	Open panel; new window	—
File · Close File	W	Close the displayed file: its history row goes and the new first row loads (R-47)	disabled while EMPTY
File · Close Window	W	Close the target window (R-47)	—
File · Close All	W	Confirmation dialog, then every history row goes and every window enters EMPTY (R-47, E-36)	disabled when history is empty

Menu · item	Shortcut	Effect	Errors / disabled
File · Print...	P	Print panel; the document re-rendered for paper (R-45, C-21)	beeps with no or an empty document, or while a print session is open
File · Edit · Edit Current File	E	Open current file in the chosen editor (R-23)	no editor: prompts to choose; launch failure: <code>NSAlert</code> “Couldn’t open in external editor” with <i>Choose Different Editor...</i> / <i>Cancel</i> (C-14)
File · Edit · Choose Editor... / Forget Editor	—	Set / clear <code>mdv6_editor_app_path</code>	—
Edit · Find...	F	Find bar, or global search when the sidebar was last focused (R-24)	—
Edit · Search History...	F	Focus global search (R-25)	—
Edit · Copy / Select All	C / A	System pasteboard group: act on the focused view’s text selection (R-22)	—
Navigate · Back / Forward	← / →	History stacks (R-18)	always enabled; no-op when the stack is empty
Navigate · Next File / Previous File	] / [(also /)	The history row below / above the displayed one, selecting route (R-48)	disabled at the last / first row and while <code>EMPTY</code> ; no wrap
View · Show/Hide Sidebar	S	Toggle history sidebar (R-20)	—
View · Show/Hide Inspector	0	Toggle the inspector (R-21); the same command as the toolbar’s <code>sidebar.right</code> button (R-30)	—
View · Zoom In / Zoom Out / Actual Size	= / - / —		Zoom In / Out disabled at the clamps; Actual Size disabled at 1.0

Menu · item	Shortcut	Effect	Errors / disabled
View · Smart Typography	—	Toggle R-17 ; label reads “(off for this theme)” and is disabled when the theme opts out	—
View · Load Remote Images	—	Toggle R-16	—
View · Show Frontmatter	—	Toggle R-44 (<code>mdv6_show_frontmatter</code>)	—
Bookmarks · Bookmark Current Spot	D	R-27	—
Bookmarks · Set Placeholder / Jump to Placeholder	0 / 0	R-28	Jump always enabled; beeps when no placeholder
Bookmarks · slot 1...5	1... 5	Open numbered bookmark (R-27)	disabled when the slot is empty; beeps when the bookmark’s file is missing (E-09)
Help · mdv6 Help Find bar	? G / G / Esc	R-31 next / previous / close (R-24)	— stepping disabled with no matches
Document pane	↓ ↑ · PgDn PgUp · Space Space · Home End	Keyboard scrolling (R-49 , K-18), without clicking into the text first	passes through while a field, the list or a focused control has first responder, or with / / held

Menu · item	Shortcut	Effect	Errors / disabled
Toolbar	—	Five icon-only buttons, trailing <code>(primaryAction)</code> , in this order: plus (Open..., O) · pencil (Edit in external editor, E) · paintpalette pop-up menu (nine themes + System, checkmark on the active one — R-29) · bookmark / bookmark.fill (Bookmark Current Spot, D; filled and accent-tinted when the displayed file has a bookmark) · sidebar.right (Toggle Inspector, 0; accent-tinted while the inspector is shown). No text labels, no visible picker control. See C-18.2	bookmark button disabled with no document

In-block controls: code blocks — hover toolbar (wrap, copy), context menu (Copy Code, Wrap Long Lines, Copy Without Prompts when applicable); Mermaid blocks — hover capsule (style menu, show source, export PNG, copy) and context menu (Copy Code, Show Mermaid Source / Show Diagram, Diagram Style, Export Diagram as PNG), with the style menu and both export items absent on a web-path diagram ([R-09](#), [R-46](#)); `diff/patch` blocks — the code-block controls ([R-50](#)); display math — context menu (Copy LaTeX). PNG export writes the diagram at natural size, 2× pixel density, to a user-chosen path; failure beeps.

5.2 CLI: `bin/mdv6`

Invocation	Behaviour	Exit
mdv6	open <app>	0
mdv6 FILE... / mdv6 DIR	Each argument resolved to an absolute path; open -a <app> <paths...> (the app receives them via LaunchServices, R-01/R-02)	0; 1 + mdv6: no such file: <arg> on stderr for the first argument that does not exist (later arguments are not checked; nothing opened)
mdv6 -	Only as the sole argument: stdin copied to \$(mktemp -t mdv6-stdin).md, then opened. mdv6 - FILE treats - as a filename (no such file: -, exit 1)	0
mdv6 -h / --help	Usage text (lines 2–9 of the script) to stdout	0
mdv6 --version	CFBundleShortVersionString from the located bundle's Info.plist	0
any (including --help/--version), bundle not found	mdv6: mdv6.app not found (set MDV6_APP or install to /Applications) on stderr — the bundle is located before the arguments are read	1

Bundle search order: \$MDV6_APP (if a directory) → /Applications/mdv6.app → ~/Applications/mdv6.app → ../build/mdv6.app and ../mdv6.app relative to the script → mdfind "kMDItemCFBundleIdentifier == 'com.mdv6.app'" (first hit).

5.3 Build and release: make

Target	Effect
make / build	deps check (Swift ≥ 5.9, macOS ≥ 13, build.sh executable) then ./build.sh debug → build/mdv6.app (C-13 ; the first build downloads and verifies the pinned mermaid.js)
release	./build.sh release
run	build + launch
install	copy to /Applications/mdv6.app, lsregister -f, then install-cli (sudo symlink /usr/local/bin/mdv6 → bin/mdv6)
uninstall	remove the symlink and /Applications/mdv6.app
register	lsregister -f build/mdv6.app
clean	remove build/, .build/, and build_icon/

Target	Effect
<code>dist</code>	<code>check-version</code> MUST require an exact <code>vX.Y.Z</code> tag on <code>HEAD</code> and MUST reject an absent, malformed, or command-line-overridden version before <code>clean</code> or any build step (R-34) → <code>clean</code> → <code>release</code> → <code>sign</code> (Developer ID, hardened runtime, timestamp; <code>codesign --verify --deep --strict</code>) → <code>zip-notary</code> → <code>notarize</code> (keychain profile) → <code>staple</code> → <code>zip-release</code> → <code>checksum</code> (<code>.sha256</code>) → <code>verify-release</code> (<code>spctl</code>)
<code>github-release</code>	upload the zip and checksum to the GitHub release for the tag
<code>icon</code>	regenerate <code>mdv6/AppIcon.icns</code> from <code>MDV6.png</code>

Release inputs (Makefile variables, overridable on the command line): `TEAM_ID` and `CERT_NAME` (Developer ID identity; `sign` exits 1 when empty — the checked-in defaults name the repository owner’s identity and MUST be overridden by any other release engineer), `NOTARY_PROFILE` (default `mdv6-notary`; `notarize` exits 1 when empty), and `NOTES_FILE` (optional release notes for `github-release`). `VERSION` is derived only from the exact tag and is not an overridable release input.

CI (`.github/workflows/build.yml`): on every push to `main`, every pull request, and manual dispatch, build debug and release on `macos-15`, verify the bundle layout, and upload `mdv6-release.tar.gz`; on a push to `main` only, publish it as the rolling `latest` prerelease.

5.4 Cross-cutting: diagnostics and failure reporting

ID	Contract
R-35 (above)	Nothing document-derived is logged.
C-14	Failures arising from document content are reported in place, never modally: unrenderable diagram / math → fallback text in the block (R-10, R-14); missing or blocked image → placeholder in the block (R-16); unreadable file → the window stays on its previous content (E-03); missing bookmark file → row marked as missing in the inspector, and opening it beeps (E-09). Failures of a user-initiated system action are reported by a beep (PNG export, R-09; P with nothing to print or a session already open, R-45) or an <code>NSAlert</code> (CLI install outcomes; external-editor launch failure, R-23) — the only two modal dialogs the application shows besides open/save panels, the editor chooser (§5.1), the <i>Close All</i> confirmation (R-47) and the system print panel (R-45). A web-path Mermaid failure is a document-content failure: the R-10 fallback in the block (R-46).

C-18 Window chrome (§5.5, normative structure)

C-18.0 Reference images. `reference/MDV-SCREEN.png` (Sevilla theme, a long technical document; from the original’s README) and `reference/MDV-ORIGINAL-SEVILLA.png` (Sevilla, 05-fixed-mortgages.md, with a placeholder and four bookmarks set) show the original application and are **normative for structure**: which elements exist, where they sit, what a row contains, what is highlighted when. They are **not** normative for pixel values — colours, faces and sizes come from `TYPOGRAPHY.md` and C-09. `reference/RECREATION-MDV6-SEVILLA.png` and `reference/RECREATION-MDV7-SEVILLA.png` are the two v0.8 recreations of this specification rendering the same document; every difference between them and `MDV-ORIGINAL-SEVILLA.png` is a gap this section closes, and an implementer SHOULD look at all three before writing any pane. The images show the product name `mdv`; the rename to `mdv6` changes no structure.

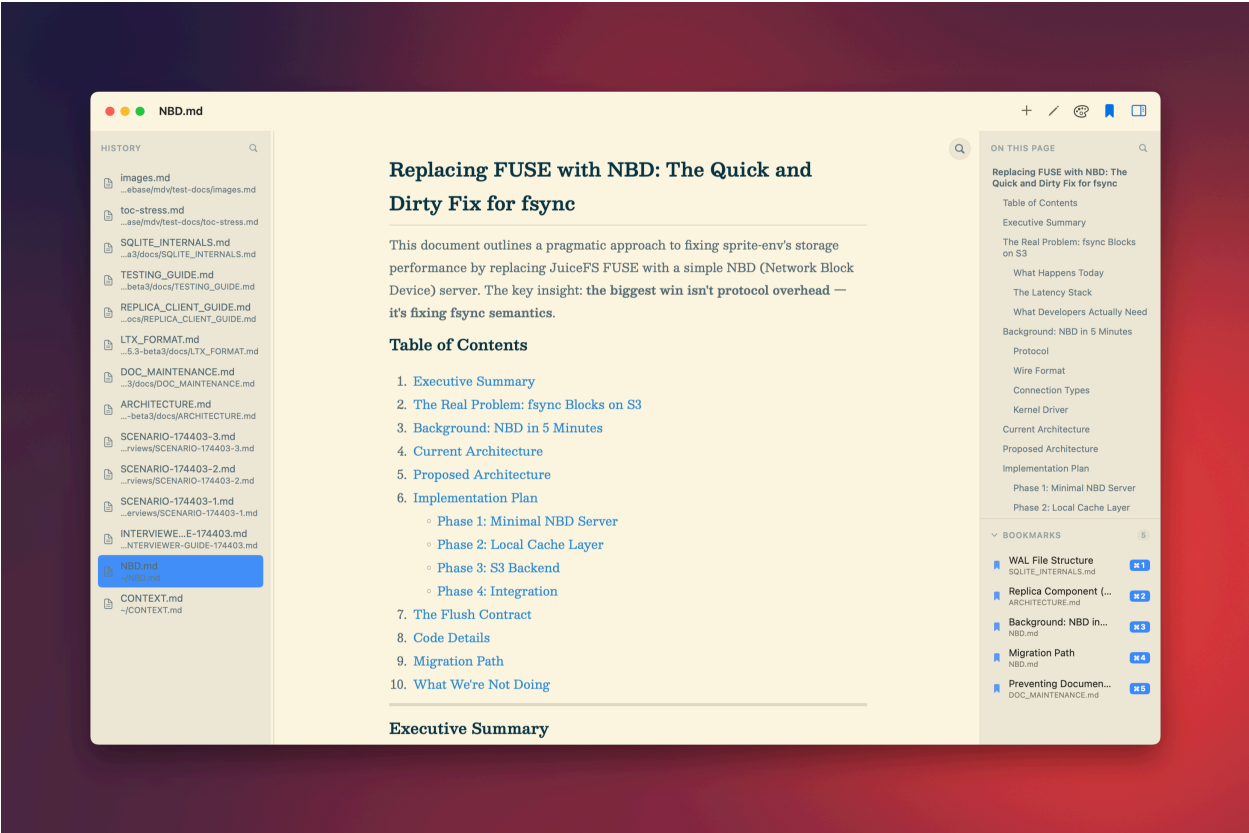


Figure 1: Reference — the original application, Sevilla, a long technical document (reference/MDV-SCREEN.png)

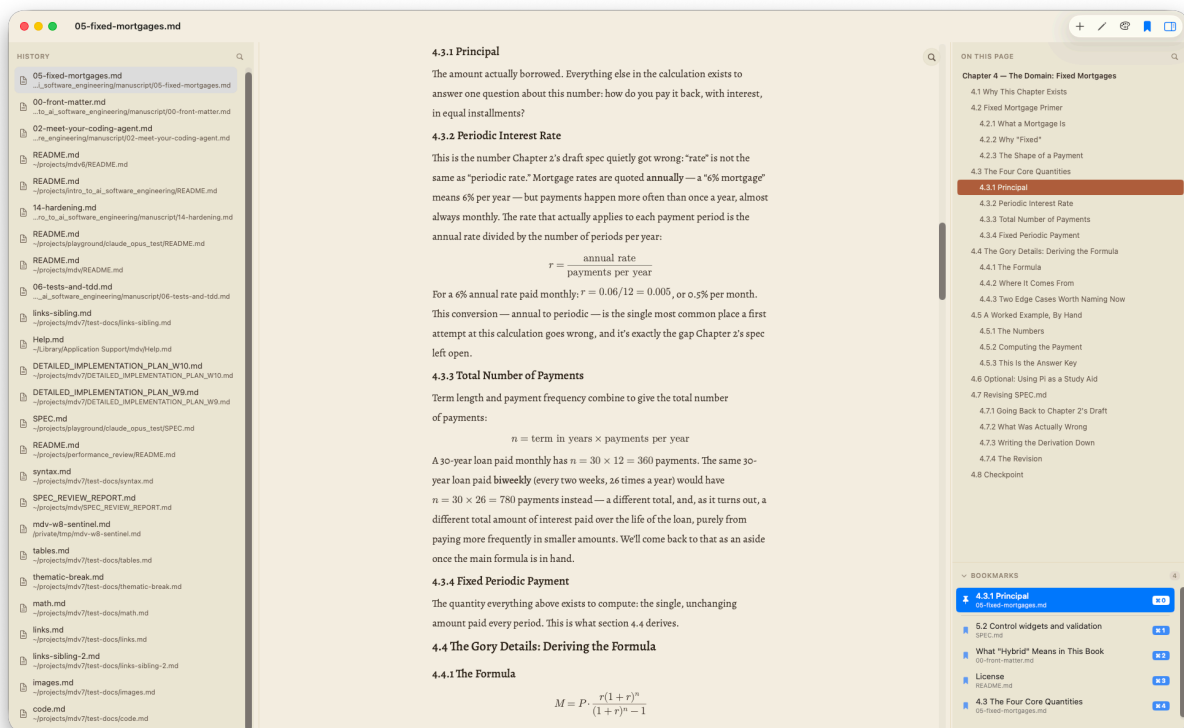


Figure 2: Reference — the original application, Sevilla, 05-fixed-mortgages.md with a placeholder and four bookmarks (reference/MDV-ORIGINAL-SEVILLA.png)

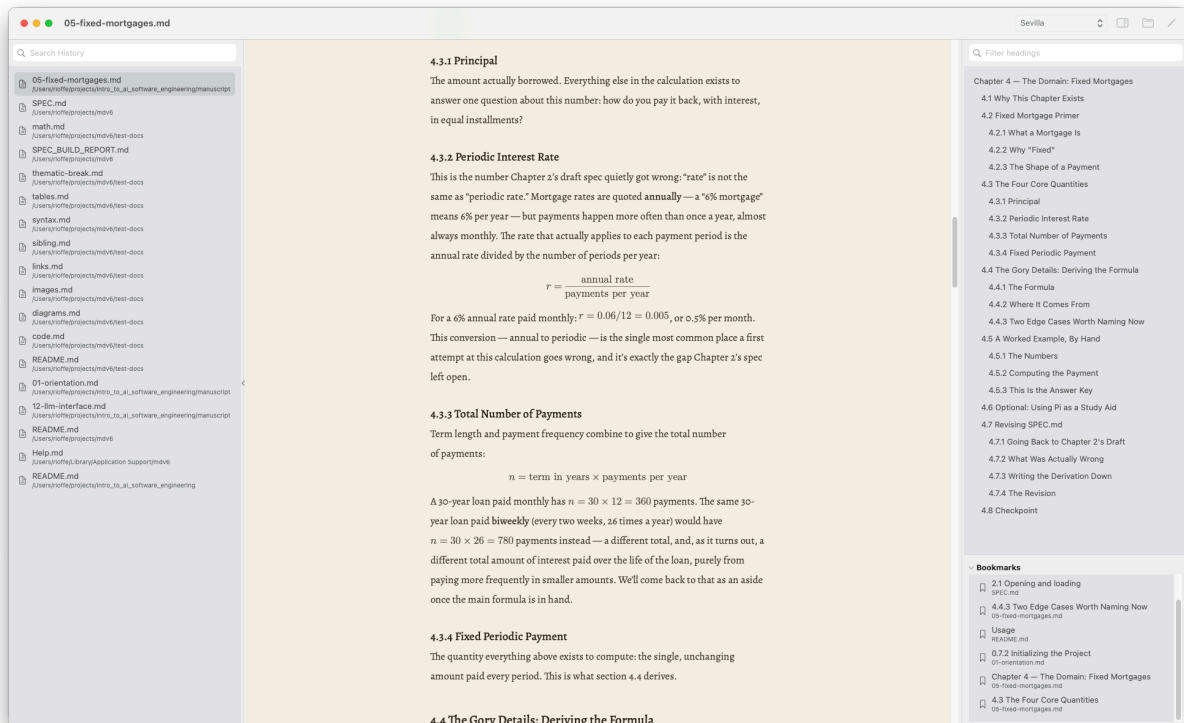


Figure 3: Gap example — v0.8 recreation mdv6: permanent search fields, full paths, no section headers, no TOC selection, no badges (reference/RECREATION-MDV6-SEVILLA.png)

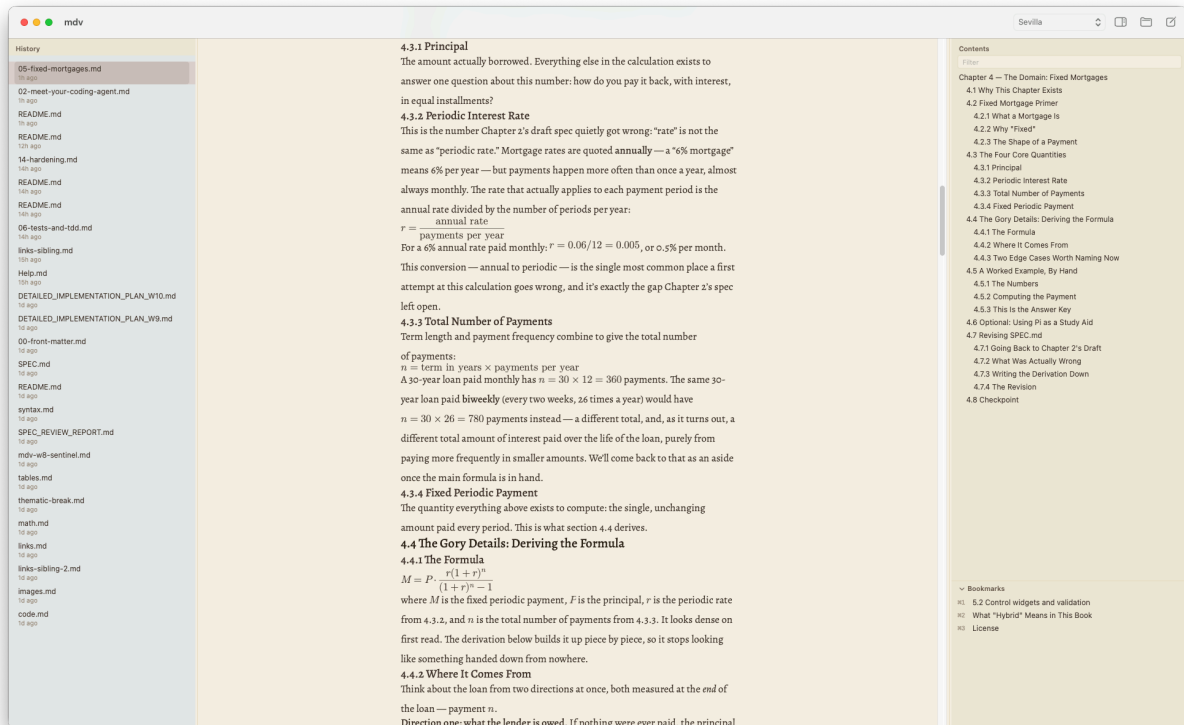


Figure 4: Gap example — v0.8 recreation mdv7: headings flush against the preceding paragraph, single-line display math leading-aligned at body size, window titled with the product name (reference/RECREATION-MDV7-SEVILLA.png)

C-18.1 Window. The window title is the displayed file’s last path component (`navigationTitle`), or the product name when the window is `EMPTY` (§3.1). The title-bar/toolbar strip is painted with the theme’s page background so title bar, article and both side panes read as one surface (see both reference images; the original does this in `ContentView` via `.toolbarBackground`). The window — title bar included — takes the theme’s colour scheme (light or dark per C-09 `isDark`), so the title is drawn in the system’s title colour for that scheme and stays legible on the strip; the application-wide appearance is not changed (the System theme keeps following macOS, R-29). The three panes are: history sidebar (left, R-20), article (centre), inspector (right, R-21); sidebar and inspector are separated from the article by a 1 pt divider in the theme’s border colour, each inside an 8 pt drag handle (§7.2). **Collapse chevron (R-20):** a `chevron.left` glyph (11 pt, tertiary colour) sits at the vertical middle of the sidebar’s handle and is visible only while the pointer is over the handle; clicking it collapses the sidebar. While the sidebar is collapsed, a `chevron.right` button of the same size sits at the left edge of the article at the same height and is always visible (there is no handle to hover); the inspector’s handle has no chevron (0 and the toolbar toggle it).

C-18.2 Toolbar. Exactly the five icon buttons of §5.1, trailing, in that order, with `.help` tooltips naming the shortcut. There is no labelled theme picker control: the theme is a pop-up under the `paintpalette` glyph. Nothing else sits in the toolbar.

C-18.3 Section headers. Each pane section has a header row: a label set in 11 pt semibold, **uppercase**, 0.6 pt tracking, in the theme’s tertiary text colour — `HISTORY` (sidebar), `ON THIS PAGE` (inspector, TOC), `BOOKMARKS` (inspector, bookmarks pane). Horizontal padding 14 pt; top 14 pt / bottom 6 pt for `HISTORY` and `ON THIS PAGE`; the `BOOKMARKS` header is a 32 pt tall row that is itself the collapse toggle (`chevron chevron.down/chevron.right`, 9 pt semibold, 10 pt wide, leading the label; R-21) and, when at least one bookmark exists, ends with a count capsule (10 pt medium, monospaced digits, tertiary text on `text.opacity(0.06)`, 5 pt/1 pt padding).

C-18.4 Reveal-on-click search. The `HISTORY` and `ON THIS PAGE` headers end with a `magnifyingglass` button (11 pt semibold, tertiary colour, 18 × 18 pt hit area, tooltip *Search history / Filter headings*). Clicking it replaces the button with the search field beneath the header (0.20 s ease-out) and focuses the field after 0.05 s; the field is not shown until then. Esc, or clearing and blurring the field, hides it again and restores the button. F while the sidebar has focus (R-24’s definition, E-18) reveals and focuses the history field. A permanently visible field (both v0.8 recreations) does not satisfy this row.

C-18.5 History rows (R-20). Each row is: `doc.text` glyph (13 pt, secondary text colour, 16 pt wide) · a two-line stack — **line 1** the file name (13 pt, text colour, one line, middle-truncated), **line 2** the file’s path with the home directory abbreviated to `~` (11 pt, secondary text colour, one line, **head-truncated** so the tail of the path stays readable) · 2 pt vertical padding. Rows sit in a `List(selection:)` with the `.sidebar` list style, so the displayed file’s row carries the system selection fill (accent when the sidebar is focused, neutral otherwise). Neither a relative time (`1h ago`) nor an untruncated absolute path satisfies this row. Empty history shows a centred `tray` glyph (24 pt light, tertiary) over *No files yet* (12 pt, secondary).

C-18.6 Global-search hits (R-25). A hit row shows the file name and path like C-18.5 plus the C-03 snippet (secondary colour); the row for the file currently displayed is filled with the accent colour with page-background-coloured text (F-109); a hovered row is `text.opacity(0.06)` (the theme’s text colour, C-09).

C-18.7 Article affordances. While the find bar is closed, a floating find button sits at the article’s top-trailing corner: a `magnifyingglass` glyph (13 pt semibold, secondary colour) on a 28 pt circle filled with the theme’s secondary background and stroked 0.5 pt in the border colour; clicking it opens the find bar (R-24). Heading rules (`showH1Rule` / `showH2Rule`, C-09) are drawn as a full-column divider in the theme’s divider colour under the heading, as in MDV-SCREEN.png. **Hovered-block stripe:** while the pointer is inside a block that is not a code or Mermaid fence — the block D and 0 would anchor (R-27, R-28) — a 3 pt vertical stripe in the theme’s accent colour is drawn along the block’s leading edge, inside the block’s 6 pt horizontal padding (§7.2, b), spanning the block’s height; it disappears when the pointer leaves. Fenced code and Mermaid blocks carry their own hover chrome (R-08, R-09) and show no stripe. **Precedence:** the stripe, the find tint (R-24) and the section flash (R-22) may all show on one block at once — the stripe is drawn in the padding beside the block, the tint beneath the block’s content, the flash above it. (TYPOGRAPHY.md calls

this the bookmark-hover stripe; the accent is chosen per theme so it does not read as system chrome.)

C-18.8 Table-of-contents rows (R-21). One row per C-02 heading, indented $14\text{ pt} \times (\text{level} - 1)$, text 12 pt (semibold for level 1, regular otherwise; level-1 rows in the text colour, deeper levels in the secondary colour), up to two lines, 8 pt/5 pt padding, 4 pt corner radius. A hovered row is filled `text.opacity(0.08)`. The row most recently chosen in the TOC — by click, or by a same-document **slug** fragment link or bookmark that lands on a TOC heading — is the **selected row** (a **line citation** never selects a row, C-19.4): filled with the theme accent, text in the page background colour, until another row is chosen or another document is loaded (E-29). Choosing a row pushes a back snapshot (R-18) and scrolls to the block.

C-18.9 Bookmark rows (R-27). Each row is: `bookmark.fill` glyph (11 pt, accent at 70 %, 16 pt wide; `exclamationmark.triangle.fill` in orange when the file is missing, E-09) · a two-line stack — the bookmark title (13 pt, tail-truncated) over the file **name** (10 pt, secondary, middle-truncated; not the path) · trailing **hotkey badge** for the five slot bookmarks: at 9 pt bold rounded followed by the slot digit at 10 pt bold rounded, page-background-coloured on `accent.opacity(0.85)`, 4 pt continuous corner radius, 5 pt/1 pt padding. Padding 8 pt/5 pt, 5 pt continuous corner radius. States: **current** (the bookmark the reader last jumped to, cleared when another document loads) — accent fill, text and glyph in the page background colour (white on every light theme with a dark accent, as in the reference images; F-109), badge inverted (accent on the page background colour); drop target while reordering — `accent.opacity(0.18)`; hovered — `text.opacity(0.06)`; missing file — row at 60 % opacity unless current. The R-28 placeholder, when set, is the **first** row of the pane, followed by a divider (8 pt horizontal inset, 2 pt vertical): `pin.fill` glyph (12 pt, accent; orange when the file is missing) · the R-27 title (**Frontmatter** for a block-0 header; 13 pt **medium**, middle-truncated) over the file name (10 pt, secondary) · the badge 0. At rest the row is tinted `accent.opacity(0.08)` with a 0.5 pt `accent.opacity(0.25)` stroke so it reads as distinct from ordinary bookmarks; hovered `text.opacity(0.06)`; *current* — accent fill, page-background-coloured text, inverted badge — from the moment it is set (0) or jumped to (0) until another document loads; missing file at 60 % opacity unless current. Clicking it is 0. Setting the placeholder expands a collapsed pane (`mdv6_bookmarks_expanded` ← true). It is not draggable and has no slot. Bookmark rows are reorderable by drag (R-27) and the pane height is draggable (R-21). **Context menus.** Right-clicking the placeholder row shows exactly one item, *Clear Placeholder* (R-28). Right-clicking a bookmark row shows, in this order: *Go to Bookmark* · *Reveal in Finder* · separator · *Move Up* · *Move Down* · *Move to Top* · *Move to Bottom* · separator · *Remove Bookmark*. *Go to Bookmark* is the row click (E-09 beep when the file is missing); *Reveal in Finder* selects the file in Finder and is disabled when the file is missing; the four *Move* items change the persisted order (R-27; the slots 1... 5 follow the new order) and *Move Up* / *Move to Top* are disabled on the first row, *Move Down* / *Move to Bottom* on the last; *Remove Bookmark* removes the row. The placeholder row is not part of this order.

C-18.10 Per-block rhythm and display math. See I-014 and C-07.1: an implementation that renders each C-02 block as its own Markdown view (the shape §3.2 implies) MUST re-apply the theme's `h{1,2,3}TopSpacing` / `h{1,2,3}BottomSpacing` and `paragraphBottomSpacing` at the block boundaries itself, because MarkdownUI's `markdownMargin` only acts between siblings inside one view (K-16 gives the numbers and tolerance). RECREATION-MDV7-SEVILLA.png shows the failure mode: every heading flush against the preceding paragraph, and `$$$` written on one line rendered leading-aligned at body size.

6. Invariants (must hold in every valid implementation)

ID	Invariant
I-001	Rendering is pure in its inputs: the same file bytes, theme, zoom, preferences, window content width, and backing scale produce the same blocks, TOC, and rendered output; no render path reads the network except the C-16 remote-image fetch gated by R-16 — the R-46 web path included, whose page loads nothing beyond itself (C-06.5); a web-path diagram’s rendering is a function of the same inputs plus the bundled mermaid.js, and its measured height may differ by less than 0.5 pt between runs (C-06.5 ignores such reports). Print output (C-21) is a function of the document, the print-relevant preferences (smart typography, <i>Show Frontmatter</i> , the native diagram style) and the paper (I-017).
I-002	The application MUST NOT terminate because of document content admitted by K-14. Every third-party parser is reached only after the applicable limit check and sanitiser (C-06.1, C-07.1/2), and every detected parse/layout failure becomes a fallback block.
I-003	Document content MUST NOT reach a log, subprocess, or network request except that the Mermaid source of a web-dispatched diagram (R-46) reaches WebKit’s web-content process, which makes no network request (C-06.5), and C-16 MAY disclose the original remote-image URL and permitted redirect URLs after the reader enables R-16. No other document bytes may enter request headers or bodies. The only other externally visible document-derived artefacts are the user’s pasteboard on explicit copy, a user-chosen PNG on export, the print job or PDF the user asks the print panel for (R-45), and mdv6.db. The in-process mdv6-math:// and mdv6-img:// schemes are not external.
I-004	The block split (C-02) is computed at most once per distinct <code>raw</code> string per load; <code>blocks[i]</code> is stable for the life of the document, so block indices used by find, TOC, heading copy, bookmarks, and scroll anchors refer to the same text.
I-005	Every mermaid raster is displayed at exactly its own point size — <code>displaySize(for:width:)</code> is the single source of both the bitmap size and the view frame — so the diagram is never resampled by the view layer.
I-006	All access to <code>mdv6.db</code> goes through one connection opened <code>FULLMUTEX</code> , in <code>WAL</code> mode; concurrent use from the history re-index queue and the main thread is serialised by SQLite, never by the caller.
I-007	Persistence writes are whole-row <code>INSERT ... ON CONFLICT DO UPDATE</code> or single-statement updates; a crash mid-write leaves the previous row, never a partial one.
I-008	Every <code>UIImage</code> handed to SwiftUI <code>Text/Image</code> for math is bitmap-backed (not drawing-handler-backed); idle CPU with math on screen is that of a static page.
I-009	Math drawn inside a Mermaid raster is drawn at a pixel-aligned origin; its ink weight, per the formula in §7.1, is at least $0.9\times$ that of the same expression typeset for the document at the same size and scale.
I-010	Heading slugs (C-11) are computed from the un-mathed heading text, so <code>#fragment</code> links written for GitHub resolve identically whether or not the heading contains <code>\$...\$</code> .
I-011	The vendored SwiftMath carries exactly the patches listed in <code>Vendor/SwiftMath/README.md</code> ; everything else is byte-identical to upstream v1.7.3.
I-012	Smart typography never changes bytes inside code spans, fences, link URLs, <code><...></code> spans, GFM tables, thematic breaks, or math.
I-013	The history list never exceeds 100 entries and never contains duplicates; the path most recently added by an adding route is first, and a selecting route never changes that order.

ID	Invariant
I-014	Vertical rhythm is a property of the document, not of the view tree: for every adjacent pair of C-02 blocks, the gap between the last ink row of block i and the first ink row of block $i + 1$ is the same, within the K-16 tolerance, whether the blocks are rendered by one Markdown view or one view per block — and in both cases it lies in the K-16 band around the value <code>TYPOGRAPHY.md</code> assigns for that pair (the margin between siblings: heading top spacing by level, paragraph bottom spacing, combined as $\max(\text{bottom}_i, \text{top}_{i+1})$ as MarkdownUI combines them). Ink rows sit inside the line box, so the ink gap exceeds the margin by the two fonts’ leading; the band, not equality, is the invariant (F-107).
I-015	Chrome structure is theme-independent: every element, row anatomy, badge and state named in C-18 is present under every C-09 theme; only the colours and faces change. A build whose panes differ structurally from <code>reference/MDV-ORIGINAL-SEVILLA.png</code> is non-conforming even when every §9.1–9.5 test passes.
I-016	The vendored MarkdownUI (<code>Vendor/MarkdownUI</code> , upstream v2.4.1) carries exactly the changes listed in <code>Vendor/MarkdownUI/README.md</code> : the <code>markdownResolvedInlineImages(_:)</code> environment hook that lets a caller hand already-resolved inline images to <code>InlineText</code> (empty by default, so on-screen rendering is unchanged and a <code>.task</code> -loaded image wins over a supplied one for the same key), plus warning-only source fixes; everything else is byte-identical to upstream.
I-017	Print is independent of the screen’s presentation state: the same document, preferences (smart typography, <i>Show Frontmatter</i> , <code>mdv6.mermaid.style</code>) and paper produce the same printed pages whatever the screen theme, zoom, window width, scroll position, find state, hover or selection.
I-018	A frontmatter header is block 0 whether <i>Show Frontmatter</i> is on or off; toggling it changes no block index, fingerprint, <code>blockLines</code> entry, TOC row, bookmark resolution, scroll anchor or find count.

7. Constraints (precise and measurable)

ID	Constraint
K-01	Platform: macOS $\geq$ 13.0, Apple Silicon or Intel; toolchain: Swift $\geq$ 5.9 (<code>swift-tools-version: 5.9</code>); no Xcode project — <code>swift build + build.sh</code> only; WebKit (system framework) for R-46.
K-02	Bundle: <code>CFBundleIdentifier com.mdv6.app</code> ; version 1.0.0 (1); not sandboxed; entitlement <code>files.user-selected.read-only</code> .
K-03	History cap 100 entries. Global search returns at most 80 hits; snippets are 14 tokens. Bookmark hot-key slots: 5.
K-04	Zoom: step 0.10, range [0.60, 2.50], default 1.0. Sidebar width [180, 400] pt (not persisted); inspector width [180, 520] pt (persisted, default 240); bookmarks pane $\geq$ 120 pt with the TOC keeping $\geq$ 80 pt.
K-05	Highlighted languages, seventeen fence words over eighteen vendored grammars: C, Go, Rust, Bash, JavaScript, YAML, TOML, Python, Ruby (the original nine), Swift and SQL (R-38), C++, JSON, Lua, OpenCL, Perl and Markdown (R-43); a <code>metal</code> fence is highlighted by the C++ grammar (D-45) and markdown uses two grammars (C-05). Grammar commits are pinned in <code>mdv6/Grammars/README.md</code> .

ID	Constraint
K-06	Live-reload event latency: 50 ms (no deferral, R-05); transient zero-byte re-read: 500 ms (D-18). Block flash (heading copy R-22, line citation C-19.3): 0.6 s. Zoom HUD: 0.9 s. Scroll-restore mtime tolerance: 1 s (the stored <code>file_mtime</code> is truncated to whole seconds, C-08); index mtime gate (R-26): whole seconds, equality. Bookmark-title heading look-back: 40 blocks; bookmark-title fallback: 60 extended grapheme clusters (R-27).
K-07	For natively dispatched diagrams (the web path’s width rule is C-06.5’s): Mermaid raster width MUST use exactly the §7.2 formula, including its 1 pt lower bound, at the screen backing scale; pinch zoom is clamped to $[0.5, 4]$; zoomed container height is ≤ 540 pt. Caches: 96 layouts, 192 rasters, 192 MB.
K-08	Math: $\$...\$$ spans typeset in <code>.text</code> style, every $\$...\$$ span in <code>.display</code> whether on its own line or mid-line (C-07.1); diagram-label math at 16 pt; message-label line pitch 13 pt; sequence row height 40 pt (library) grown by $(n - 1) \times 13 + 4$ pt for n -line labels. Cache: 2048 images.
K-09	Fingerprints: 80 extended grapheme clusters after the C-08 normalization. FTS tokenizer <code>unicode61 remove_diacritics 2</code> .
K-10	Typography defaults: body 16 pt, line spacing 0.30 em, heading scales $h_1 = 1.75$, $h_2 = 1.40$, and $h_3 = 1.15$, article max width 860 pt (the cap on the padded article frame, K-13), gutter 40 pt (per-theme overrides in <code>TYPOGRAPHY.md</code>).
K-11	Release artefacts: <code>dist/mdv6-<version>-macos.zip + .sha256</code> , Developer ID signed with hardened runtime and timestamp, notarised and stapled; <code><version></code> equals the tag without <code>v</code> .
K-12	Ad-hoc-signed development bundles MUST pass <code>codesign --verify --deep --strict</code> ; nothing may be placed at the bundle root besides <code>Contents/</code> .
K-13	<i>Column width</i> w_{col} (used by R-11, K-07): the width a block’s content is laid out in, per the formula in §7.2. The cap applies to the padded article frame before the paddings are subtracted, so with the K-10 defaults a wide window gives $w_{\text{col}} = 860 - 80 - 12 = 768$ pt and a Mermaid raster of 732 pt.
K-14	Untrusted-content ceilings, measured before the named operation: UTF-8 document file 64 MiB; one Mermaid source 1 MiB; one LaTeX span 64 KiB; encoded or compressed local/data/remote image 32 MiB; decoded image 64 megapixels, 256 MiB, and 16,384 pixels on either axis. MiB and KiB are binary units. C-16 additionally limits redirects and time. Content at the ceiling is admitted; content above it follows E-28.
K-15	Idle-math CPU protocol: after <code>test-docs/math.md</code> has been visible and untouched for 5 s, collect 30 one-second process-CPU samples with no pointer, keyboard, window, appearance, or file activity on an otherwise idle <code>macos-15</code> host. The median MUST be $\leq 1\%$ and the nearest-rank 95th percentile MUST be $\leq 3\%$.
K-16	Chrome metrics (C-18): section header 11 pt semibold uppercase, tracking 0.6 pt; history row 13 pt / 11 pt; TOC row 12 pt, indent 14 pt per level; bookmark row 13 pt / 10 pt, badge 9 pt + 10 pt bold rounded; search button 18×18 pt; floating find button 28 pt circle, 0.5 pt stroke; hovered-block stripe 3 pt wide in the accent colour; row corner radii 4 pt (TOC) and 5 pt (bookmarks); reveal animation 0.20 s. Rhythm (I-014): let v be the theme’s margin for a block pair — Sevilla <code>h2Top</code> 32 pt, <code>h3Top</code> 22 pt, paragraph bottom 14 pt; defaults 24 / 24 / 16 (<code>TYPOGRAPHY.md</code> §Per-element vertical spacing) — and f the larger of the two blocks’ font sizes in points (body size, or body size $\times$ the heading’s em factor, C-09). The inter-block ink gap g measured on a $2 \times$ raster MUST satisfy $v \leq g \leq v + 0.6f$, and the per-block rendering’s g MUST be within ± 2 pt of the single-Markdown-view rendering’s g for the same pair (F-107: a heading drawn flush against its paragraph fails the lower bound; a doubled padding fails the upper).

ID	Constraint
K-17	Print (C-21): margins 54 pt on every side; type scale $s_p = \min(1, w_c / \text{articleMaxWidth})$ (Letter 0.586); inter-block gap $s_p \times$ the screen gap for the pair (C-21.2); inline-math raster density 12 px/pt (864 ppi); diagram fallback raster density 4 px/pt (288 ppi); diagram chrome insets $18 s_p$ pt horizontal, $12 s_p$ pt vertical; standalone-formula padding $4 s_p$ pt; web-diagram height wait 5 s, polling 50 ms until three equal reads or 3 s, 200 ms settle before capture; <code>heightAdjustLimit</code> 0.9.
K-18	Keyboard scrolling (R-49): line step 40 pt; page step $\max(0.875 u, 40)$ pt where u is the clip view’s height minus its top and bottom content insets; <i>End</i> re-aims at the bottom every $1/60$ s up to 10 times, stopping as soon as the landing position stops changing.
K-19	Mermaid web path (R-46, C-06.5): mermaid.js 11.4.1, SHA-256 <code>a43bc1afd446f9c4cc66ac5dd45d02e8d65e26fc5344ec0ef787f88d6ddb6f9e</code> ; minimum block height 60 pt; padding 12 px vertical / 18 px horizontal (24 pt added to the SVG height); height reports within 0.5 pt of the last are ignored.

7.1 Ink-weight metric (I-009, T-17)

Let P be the pixels of a crop, at $2\times$ backing scale, whose bounds are the math image’s rectangle enlarged by 4 px on each side, composited on white; $g(p) \in [0, 255]$ the luminance of pixel p ; and $D = \{p \in P : g(p) < 200\}$ the inked pixels. Then

$$\text{ink}(P) = \frac{1}{|D|} \sum_{p \in D} (255 - g(p)), \quad \text{ink}(P) = 0 \text{ when } D = \emptyset.$$

I-009 holds when $\text{ink}(P_{\text{node}}) \geq 0.9 \cdot \text{ink}(P_{\text{doc}})$ for the same LaTeX at the same font size; an empty D on either side is a failure.

7.2 Column width (K-13, R-11, T-18)

Let w_{area} be the window’s content width; w_{side} and w_{insp} the history sidebar and inspector widths **plus their 8 pt drag handles** when shown, and 0 when hidden; w_{max} the theme’s `articleMaxWidth`, or ∞ when the theme sets none; p the theme’s `articleHorizontalPadding`; and $b = 6$ pt the per-block horizontal padding. Then

$$w_{\text{col}} = \min(w_{\text{area}} - w_{\text{side}} - w_{\text{insp}}, w_{\text{max}}) - 2p - 2b.$$

A Mermaid raster is drawn at $\lfloor \min(\text{natural}, \max(w_{\text{col}} - 36, 1)) \rfloor$ pt (R-11, K-07).

8. Edge cases and failure semantics

ID	Case	Semantics
E-01	Mermaid node listed in two subgraphs (A <code>--></code> B inside <code>subgraph X</code> , B declared in <code>subgraph Y</code>).	Ownership normalised to the last subgraph before layout (C-06.2); renders. Without this the ELK importer’s <code>assert</code> aborts the process — the historical launch-crash.

ID	Case	Semantics
E-02	Mermaid diagram type the library lacks (<code>timeline</code> , <code>gantt</code> , <code>pie</code> , <code>mindmap</code> , <code>gitGraph</code> , ...); a parse or layout error on either path.	A type the library lacks is dispatched to the web path and renders there (R-46, C-06.4); only a failure — the native library rejects a natively dispatched source, or mermaid.js rejects a web-dispatched one, produces no SVG or reports no positive height — shows the fallback block: “Mermaid diagram could not be rendered” + source. A natively dispatched source the library cannot parse is not retried on the web path. In the C-17 harness a web-dispatched case reports web .
E-03	File unreadable (permissions, not UTF-8, vanished between open and read) — on any route, including a sidebar row or ← whose file was deleted since.	Load aborted; with a previous document, the window keeps that document, selection, and watcher and no history entry is added or moved. With no previous document, the window enters EMPTY ; an initial history-head row remains in history and later rows are not tried (R-40).
E-04	Directory with no Markdown files.	Nothing loads; no history change.
E-05	Link to a local Markdown path that does not exist.	Handed to the system opener (which reports the failure); no navigation.
E-06	Fragment with no matching heading slug, or invalid UTF-8 percent encoding.	A same-document fragment is a no-op. A local path plus fragment still loads the target but remains at its top; R-06 restoration is suppressed (R-19). No error is shown. A fragment that matches C-19.1’s grammar is a line citation and never reaches this row (E-31).
E-07	\$ in prose that is not math: \$5 and \$10, \$5-\$10, \$100/month, \ \$x\ \$, \$HOME in code, an empty \$\$.	Left literal by the delimiter rules of C-07.1 (opening followed by space, closing before space or followed by a digit, backtick crossing, escape, empty span).
E-08	Bookmark or scroll anchor whose block moved or changed.	Resolve by fingerprint first, then clamped index (C-08); a scroll position is discarded (start at top) if the file’s mtime differs by more than 1 s or the index is out of bounds.
E-09	Bookmark whose file no longer exists.	Row shown as missing in the inspector; opening it by row or numbered shortcut beeps and does nothing else; the row remains until removed.
E-10	LaTeX SwiftMath rejects (unknown command, unbalanced braces).	Source shown in monospace at 0.9× size (inline) or with the parser’s message (display); never blank.
E-11	Remote image loading disabled; or enabled but redirect, timeout, status, media type, byte ceiling, decode, or image-dimension validation fails.	Disabled: “Remote image blocked” placeholder that reveals the View menu item. Enabled failure: an explicit failure placeholder; partial bytes are discarded and no response body is persisted (C-16).
E-12	mdv6.db cannot be opened or a statement fails.	NSLog("[mdv6] ..."); the feature degrades (no search hits, no bookmarks, no scroll restore); viewing continues.
E-13	Mermaid in a sequence message that makes the label wider than its actors’ gap, or taller than a row.	Gap widened and rows expanded per C-06.2; labels never cross a lifeline they don’t span and never overlap the previous arrow.

ID	Case	Semantics
E-14	Mermaid <code>style ... fill:#eee / fill:white.</code>	Normalised to 6-digit hex (C-06.1) — rendered as the intended colour, not black.
E-15	<code>xchart</code> with line <code>"name" [...]</code> .	Series rendered; legend shows Line n (the library has no series name field).
E-16	Math that is the <i>entire</i> content of a table cell or list item.	Rendered through the block-image path at body size (the mode still follows the delimiter, K-08), leading-aligned, not centred (MarkdownUI routes image-only paragraphs there).
E-17	The find query matches inside a code fence, a <code>\$\$</code> math fence, a GFM table, or any block that contains <code>!</code> .	Block tinted as a whole; no character-level highlight. Every other block is inline-highlighted per R-24, so an occurrence that lies only inside markup (<code>**bold**</code> 's asterisks, a link URL, a <code>#</code> marker) is counted in <i>m</i> but not marked, and a heading containing <code>\$...\$</code> shows its LaTeX source while the find bar is open.
E-18	F while the history sidebar has focus (first responder inside the sidebar, R-24).	Routes to global search (R-24): the HISTORY search field is revealed if hidden and focused (C-18.4); the document find bar does not open.
E-19	Reload of the current file while text is selected.	Scroll position kept; the text selection is not preserved.
E-20	Same file opened in two windows and edited on disk.	Each window's watcher reloads independently; scroll positions are per path, last writer wins.
E-21	The displayed file is deleted, replaced by an atomic-rename save, or read mid-save as empty / not UTF-8.	Rename: reload with the new content (R-05). Delete without re-creation, or an undecodable read: content and scroll position kept, no error, watch stays armed indefinitely; a later re-creation or completed write reloads. A zero-byte read while content is displayed: page kept, re-read after 500 ms, and that read is shown — so a file that really was emptied appears empty 500 ms after the event.
E-22	A <code>slug #fragment</code> whose target is an h4–h6 heading or a setext (<code>===/---</code>) heading; a click or hover on such a heading.	Slug fragment: no-op — only <code>#-###</code> single-line ATX headings are slug targets (C-02, D-17). A line citation is unaffected: it resolves by line whatever heading level that line falls in (C-19.3). Click/hover: the heading is prose for R-22 — text-selectable, arrow pointer, no section copy.
E-23	Fence closed by a longer/shorter backtick run than the opener; a fenced block that is never closed; an indented (four-space) code block containing a blank line.	Splitter semantics of C-02 (fence, math-fence, and indented-code rules): closes on any run of the same three-character marker; runs to end of input; the indented block is split into two blocks and renders as two.
E-24	An empty or whitespace-only global search query.	No search is performed; the results list is empty (C-03).
E-25	Rendering in flight (diagram layout, math typesetting) when the document changes.	The in-flight task is cancelled; its result is discarded, never shown for the new document.

ID	Case	Semantics
E-26	Two windows open (<code>O</code> is the only way to get a second one; E-30) and a menu command (<code>O</code> , <code>F</code> , <code>D</code> , <code>←</code> , <code>E</code> , ...) or an open event (Finder, <code>open -a, bin/mdv6 FILE</code>) arrives.	Only the key window acts (R-01); the other window is unchanged — no second open panel, no second bookmark, no navigation, no file load. The one exception is <i>Close All</i> , which empties every window because history is shared (R-47, E-36, D-50).
E-27	<code>0</code> (placeholder) or <code>←/→</code> (snapshot) whose file no longer exists on disk; a snapshot whose history row was removed.	Missing file: <code>0</code> beeps and keeps the placeholder (as E-09); <code>←/→</code> beeps, discards that snapshot, and the view is unchanged. Removed row: the snapshot was already dropped (R-18), so <code>←</code> skips to the next one.
E-28	Untrusted content exceeds a K-14 ceiling.	Reject it before the applicable third-party parser or decoder. An oversized document load aborts exactly as E-03. Oversized Mermaid or LaTeX shows the R-10/R-14 source fallback with “input exceeds limit”. An oversized image shows the E-11 failure placeholder. A remote request is cancelled as soon as its body crosses the ceiling; partial bytes and partial decoded output are discarded. Viewing other blocks continues.
E-29	The TOC selected row (C-18.8) after scrolling away, after a reload (R-05), and after loading another document.	Scrolling does not move or clear it (selection is choice-driven, not viewport-driven — D-41). A reload keeps it when the heading still exists at that block index, else clears it. Loading a different document clears it; a cross-file slug fragment (R-19) that lands on a TOC heading selects that row. A line citation never selects a row, including one that lands on a TOC heading (C-19.4).
E-30	Quit with two or more windows open (<code>O</code>), then launch again.	Exactly one window opens, and it follows R-40 (the history head, or the cold-start argument). Windows are not restorable (<code>NSWindow.isRestorable</code> false) and the application does not opt into state restoration; the second window’s document is reachable through history.
E-31	A line citation whose line does not exist in the target: past <code>lineCount</code> , inside a gap the C-02 split removed, before the first block, or the non-citation <code>#L0</code> .	#L0 does not match the C-19.1 grammar, is not a line citation, and falls through to C-11 slug matching (E-06). A line past <code>lineCount</code> or in a gap resolves to the last block whose first line is at or before it; a line before the first block resolves to the first block (C-02 rule 8). Both are scrolled to and flashed like any other citation — no error, and never a refusal to load the target file. A line citation into a document with no blocks scrolls to the top and flashes nothing.

ID	Case	Semantics
E-32	A document that opens with <code>---</code> but is not frontmatter: a thematic break followed by a paragraph and another <code>---</code> ; a first line <code>---</code> with a trailing space or <code>---</code> indented; an opening fence with no closer; a YAML candidate with one prose line (<code>TODO: reprint</code> passes the colon test, but the line after it, a sentence with no mapping colon, fails); a candidate with no rows — <code>---</code> , blank, <code># Title</code> , blank, <code>---</code> , or <code>---</code> directly followed by <code>---</code> .	Not a header (C-20.1): the document renders exactly as it would without frontmatter support — rules, paragraphs, a setext heading where CommonMark makes one — and <code>frontmatter</code> is nil. A TOML candidate (<code>+++</code>) with any content is a header once it has its closer. <code>test-docs/frontmatter-negative.md</code> is the fixture.
E-33	P with no document (EMPTY) or one with no blocks; a block whose inline formulas C-21.4 cannot place; P while a print panel is up; a block taller than one printable page; a Mermaid fence that fails on both print routes; a remote image; a find bar open with matches.	Beep, nothing else, for the first and third (R-45). The unplaced block prints its inline formulas as 12 px/pt bitmaps and the rest of the page is unaffected. The tall block is sliced at AppKit’s break (C-21.5). The failed fence prints its source as a plain code block (C-21.3). Remote images print as the blocked placeholder without any fetch, whatever <i>Load Remote Images</i> says. Find tint and marks never print (I-017).
E-34	A web-path diagram: mermaid.js rejects the source; the page never reports a height; the source contains <code>click</code> callbacks, links, <code></code> or other HTML in labels; the network is unavailable.	Rejection or no SVG → R-10 fallback. No report → the block keeps its spinner on screen at 60 pt (there is no on-screen timeout), and in print the 5 s wait ends in the source fallback (C-21.3). <code>securityLevel: 'strict'</code> sanitises labels and disables callbacks, and every load beyond the page itself is refused (C-06.5), so nothing is fetched or navigated to. Rendering needs no network: the script is bundled.
E-35	<code></code> edge cases: no <code>src</code> or an empty one; <code>width="50%", width="0", width="auto"</code> ; <code>src</code> with a scheme other than <code>file</code> , <code>data</code> , <code>http(s)</code> ; an <code></code> in a code span or fence; an <code><img</code> with no closing <code>></code> ; a <code>\$</code> or backtick inside <code>alt</code> .	No or empty <code>src</code> → the tag stays literal text. A width or height that is not a positive number after dropping <code>px</code> is absent (the other attribute or the natural size decides). Other schemes follow R-16’s classification for an image URL (not fetched; the failure placeholder). Code keeps the tag literal; a missing <code>></code> leaves the <code><</code> and the rest literal. <code>alt</code> travels base64url-encoded, so it can never open a math span or code span (C-22.1, §3.2).

ID	Case	Semantics
E-36	<i>Close</i> All confirmed with two windows open (O), with a load in flight in another window, or while a print panel is up.	History, the index and scroll positions are shared, so every window loses its row: an in-flight load or reload in any window is cancelled first and its result discarded (as E-25 discards a render), so no adding route can re-add a row after the clear; then every window's stacks are cleared and every window enters EMPTY (R-47). A print already handed to the panel keeps its pages; the job is unaffected. Bookmarks survive and still open their files (adding route).
E-37	Keyboard-scroll keys while the find field, the history search field or the TOC filter is focused; while the history list has focus; while a code block's horizontal scroller is under the pointer; with held (↓ is not a scroll key here).	The focused control keeps the key (R-49): the field edits, the list steps its selection. Pointer position is irrelevant — the document's vertical scroller is found from the article, never from the view under the pointer. A held / / passes the key through untouched.
E-38	No document window exists — the reader closed the last one (W or the close button) and the application is still running — and an open event (Finder, open -a, bin/mdv6 FILE), O, a bookmark, 0, ?, or a Dock click arrives.	Exactly one window is created and becomes the target (R-01), and only when the action has a file to load: an open event, O's chosen file, a bookmark's or the placeholder's file, or Help loads into it as its route says; a cancelled O panel, 0 with no placeholder, or a bookmark or placeholder whose file is missing creates no window (the latter two beep); a Dock click's window follows R-40 (the history head, or EMPTY). Window-scoped commands are disabled. Two open events arriving together create one window, not two; the last URL is displayed (R-01).

9. Acceptance criteria, tests, and evals

9.0 Status and target

The repository carries the automated suite of R-37 (`swift test: Tests/mdv6Tests, Tests/mdv6RenderTests`) and the C-17 harness with its corpus and manifest (R-39: `tools/render-harness/, test-docs/mermaid/, test-docs/render-cases.json`). Every test below names the ids it proves and cites them literally in code, so `tools/speccheck.sh` (the grep walk standing in for `speccheck`, which is Python-only) can join ids to tests. “Renders” means: the C-17 case reaches its declared `render` expectation, produces the required output, and leaves no crash report in `~/Library/Logs/DiagnosticReports/mdv6-*.ips`.

The shape of the suite — which group each test lives in:

Group	Target	What moves there	Runs
Visual parity (manual against reference /, plus two scripted rhythm checks)	window chrome (C-18), block-boundary rhythm (I-014), single-line display math (C-07.1)	T-44 , T-45 , T-46 , T-47 , T-48	T-45/T-46 in swift test via the C-17 harness; T-44/T-47/T-48 by eye on an isolated store (T-48 's menu order, enablement and reorder also in swift test), before any conformance report is written

The app code is the library target `mdv6Core` with a thin executable (`App/main.swift`), so the tests `@testable import mdv6Core` and the harness links the pipeline instead of copying it; `Database(url:)` and `AppModel.bootstrap` inject the store and defaults.

9.1 Build, bundle, launcher (scripted)

ID	Test
T-01	Fresh clone, <code>make</code> → <code>build/mdv6.app</code> exists with every file in C-13 present; <code>codesign --verify --deep --strict build/mdv6.app</code> exits 0; <code>otool -l</code> shows a minimum OS of 13.0 and <code>Info.plist</code> the identifier/version of K-02 . Proves R-34 , K-01 , K-02 , K-12 , C-01 , C-13 .
T-02	On an untagged commit, both <code>make dist</code> and <code>make dist VERSION=9.9.9</code> exit non-zero at <code>check-version</code> before <code>clean</code> or any build command runs. Proves R-34 and the negative gate of K-11 .
T-03	<code>bin/mdv6 --version</code> prints 1.0.0; <code>bin/mdv6 nope.md</code> prints <code>mdv6: no such file: nope.md</code> to stderr and exits 1; <code>echo '# hi' \</code> <code>bin/mdv6 -</code> opens a window showing “hi”; <code>MDV6_APP=/nonexistent</code> <code>bin/mdv6</code> falls through the search order. <code>bin/mdv6 a.md b.md</code> : both appear in history, <code>b.md</code> is displayed. Proves R-01 , R-33 , §5.2.
T-04	<code>open build/mdv6.app test-docs/</code> loads <code>README.md</code> and the sidebar lists the other <code>.md</code> files. <code>chmod 000</code> a file and open it: the window keeps its previous document and history is unchanged. Drag a <code>.mkd</code> file onto the window: it opens; drag a <code>.pdf</code> : nothing happens; drag <code>a.md</code> and <code>b.md</code> together: only the first opens. A directory holding <code>B.md</code> and <code>a.md</code> and no <code>README</code> opens <code>a.md</code> ; a hidden <code>.notes.md</code> is neither opened nor listed. Proves R-02 , R-03 , E-03 , E-04 (an empty directory changes nothing).

ID	Test
T-43	In a disposable release checkout whose HEAD has exact tag v1.2.3, with valid signing and notarization credentials, <code>make dist</code> produces <code>dist/mdv6-1.2.3-macos.zip</code> and its <code>.sha256</code> ; the checksum verifies, <code>codesign --verify --deep --strict</code> and <code>spctl</code> succeed on the extracted app, <code>stapler validate</code> succeeds, and no artifact name contains a command-line version. Proves R-34 and K-11 .

9.2 Rendering (manual, test-docs/)

ID	Test
T-05	<code>test-docs/syntax.md</code> : every GFM construct renders (tables, task lists, footnotes, strikethrough). Proves R-07 .
T-37	<code>test-docs/code.md</code> gains a <code>swift</code> block (a <code>struct</code> with a <code>@Published</code> property, a <code>guard let</code> , a string interpolation, a <code>// MARK:</code> comment) and a <code>sql</code> block (<code>CREATE TABLE</code> , a <code>SELECT ... JOIN ... WHERE</code> with a string literal, a <code>-- comment</code>): keywords, strings, comments, numbers, types and function names are each coloured differently from plain text, and a <code>postgresql</code> -tagged block highlights identically to <code>sql</code> . Proves R-38 , C-05 , K-05 .
T-51	<code>test-docs/code.md</code> gains a <code>cpp</code> block (a <code>struct</code> , a <code>std::uint32_t</code> field, a string literal, a <code>// comment</code>), a <code>metal</code> block (<code>kernel void</code> with a <code>[[buffer(0)]]</code> attribute and a <code>float4</code> type), an <code>opengl</code> block (<code>__kernel void</code> with <code>__global const float *</code>), a <code>json</code> block (a key, a string, a number, <code>true</code>), a <code>lua</code> block (<code>local function</code> , a <code>-- comment</code> , a string), a <code>perl</code> block (<code>my</code> , <code>sub</code> , a <code># comment</code> , a string) and a <code>markdown</code> block (a heading, a paragraph with strong emphasis, emphasis, a code span and a link, a list item, a quote): in each, keywords, strings, comments, numbers, types and function names — where the language has them — are coloured differently from plain text; in the <code>markdown</code> block the heading text, the strong emphasis, the emphasis, the code span, the link label and the link destination are each coloured, which needs the inline grammar; and a <code>c++/cc/metal</code> block highlights identically to <code>cpp</code> , a <code>c1</code> block to <code>opengl</code> , a <code>pl</code> block to <code>perl</code> , an <code>md/gfm</code> block to <code>markdown</code> . Proves R-43 , C-05 , K-05 .
T-06	<code>test-docs/code.md</code> : each of the nine languages is coloured; an unknown fence (<code>```brainfuck</code>) is plain monospace with the label shown; a <code>bash</code> block with <code>\$</code> prompts and one output line offers <i>Copy Without Prompts</i> ; the copy has no prompts, the output line is present, and the line count is unchanged; the same block tagged <code>console</code> and <code>fish</code> offers it too (plain-highlighted), tagged <code>shell-session</code> or <code>powershell</code> it does not. Proves R-08 , C-05 , K-05 .
T-07	<code>test-docs/math.md</code> : inline, display, <code>cases/pmatrix/aligned</code> , math in lists/quotes/tables/headings, <code>\boxed</code> , registered symbols; the “must NOT become math” section stays literal; the “Errors” section shows source + message. Proves R-12..R-14 , C-07 , E-07 , E-10 , E-16 .
T-08	Same file: the TOC shows Heading with Σ in it and <code>at h2 size</code> , <code>a/b</code> too (Unicode, no <code>\$</code>); the <code>##</code> heading’s <code>is</code> is visibly larger than body <code>.</code> . A heading <code>## _Draft_ notes</code> shows in the TOC, and as a <code>D</code> title, as <code>Draft notes</code> (no underscores); <code>## snake_case_name</code> keeps its underscores. Proves R-13 , R-21 , R-27 , C-07.3 , C-12 , I-010 .

ID	Test
T-09	<code>test-docs/images.md</code> : relative image renders; missing file shows the named placeholder; a <code>data:</code> image renders; an <code>https:</code> image shows “Remote image blocked” until View → Load Remote Images, then loads. Proves R-16 , E-11 .
T-10	<code>test-docs/thematic-break.md</code> and <code>tables.md</code> with Smart Typography on: rules and tables render; inline <code>--flag</code> and code spans keep straight characters; prose quotes curl. Then switch to Phosphor: the menu item reads “(off for this theme)” and is disabled. Proves R-17 , C-10 , I-012 .
T-11	Zoom = five times from 1.0: body text, headings, inline code, fenced code blocks, and math grow together; the post-change HUD shows 150 %; Actual Size resets to 100 %; relaunch restores the saved factor. Write 1.25 into <code>mdv6_font_scale</code> , relaunch, then press <code>=</code> : R-30 rounds 1.25 to 1.3, adds 0.1, stores 1.4, and the only post-change HUD reads 140 %. Proves R-30 , K-04 , K-10 , C-04 , C-05 .
T-12	Choose System theme; toggle macOS appearance: the article switches high-contrast twilight live. Proves R-29 .

9.3 Mermaid (scripted via the harness, then manual)

ID	Test
T-13	Run <code>swift run --package-path tools/render-harness render-harness --scan test-docs/mermaid --output-dir "\$TMPDIR/mdv6-scan"</code> : discovery includes every raw <code>.mmd</code> file and any Mermaid fence in <code>.md</code> files in C-17 order; exit is 0, every case C-06.4 dispatches to the web path is reported <code>web</code> , every other case is <code>pass</code> , and no crash report is created. Repeat with networking disabled; outputs and JSON records are identical. In the app, switch documents while a large diagram is laying out: the new document never shows the old diagram. Proves R-10 , R-39 , C-17 , I-001 , I-002 , E-01 , E-02 , E-25 .
T-14	The diagram of E-01 (two subgraphs claiming PD) renders with PD in the <i>last</i> subgraph. Proves C-06.2 , E-01 .
T-15	A diagram with front matter, <code></code> labels, <code>[/parallelogram/]</code> , <code>style X fill:#eee</code> and <code>fill:white</code> : renders with clean labels and light-grey/white fills. Proves C-06.1 , E-14 .
T-16	<code>xychart-beta</code> with line <code>"a" [...]</code> : two curves visible. Proves E-15 .
T-17	Run <code>swift run --package-path tools/render-harness render-harness --check test-docs/render-cases.json --case mermaid-math-ink</code> : exit is 0; the whole- <code>\$\$</code> node is typeset, mixed and edge labels use C-07.3 Unicode, the label is 16 pt, and the §7.1 ink threshold passes at 2×. Proves R-15 , R-39 , C-17 , I-009 , K-08 .
T-18	Resize a window across a diagram wider than the column: labels remain sharp, the diagram never exceeds natural width, and ordinary widths equal the §7.2 formula (with <code>high-contrast</code> in a window wider than the cap: 732 pt). Force $w_{\text{col}} < 37$ pt: the raster width is exactly 1 pt and no negative or zero size reaches the rasterizer. Proves R-11 , I-005 , K-07 , K-13 , §7.2 .

ID	Test
T-19	Run <code>swift run --package-path tools/render-harness render-harness --check test-docs/render-cases.json --case sequence-layout</code> : exit is 0; no label crosses a lifeline it does not span or overlaps an arrow/block header, the final note is enclosed, autonumber discs $1, \dots, n$ appear, and a 3-line row is 30 pt taller than a 1-line row. Proves R-39 , C-06.2 , C-17 , E-13 , K-08 .
T-20	A <code>stateDiagram-v2</code> with several ID: line descriptions and <code>classDef</code> colours: each state shows all its lines and its colours. Proves C-06.1 rule 4, C-06.2 .
T-21	Mermaid block controls on a natively dispatched diagram: style menu switches and persists after relaunch; Show Source toggles; Export PNG writes a file whose pixel size is $2\times$ the natural point size; pinch zoom clamps at $4\times$ and $0.5\times$. On a <code>gantt</code> block the hover capsule and context menu offer Show Source and copy but no style menu and no export. Proves R-09 , K-07 .

9.4 Navigation, find, search, bookmarks (manual)

ID	Test
T-22	<code>test-docs/links.md</code> : a sibling path navigates in-app and <code>←</code> returns; same-document <code>#fragment</code> , a percent-encoded UTF-8 fragment, and <code>sibling.md#fragment</code> reach the first matching C-11 slug, with the cross-file fragment overriding saved scroll position. A missing same-document fragment is a no-op; a missing cross-file fragment loads the file at its top. <code>https:</code> opens the browser; a broken local link does not navigate. Duplicate <code>### Example</code> headings resolve <code>#example</code> to the first and leave <code>#example-1</code> unmatched; h4 and setext targets remain unmatched. <code>#a---b</code> , <code>#c--rust</code> , and <code>#draft-notes</code> reach the named GitHub-style headings. The same sibling links delivered as resolved <code>file:</code> URLs (the form the renderer produces) navigate in-app; a <code>file:</code> URL to a <code>.txt</code> or a missing file goes to the system opener (F-117). A sibling path load and a same-document TOC/fragment jump create back snapshots. A bookmark in another file loads it but <code>←</code> does not return to the pre-bookmark file. A fragment that parses as a line citation reaches C-19 rather than a slug (T-49), so a heading slugged <code>l10</code> is not reachable by <code>#l10</code> . Proves R-18 , R-19 , R-21 , R-27 , C-11 , C-12 , E-05 , E-06 , E-22 .
T-23	Immediately after <code>F</code> , the empty query shows “No matches” and disables <code>G</code> ; a whitespace-only non-empty query is matched verbatim. For “the” in one block three times, <code>m</code> rises by three and <code>G</code> advances <code>n</code> three times while the block stays in view with all three occurrences in the stronger tint. An inline-image paragraph, code fence, and <code>\$\$</code> display block are tinted; ordinary prose is highlighted per character. Query <code>**</code> on <code>**bold**</code> counts 2 and marks nothing. <code>G/</code> <code>G</code> wrap; <code>Esc</code> closes. With sidebar focus, <code>F</code> focuses global search. Proves R-24 , E-17 , E-18 .

ID	Test
T-24	F “auth” includes a file containing “authentication” with the term highlighted; choosing it opens the file. “résumé” matches “resume”; an empty or whitespace query returns none. Editing without changing mtime leaves old indexed content; touching then opening by O re-indexes, while selecting from the sidebar does not. Insert 81 files with identical FTS rank in reverse path order: results are the first 80 by <code>path COLLATE NOCASE</code>, then binary path, exactly as C-03; repeated queries return the same sequence. Proves R-01, R-25, R-26, C-03, K-03, K-09, E-24.
T-25	Open 101 distinct files: the sidebar shows the newest 100, most recent first, no duplicates, and F for a word unique to the first file finds nothing (evicted from the index); click the third row: it loads and the order is unchanged; O the same file: it moves to the top. Swipe-delete removes one and F no longer finds that file, and re-opening it starts at the top (scroll position removed); swipe-delete the displayed row: the next row’s file loads and ← does not return to the deleted one; with one row left, delete it: the window shows the empty drop target; relaunch preserves the list; the stored <code>mdv6_history</code> value decodes per C-15. Proves R-01, R-18, R-20, R-26, I-013, K-03, C-15, §3.1.
T-26	Hover a paragraph and D: the bookmark anchors there and uses the preceding heading; with no hover it uses the topmost visible block. Duplicate bookmarks create duplicate rows. Beyond the 40-block look-back, the stripped first line is truncated to 60 extended grapheme clusters; an empty stripped line becomes (<code>line n</code>). Insert content above an anchor: fingerprint resolution still lands on it; delete it: the clamped index is used. Delete the file: the row is marked missing and its shortcut beeps. Unit cases verify that tabs, CRLF, NBSP, and repeated spaces normalize to U+0020; non-ASCII case uses locale-independent lowercase; canonically equivalent but byte-distinct combining sequences remain distinct; and truncation counts 80 extended grapheme clusters, including emoji. Proves R-27, C-08, E-08, E-09, K-06, K-09.
T-27	With the bookmarks pane collapsed, 0: the pane expands and a first row appears above a divider with <code>pin.fill</code>, the R-27 title of the block, the file name and a 0 badge, filled with the accent; scroll away, 0 returns and the row is still current; click the row: same as 0. 0 elsewhere: the row’s title changes, no second row appears. Then: 0, scroll away, 0 returns; open another file, 0 loads the first file and returns; 0 then ← does not go back to the pre-0 spot; delete the placeholder’s file, 0: beep, view unchanged; follow a link to B and delete A, ←: beep, still on B; relaunch: 0 beeps. Proves R-18, R-28, E-27, C-18.9.

ID	Test
T-28	Scroll to the middle, quit, and relaunch with no argument: the same readable history head is displayed at the same position; modify it externally before relaunch and the document starts at the top. Make the persisted head unreadable while a later row remains readable: launch enters <code>EMPTY</code>, retains both rows, and does not try the later row. With empty history, launch enters <code>EMPTY</code>. Quit with A as the head, then cold-start with <code>bin/mdv6 B.md</code>: B is displayed without first restoring A, and <code>←</code> does not show A. Open a second window with <code>O</code>, quit with both open, relaunch: exactly one window, showing the head (E-30). Proves R-06, R-18, R-40, C-08, E-03, E-08, E-30, K-06, §3.1.
T-29	With the file open, save it from an editor five times within 50 ms (script): at most two reloads, the final content is displayed, scroll position kept. Save via <code>mv tmp file</code> (atomic rename): reloads. <code>rm file</code>: content stays, no error; re-create it: reloads. Write invalid UTF-8 over it: content stays; write valid content: reloads. Truncate the file to zero bytes and write it back 100 ms later: no blank frame is shown. Leave it empty for 1 s: the page shows empty. Proves R-05, K-06, E-19, E-21.
T-30	Single-click a heading: the section flashes and the pasteboard holds its Markdown source ending at the next same-or-higher heading; click again: it flashes again; <code>-</code>click behaves the same. Drag across a paragraph: text is selected and <code>C</code> pastes rendered text; dragging on a heading selects nothing. Hover and click a <code>####</code> heading: arrow pointer, text selectable, nothing copied. Throughout, the TOC row, find match, and bookmark for one paragraph all address the same block index. Proves R-22, C-12, E-22, I-004.
T-31	Drag the inspector's left edge to 520 pt and 180 pt (clamps), relaunch: width kept; drag the sidebar divider: clamps at 180/400. Proves R-20, R-21, K-04.
T-42	Set every C-04 key to a non-default valid value, relaunch, and verify each visible behavior/value is restored. Then, one key at a time, store a wrong type, an out-of-range number, malformed <code>mdv6_history</code> JSON, and unknown theme/Mermaid ids; relaunch and verify the exact C-04 default or clamp without a crash. Covers smart typography, remote images, show frontmatter, sidebar collapse, inspector visibility/width, bookmark expansion/height, editor path, history, theme, font scale, and Mermaid style. Proves R-32 and C-04.

ID	Test
T-49	In <code>test-docs/links-sibling.md</code> (11 lines — <code># Sibling</code> on 1, a paragraph on 3, <code>## Second heading</code> on 5, a paragraph on 7, <code>## Third heading</code> on 9, a blank on 10, a paragraph on 11): <code>[cite](links-sibling.md#L7-L9)</code> loads the sibling, scrolls the paragraph containing line 7 to the top of the viewport, and flashes that one block for 0.6 s in the R-22 accent tint; the flash restarts on a second click. <code>#L1</code> reaches <code># Sibling</code> and <code>#L11</code> the paragraph under the third heading. A same-document <code>#L1</code> does the same without a load and without pushing a back snapshot, while <code>#second-heading</code> still pushes one. <code>#17-19</code>, <code>#L7-9</code>, and <code>#L9-L7</code> resolve identically to <code>#L7-L9</code> (case-insensitive; a reversed range is read as named). <code>#L10C5</code>, <code>#L</code>, <code>#L0</code> and <code>#nope</code> are not citations and are E-06 no-ops — the file has no such heading, so nothing flashes. <code>#L10</code> names a line inside the removed blank-line gap and flashes <code>## Third heading</code>, the block preceding the gap; <code>#L12</code> is past <code>lineCount = 11</code> and flashes the last block, the paragraph under the third heading (E-31). A citation into a document with no blocks — the zero-byte <code>.md</code> of T-39 — scrolls to the top and flashes nothing. The TOC selected row is set by a slug fragment and not by a line citation (E-29, C-19.4). Proves R-19, C-19, E-06, E-31.

9.5 Robustness and resources (scripted)

ID	Test
T-32	On an otherwise idle <code>macos-15</code> host, open <code>test-docs/math.md</code>, wait 5 s without input, then collect 30 one-second process-CPU samples. Compute the median and nearest-rank 95th percentile over exactly those samples: they are respectively $\leq 1\%$ and $\leq 3\%$. Proves I-008 and K-15.
T-33	Corrupt <code>mdv6.db</code> (truncate the file) and launch: the app opens, documents render, <code>NSLog</code> shows the <code>[mdv6]</code> failure line; bookmarks and search are empty; no crash. Kill the app mid- <code>D</code> (kill <code>-9</code> in a loop): on relaunch every bookmark row is either complete or absent. Proves E-12, I-006, I-007.
T-34	<code>diff -r</code> between <code>Vendor/SwiftMath/Sources</code> and upstream <code>v1.7.3 Sources/SwiftMath</code> shows only the files and hunks listed in <code>Vendor/SwiftMath/README.md</code>. Proves I-011.
T-35	Open a document while the same path is open in a second window, edit it on disk: both windows reload. Proves E-20.
T-40	With two windows open showing different files, make window 2 key: <code>open -a build/mdv6.app c.md</code> loads <code>c.md</code> into window 2 only; <code>O</code> shows one open panel; <code>D</code> adds exactly one bookmark (window 2's); <code>←</code> navigates window 2 only; <code>F</code> opens window 2's find bar only. Proves R-01, R-18, E-26.
T-38	Choose an editor via <code>File → Edit → Choose Editor...</code>, <code>E</code>: the file opens there; <code>Forget Editor</code>, <code>E</code>: the chooser appears. <code>?</code>: <code>Help</code> opens, <code>~/Library/Application Support/mdv6/Help.md</code> exists, and <code>D</code> inside it creates a bookmark with that path. Proves R-23, R-31.

ID	Test
T-39	Fence edge cases (E-23): a <code>````</code> block containing a <code>```</code> line, an unclosed fence at EOF, and an indented code block with a blank line render per C-02. CRLF and LF copies produce the same block count and TOC. An ISO-8859-1 file does not open, keeps the previous document, and adds no row; a zero-byte <code>.md</code> opens as an empty article. With find open on “the”, save one additional occurrence: <i>m</i> rises by one and the bar reads “1 of <i>m</i> ”. Proves R-04, R-24, C-02, E-03, E-23.
T-36	Grep the built binary’s log output during T-05..T-31 and T-54 (<code>log stream --process mdv6</code> , and <code>log stream --process com.apple.WebKit.WebContent</code> for the Mermaid pages, C-06.5): no line contains document text, a query string, or a path, except the <code>[mdv6]</code> failure message and lines beginning <code>"mathFonts bundle resource:</code> (the SwiftMath font-registration lines R-35 permits). Proves R-35, I-003.
T-41	Generate each K-14 payload at the exact ceiling and one unit above it. Exact-ceiling inputs reach their normal renderer; oversized document, Mermaid, LaTeX, compressed image, decoded-pixel, decoded-byte, and image-axis cases produce the E-28 outcome without a crash or parser/decode invocation. With a local recording HTTP server, remote loading off sends no request; on sends only unauthenticated GET with no Cookie/Authorization/Referer, follows five <code>http(s)</code> redirects but rejects the sixth or a non-HTTP target, fails at 15 s connection or 30 s resource timeout, cancels above 32 MiB, persists no body, and cancels when the preference turns off. Proves R-16, R-36, R-41, C-16, I-001..I-003, K-14, E-11, E-28.
T-50	Assert the C-02 rule 8 map directly, in <code>swift test</code> : for a document of a heading, a two-line paragraph, a blank-line-separated paragraph, and an unclosed fence at EOF, <code>blockLines</code> is half-open and <code>blockLines[i]</code> covers exactly <code>blocks[i]</code> ’s source lines; <code>lineCount</code> ignores a trailing newline; a file opening with two blank lines resolves #L1 to the first block; a file ending without a trailing newline resolves its last line to the last block; and a CRLF copy yields the same <code>blockLines/lineCount</code> as its LF equivalent. For <code>test-docs/links-sibling.md</code> , <code>lineCount</code> is 11 and line 10 resolves to the block of line 9. Proves C-02 rule 8, C-19.3, E-31.
T-62	<code>diff -r</code> between <code>Vendor/MarkdownUI/Sources/MarkdownUI</code> and upstream v2.4.1 (5f613358) <code>Sources/MarkdownUI</code> shows only the files and hunks listed in <code>Vendor/MarkdownUI/README.md</code> ; <code>Documentation.docc</code> is absent. Proves I-016.

9.6 Visual parity (C-18, I-014, I-015)

A conformance report for this specification MUST include the outcome of T-44, T-47, T-48 and T-49 — and, since v0.14.1, of T-52, the manual half of T-53, T-54 and T-58 (F-163), and of T-59 (F-180) — as observed on a screen by a person, or in a screenshot the person has looked at (T-48’s menu order, enablement and reorder are also proved in `swift test`; T-49’s scroll position and flash are visible only on screen — its map half is T-50 in `swift test`); a build whose §9.1–9.5 tests are green but whose T-44 or T-49 was not observed is *verification pending*, not conforming (D-40). Where a host cannot show a window, the report says so and the status stays pending. Observed tests — and every §9 test that opens, deletes or reorders history, bookmarks or the placeholder (T-25, T-27, T-47, T-48, T-49) — run against an **isolated store** seeded from a fixture (`MDV6_SUPPORT_DIR`, `MDV6_DEFAULTS_SUITE`, §10), never against the reader’s own.

ID	Test
T-44	Open <code>test-docs/math.md</code> in Sevilla with four bookmarks and a placeholder set, sized like <code>reference/MDV-ORIGINAL-SEVILLA.png</code> , and compare pane by pane against that image and <code>reference/MDV-SCREEN.png</code> : window title is the file name; toolbar is the five glyphs of §5.1 and nothing else; HISTORY, ON THIS PAGE, BOOKMARKS headers with the C-18.3 styling; history rows show name over ~-abbreviated head-truncated path; clicking the header magnifier reveals and focuses the field, Esc hides it; TOC rows indent by level and the clicked row takes the accent fill; bookmark rows show title over file name with 1–5 badges, the placeholder row first with <code>pin.fill</code> and 0, the current one filled with the accent; the BOOKMARKS header shows the count capsule and collapses on click; the floating find button sits at the article’s top-trailing corner. Repeat the title, toolbar and header checks in Charcoal and Twilight: the structure is unchanged and the title is legible on the dark strip (I-015, C-18.1). Hover the sidebar’s handle: the collapse chevron appears; move away: it disappears. Proves R-42, C-18, I-015, K-16.
T-45	Render, through the C-17 harness at 2× and 860 pt, a document of exactly <code># A\n\nfirst paragraph.\n\n## B\n\nsecond paragraph.\n\n### C\n\nthird paragraph.</code> under Sevilla and under Charcoal. For each adjacent block pair, locate the last ink row of the earlier block and the first ink row of the later one (a row is ink when any channel differs from the page colour by more than 8/255) and divide by 2: for the four boundaries <code>paragraph→## B, ## B→paragraph, paragraph→### C, ### C→paragraph</code> the gaps g satisfy the K-16 band $v \leq g \leq v + 0.6f$ with $v = \text{h2TopSpacing}, \text{paragraphBottomSpacing}, \text{h3TopSpacing}, \text{paragraphBottomSpacing}$ for the theme, and the <code>paragraph→heading</code> gap is strictly larger than the <code>heading→paragraph</code> gap. The same document rendered as a single Markdown view gives the same four gaps within ± 2 pt (the single view is the oracle: MarkdownUI applies the theme margins between siblings). Proves I-014, K-16, C-18.10, R-42.
T-46	Render <code>\$\$r = \frac{a}{b}\$\$</code> as its own paragraph (one source line) and the three-line fence form <code>\$\$\nr = \frac{a}{b}\n\$\$</code> through the harness at 860 pt: both rasters are identical, the ink bounding box is horizontally centred in the column within 2 pt, and its height is the <code>.display</code> height of C-07.2, not the body-size E-16 height. Text <code>\$\$x\$\$ text</code> on one line remains an inline image inside the sentence. Proves C-07.1, R-12, C-18.10.
T-47	Launch with <code>test-docs/syntax.md</code> : the window title reads <code>syntax.md</code> ; O a second window and open <code>tables.md</code> into it: its title reads <code>tables.md</code> and the first is unchanged; delete the only history row (T-25): the title reverts to the product name. Proves C-18.1, R-42.

ID	Test
T-48	With <code>test-docs/math.md</code> open and five bookmarks plus a placeholder set: move the pointer into a paragraph — a 3 pt accent stripe appears at its leading edge and leaves with the pointer; into a code fence and a Mermaid fence — no stripe. Right-click the placeholder row: the menu is exactly <i>Clear Placeholder</i> ; choose it: the row and its divider disappear, the pane shows the bookmarks only, 0 beeps. Right-click the third bookmark: the nine-entry menu of C-18.9 in order; <i>Move Up</i> makes it the second row and 2 opens it; <i>Move to Bottom</i> makes it the last and 5 opens what was fourth; on the first row <i>Move Up</i> and <i>Move to Top</i> are disabled, on the last <i>Move Down</i> and <i>Move to Bottom</i> ; <i>Reveal in Finder</i> on a bookmark whose file was deleted is disabled; <i>Remove Bookmark</i> removes the row. Relaunch: the order persists. Proves C-18.7 (stripe), C-18.9 (menus), R-27, R-28, K-16.

9.7 v0.14 features (frontmatter, print, web diagrams, file commands, scrolling, diffs,)

Reference images (F-163). Four images of the original’s build at 68aa008 (2026-09-27), in **high-contrast** at zoom 1.0 with the pointer outside the window (no hover chrome), are normative for **structure** under D-42’s rule and are cited by the contracts they illustrate. The three window captures (`screenshot -1`, 2× backing) show the reader’s own history and bookmarks in the side panes, which are not part of what they illustrate.

- `reference/ORIGINAL-FRONTMATTER.png` — `frontmatter.md`: the C-20.3 properties table above the first heading.
- `reference/ORIGINAL-GANTT.png` — `gantt.md`: a web-path diagram (C-06.5) on the code-block background.
- `reference/ORIGINAL-DIFF.png` — `diff.md`: C-05.1’s tints, italic hunk headers, semibold metadata, the italic \ No newline note, and the removed --- stale comment line tinted, not bolded.
- `reference/ORIGINAL-RAW-HTML-IMAGES.png` — `raw-html-images.md` down to, not including, its final *departs from the original* section, rendered by the original (View → Load Remote Images off) and stitched from four window captures of consecutive slices of the file — the original’s window cannot be scrolled by script — cropped to the article column, at 1 pixel per point (the slices’ left edges differ by a few points; the image is normative for structure only): the C-22.2 sizes, the inline baseline placement, the block `alt` for a missing file and the empty inline slot, the blocked remote placeholder, and the literal cases.
- `reference/ORIGINAL-PRINT-MATH.png` — `math.md` through the original’s print pipeline at the Letter column width ($w_c = 504$ pt), rasterised at 144 ppi. It comes from the original’s MDV_PRINT_TEST self-test, which writes the composited block column as **one** PDF page (504×1465 pt) **before** AppKit paginates it, so it shows no page margins, no header or footer and no page breaks. It is normative for the printed blocks’ structure (type scale, formula placement and size, diagram chrome, soft-wrapped code; whether a formula is vector is T-53 (2)’s to show, not a raster’s) and **not** for pagination or the rhythm: it shows the flat $12 s_p$ gap — headings flush against the paragraph above — that C-21.2 departs from (F-160, D-47).

Fixtures `frontmatter.md`, `frontmatter-toml.md`, `frontmatter-ellipsis-close.md`, `frontmatter-negative.md`, `gantt.md`, `mermaid-web-fallback.md`, `diff.md` and `raw-html-images.md` are carried into `test-docs/` from the original’s tree at 68aa008 (in `raw-html-images.md`, `../MDV.png` is repointed to `../MDV6.png` and `images/icon.png` to `assets/local.png`). Every test here that opens, closes or steps history runs on an isolated store (§9.6).

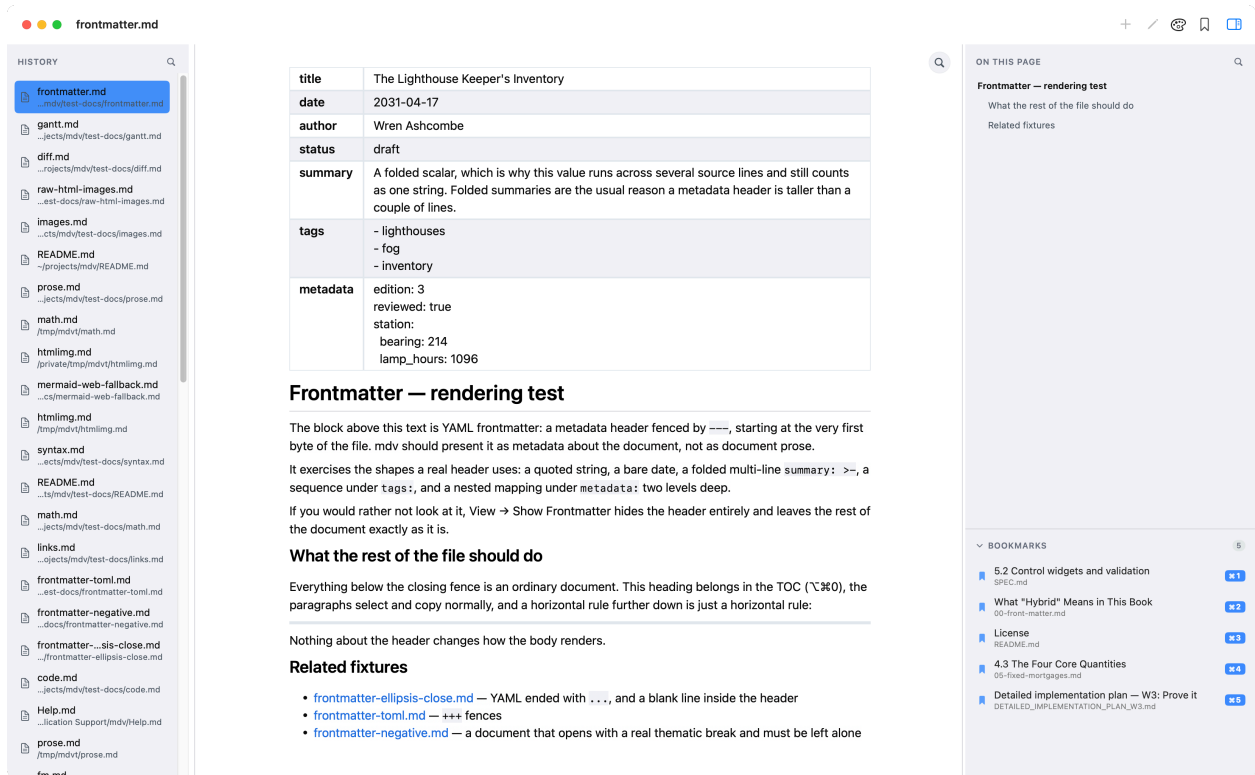


Figure 5: Reference — the original, `frontmatter.md` in high-contrast (reference/ORIGINAL-FRONTMATTER.png)

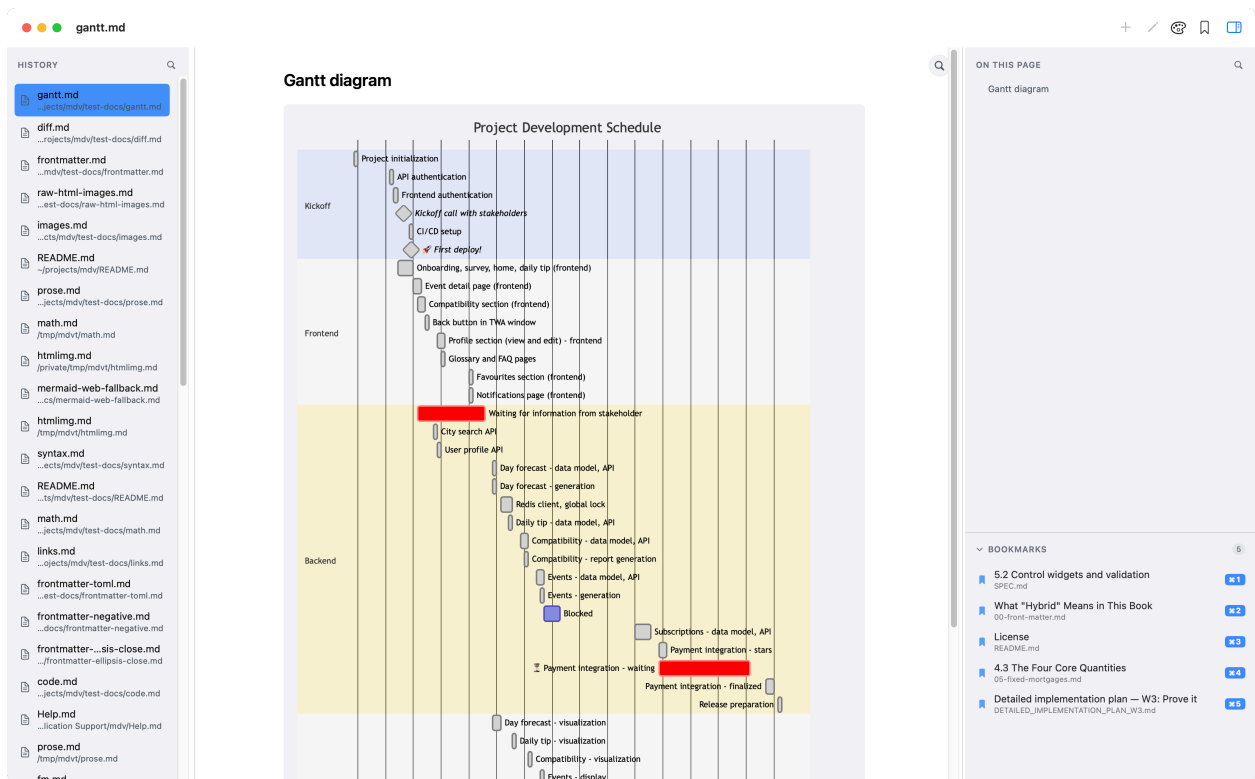


Figure 6: Reference — the original, `gantt.md` rendered by the web path (reference/ORIGINAL-GANTT.png)

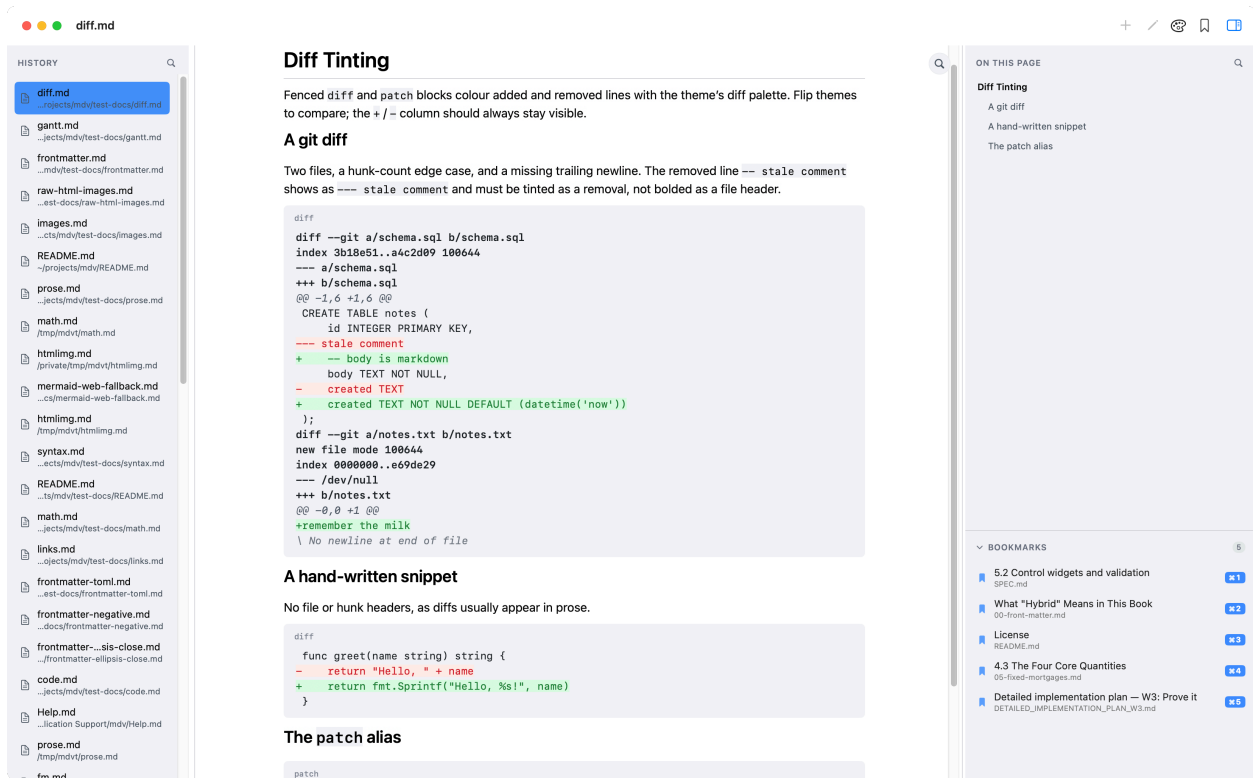


Figure 7: Reference — the original, `diff.md` (reference/ORIGINAL-DIFF.png)

ID	Test
ID	Test
T-52	Open frontmatter.md: a two-column table sits above the first heading, keys semibold, alternating fills, one border colour, a folded > value reads as one paragraph, a \ value keeps its line breaks, a nested list shows indented; no --- rule and no setext heading appear. frontmatter-toml.md: [section] rows and a multi-line array render; frontmatter-ellipsis-close.md: a YAML header closed by ... renders as a table. frontmatter-negative.md renders as two rules around a paragraph — no table. Toggle View · Show Frontmatter off: the table disappears and nothing else moves in the TOC; D on the first paragraph then back on: the bookmark still lands there; relaunch: the setting persists. With it off, F for a word only in the header: $m = 1$ and the header's source appears verbatim with the match highlighted until the bar closes — also for a header containing image: ![x](y), which is never tinted as a whole. Hidden: the first heading sits at the top with no gap above it; D with the article scrolled to the top anchors the first body block, not the header; #L2 scrolls to the top and flashes nothing. A file that is only a hidden header: D and 0 beep and add nothing. Shown: hovering the table shows the stripe, its values can be selected and copied, and a bookmark made on it is titled Frontmatter. A file opening ---, blank, # Title, blank, --- shows two rules around Title, and Title is in the TOC. A BOM-prefixed copy of frontmatter.md renders the same table. Unit: frontmatterSpan accepts and rejects the C-20.1 cases (byte-0 fence, exact closers, indented ---, the colon rule with http://x, a sequence item, a comment, a TOML array closing at column 0) and CRLF input splits like LF; frontmatterRows returns the C-20.2 rows for each fixture, including quote stripping, folding with blank lines, and bracket depth ignoring "Tower [12]"; the C-02 rule 9 split puts a header with an internal blank line in one block and keeps blockLines in whole-document lines; a zero-row candidate is rejected; a leading U+FEFF is removed before the split. Proves R-04, R-24, R-27, R-44, C-02 rule 9, C-04, C-20, I-018, E-32.

ID	Test
T-53	Scripted, through <code>render-harness ... --print-pdf</code> (C-17) on Letter, with its JSON as the oracle: (1) vector text — for <code>test-docs/rhythm.md</code> (plain headings and paragraphs, so the oracle needs no GFM normalisation; GFM rendering is T-05's) with smart typography off, the whitespace-split tokens of every prose block's C-12-stripped source appear, in order, in PDFKit's extracted text of that block's <code>rect</code> (F-175); (2) vector formulas — for <code>test-docs/math.md</code>, no <code>formula</code> block and no <code>rect</code> in any block's <code>formulas</code> list of a span <code>SwiftMath accepts</code> intersects an entry of that block's <code>images</code> other than a 1×1-pixel spacer (every accepted formula, standalone and inline, is glyphs; F-184); (2b) each span <code>SwiftMath</code> rejects (the fixture's <i>Errors</i> section) has an image in its <code>rect</code> — its E-10 fallback printed, not dropped (F-176); (3) pagination — every block no taller than the printable page height reports exactly one object (one page), and a synthetic block taller than a page reports two; (4) rhythm — for <code>test-docs/rhythm.md</code> the gap between consecutive block <code>rects</code> is s_p times the screen's gap for that pair within ± 1 pt (C-21.2, I-014); (5) wrapping — no <code>code</code> block's extracted text lies outside the page's content width; (6) diagrams — each raw <code>.mmd</code> in <code>test-docs/mermaid/</code> whose C-06.4 dispatch is native prints a <code>diagram</code> block with no <code>images</code> entry and with extractable label text; (7) frontmatter — <code>frontmatter.md</code> prints a <code>frontmatter</code> block, and with <code>mdv6_show_frontmatter</code> false prints none; (8) independence — printing <code>math.md</code> with the screen at zoom 1.0 and 1.5 and under <code>sevilla</code> and <code>twilight</code> yields identical extracted text and identical block <code>rects</code> within 0.5 pt (I-017). Manual, observed (§9.6): P on <code>math.md</code> in <code>Twilight</code> at 150 % zoom opens the panel with a light, high-contrast preview at ~ 9.4 pt body on Letter, header and page numbers; <i>PDF</i> $\rightarrow$ <i>Save as PDF</i> writes a file whose inline formulas stay sharp at 800 % in Preview; <code>gantt.md</code> prints its chart, and when the PDF route succeeds (C-21.3) no window appears on screen during the job; a deliberately broken fence prints as source; P again while the panel is up beeps; P in <code>EMPTY</code> beeps. Proves R-45, C-21, I-017, K-17, E-33.

ID	Test
T-54	Open <code>gantt.md</code> and <code>mermaid-web-fallback.md</code>: every <code>gantt</code>, <code>pie</code>, <code>timeline</code>, <code>journey</code>, <code>quadrantChart</code>, <code>requirementDiagram</code>, <code>mindmap</code> and <code>gitGraph</code> block renders as a diagram after a brief spinner, on the code-block background, in mermaid's <code>default</code> theme under <code>high-contrast</code> and <code>dark</code> under <code>Twilight</code>; the document reflows once per diagram, not repeatedly; a flowchart preceded by <code>%%{init: ...}%%</code> and a <code>%%</code> comment, one preceded by a multi-line <code>%%{ ... }</code> directive, and one preceded by a <code>---</code> config block, render natively (style menu present). Right-click a <code>gantt</code> chart: the block menu (<i>Copy Code</i>, <i>Show Mermaid Source</i>) appears, never WebKit's; scroll with the pointer over it: the article scrolls; click it, then press <code>↓</code>: the article scrolls (R-49); drag across its labels: nothing is selected. A <code>gantt</code> with a syntax error shows the R-10 fallback. With networking disabled every web diagram still renders, and a recording proxy sees no request from the process during the whole test. VoiceOver reads each web diagram as <i>Mermaid diagram</i>. Build with a corrupted <code>Vendor/mermaid/mermaid.min.js</code>: <code>build.sh</code> exits 1 naming both digests. Unit: C-06.4 dispatch for <code>graph</code>, <code>graph\tLR</code>, <code>graphfoo</code>, <code>flowchart-elk</code>, <code>stateDiagram-v2</code>, <code>xychart-beta</code>, a multi-line <code>%%{ ... }</code> directive, and an empty source; C-06.1 rule 0 removes exactly the lines C-06.4 skipped. Proves R-09, R-10, R-46, R-49, C-06.1, C-06.4, C-06.5, C-13, I-001, I-003, K-19, E-02, E-34.
T-55	With A, B, C in history (C newest) and C displayed: W — C's row, index row and scroll position go (F finds nothing unique to C; re-opening C later starts at the top, R-06), B loads, <code>←</code> does not return to C; W until one row is left, then W: EMPTY, and File <code>·</code> Close File is disabled. Re-open three files, O a second window, W: the dialog reads <i>Close all files? / This clears the sidebar history and the search index; Cancel</i> changes nothing; W → Close All: both windows show the empty drop target, the sidebar is empty, F finds nothing, bookmarks remain and 1 reopens its file; W is now disabled. W closes only the key window. With a large file loading in window 2, confirm Close All in window 1: the load is abandoned and history stays empty. Proves R-06, R-47, R-26, R-18, §3.1, E-26, E-36.
T-56	History A, B, C, D (A newest), B displayed: <code>]</code> shows C, <code>]</code> shows D, <i>Next File</i> is then disabled and <code>]</code> does nothing; <code>[×3</code> shows A and <i>Previous File</i> is disabled; the sidebar order is unchanged throughout and no row is re-indexed; <code>←</code> walks back D, C, B. <code>/</code> do the same; with the find field focused, Tab and <code>Tab</code> move focus as usual and still steps files. With two windows, stepping acts on the key window only. Proves R-48, R-01, R-18, E-26.
T-57	Open <code>math.md</code>, click nowhere: <code>↓</code> scrolls 40 pt, <code>↑</code> back; Page Down and Space scroll <code>max(0.875u, 40)</code> pt with a visible overlap line; Space and Page Up return; End reaches the true bottom of the lazily laid-out document within 10 frames, Home the true top with the first heading clear of the title bar; no key beeps. Click into a paragraph (selectable text): the keys still scroll. Focus the find field: <code>↑/↓</code> and Space edit the field. Focus the history list: <code>↑/↓</code> step the selection. <code>↓</code> does not scroll through this path. Proves R-49, K-18, E-37.

ID	Test
T-58	Open <code>diff.md</code>: added lines green-on-green-tint and removed lines red-on-red-tint in <code>high-contrast</code> with the C-05.1 hex values, @@ headers italic secondary, <code>diff --git/index/---/+++</code> headers semibold, \ No newline at end of file italic comment-coloured; a removed SQL comment line <code>--- old comment</code> inside a hunk is red, not a header; a <code>patch</code> fence and a header-less <code>+/-</code> snippet colour the same way; switch to Phosphor: brightness, not hue, separates the two. Unit: C-05.1 over the fixture's hunks, including omitted counts (@@ -1 +1 @@), a context line reduced to empty, and a malformed @@ x y @@ read as meta. Proves R-50, C-05, C-05.1.
T-59	Open <code>raw-html-images.md</code> with View → Load Remote Images off: the header <code></code> draws 320 pt wide; width-only, both-attribute (fits inside 240 × 80 without stretching), height-only and attribute-less cases follow C-22.2 and match <code>reference/ORIGINAL-RAW-HTML-IMAGES.png</code>; the inline <code></code> sits on its sentence's baseline at 120 pt with only that line grown, and a plain inline <code>![] (assets/local.png)</code> loads from beside the document; single-quoted and bare attribute values work; the block missing-file tag shows its <code>alt</code> text, and the inline one draws nothing — its parentheses are empty; the remote <code></code> shows the clickable <i>Remote image blocked</i> placeholder and a recording server on 127.0.0.1:8765 (T-41's kind) receives no request. Turn Load Remote Images on with <code>python3 -m http.server 8765</code> running in <code>test-docs/assets/</code>: the remote image loads at 120 pt; stop the server and reopen the file: the failure placeholder appears. The fenced and code-span tags and the <code><a></code> tag stay literal. In the fixture's final section (departures from the original): <code>alt="a > b"</code> does not end its tag and the image draws at 60 pt; the inline remote <code></code> shows the non-clickable inline <i>Remote image blocked</i> text image while the preference is off (R-16); the ### Logo <code></code> heading renders its image and its TOC row reads Logo. Unit: <code>RawHTMLImages.rewrite</code> output for each case in E-35 and the <code>mdv6-img://</code> round trip; <code>displaySize</code> for the four attribute combinations and a natural size smaller than the cap. Proves R-51, R-16, C-12, C-22, E-35.
T-60	In Sevilla at 125 % zoom, F for a word that occurs in an # heading, a ### heading, a paragraph with <i>bold</i>, <code>_em_</code>, <code>`code`</code> and a link, and a ##### heading: as the matches appear, no block changes face, size, weight, colour or leading; the h1 keeps its rule; the inline styles stay as rendered. Unit: the C-09.1 style for $\ell = 1 \dots 6$ and body under two themes. Proves R-24, C-09.1.

ID	Test
T-61	With the application inactive behind Terminal, <code>open -g -a build/mdv6.app c.md</code> : <code>c.md</code> loads into the frontmost document window and Terminal stays frontmost; <code>open -a build/mdv6.app d.md</code> brings mdv6 forward with <code>d.md</code> displayed. With two windows, window 2 frontmost and the application inactive, an open event loads into window 2. With the application active, the target window becomes key and comes to the front. Close the only window with <code>W</code> : the application keeps running; <code>P</code> and <code>W</code> are disabled; press <code>O</code> and cancel the panel: no window appears; press <code>0</code> with no placeholder: a beep and no window; double-click <code>c.md</code> in Finder: exactly one window opens showing <code>c.md</code> ; close it, <code>1</code> : one window opens on the bookmark; close it, click the Dock icon: one window opens on the history head. Proves R-01 , E-26 , E-38 .

10. Dependencies and environment

Dependency	Version / pin	Role
macOS	≥ 13.0 (built and tested on 15)	platform
Swift toolchain	≥ 5.9 (<code>swift-tools-version: 5.9</code>); CI uses the <code>macos-15</code> runner's Xcode	build
<code>gonzalezreal/swift-markdown-ui</code>	vendored at <code>Vendor/MarkdownUI</code> from v2.4.1 (5f613358), <code>Sources/</code> only (no <code>Documentation.docc</code>), declared as a target in the root <code>Package.swift</code> , with the changes listed in its <code>README.md</code> (I-016)	GFM $\rightarrow$ SwiftUI; image-provider and code-highlighter hooks; the resolved-inline-images hook print needs (C-21.4)
<code>swiftlang/swift-cmark</code> (from 0.4.0), <code>gonzalezreal/NetworkImage</code> (from 6.0.0)	direct package dependencies of the vendored MarkdownUI target	<code>cmark-gfm</code> ; default remote image loader
<code>mermaid.js</code>	11.4.1, <code>mermaid.min.js</code> from <code>jsDelivr</code> , SHA-256-pinned (C-13 , K-19); not committed	the R-46 web path, on screen and in print
WebKit	system framework (<code>WKWebView</code>)	hosts <code>mermaid.js</code> (C-06.5)
<code>ChimeHQ/SwiftTreeSitter</code>	from 0.8.0, resolved 0.25.0 (<code>tree-sitter 0.25.10</code>)	tree-sitter runtime
tree-sitter grammars (18)	commits in <code>mdv6/Grammars/README.md</code> , vendored C sources compiled as target <code>CGrammars</code>	code highlighting
<code>lukilabs/beautiful-mermaid-swift</code>	from 1.0.4 (<code>elk-swift 1.0.2</code>)	Mermaid parse/layout/render

Dependency	Version / pin	Role
SwiftMath (mgriebing/SwiftMath 1.7.3)	vendored at Vendor/SwiftMath with the patches listed in its README.md (font-bundle resolution, public MTMathAtom.init, \boxed, trimmed font bundle)	LaTeX typesetting; font latinmodern-math.otf (GUST licence)
SQLite	system libsqlite3 (linked via linkerSettings), FTS5	persistence
Fonts	Alegreya, Besley, OpenDyslexic (mdv6/Fonts, registered at launch)	themes
Release tooling	codesign, notarytool (keychain profile), stapler, spctl, gh	make dist, github-release

Environment variables: MDV6_APP (launcher bundle override); MDV6_SUPPORT_DIR (a directory that replaces ~/Library/Application Support/mdv6 as the location of mdv6.db, created if absent) and MDV6_DEFAULTS_SUITE (a UserDefaults suite name that replaces the standard domain) — set by the launching user, never by document content, so a live check runs against an isolated store (§9.6); an empty value is unset; the test bootstrap takes precedence over both; release-time Makefile inputs per §5.3. Build-time files: Vendor/mermaid/mermaid.min.js (downloaded on first build, C-13). Runtime files: ~/Library/Application Support/mdv6/{mdv6.db, Help.md}, UserDefaults domain com.mdv6.app. Install and run: make install; run tests: swift test (R-37) and the C-17 swift run --package-path tools/render-harness render-harness ... commands.

11. Traceability matrix (id → where realized)

Statuses: a plain row is realised and verified as written; *not yet realised* marks specified work with no implementation; **open defect** marks a known code violation; *verification pending* marks an implemented path whose revised behavior has not yet been observed. Every row is plain except the three marked *verification pending* (R-01, R-45, R-46), whose clauses need a person at an unlocked console. C-19, E-31, T-49 and T-50 were built by W8 (e61c60f); the v0.14 rows and amendments were built by W10–W15 (a78d2ec...) — SPEC_BUILD_REPORT.md carries the per-id evidence. The v0.8 *not yet realised* and **open defect** rows were built and fixed by the v0.8 build, the v0.9/v0.10 chrome rows by the v0.9–v0.10 build, and the observed rows (T-44..T-49) were seen on screen — SPEC_BUILD_REPORT.md carries the per-id evidence. *Where realised* names this repository’s files (mdv6/...).

Spec id	Where realized	Verified by
R-01	mdv6App.swift, ContentView.loadFile/select, addressed notifications; frontmost-document-window targeting and background opens (v0.14); <i>verification pending</i> : open -g activation unobserved (locked console, W15)	T-03, T-04, T-22, T-24, T-26, T-40, T-61
R-02	ContentView.loadDirectory	T-04
R-03	ContentView.handleDrop	T-04
R-04	ParsedDocument, ContentView.readDocument/loadCurrentEntry; BOM removal (v0.14)	T-30, T-39, T-52
R-05	FileWatcher, ContentView watcher hookup	T-29
R-06	persistScrollPosition, Database.scroll_positions; removed-row exception (v0.14)	T-28, T-55

Spec id	Where realized	Verified by
R-07	MarkdownUI via	T-05
R-08	ThemeManager.markdownTheme	
R-08	CodeRenderer, CodeBlockChrome; diff	T-06, T-58
R-09	exception (v0.14)	
R-09	MermaidCodeBlockChrome,	T-21, T-54
	MDVMermaidDiagramView; native-only style	
	menu/export (v0.14)	
R-10	MDVMermaidPipeline,	T-13, T-15, T-54
	MermaidFallbackView; C-06.4 dispatch	
	(v0.14)	
R-11	MDVMermaidDiagramView.displayWidth,	T-18
	raster cache; native-only scope (v0.14.1)	
R-12	MathMarkdown, math image providers/views	T-07
R-13	heading math scales,	T-08, T-11
	MDVTheme.headingSizeEms	
R-14	MathImageCache, MathSymbols	T-07
R-15	Mermaid math substitution/raster	T-17
	composition	
R-16	ImageProviders, ImageLoading	T-09, T-41, T-59
	(RemoteImageLoader, C-16; ImageDecoding,	
	K-14), View menu toggle; the inline-image	
	clause (v0.14)	
R-17	smartenMarkdown, ContentView.blockView	T-10
	ordering	
R-18	per-window stacks, snapshot	T-22, T-25, T-27, T-28, T-40
	push/apply/drop paths	
R-19	DocumentSession.handleLink (resolve,	T-22, T-49
	then classify; fragments per C-11 and C-19)	
R-20	HistoryManager, sidebar views	T-25, T-31
R-21	inspector, TOC parsing/views	T-08, T-31
R-22	selection, copySection, heading interaction	T-30
R-23	editor picker and launcher	T-38
R-24	find state, match	T-23, T-39, T-52, T-60
	counting/highlighting/routing; C-09.1	
	typography and the hidden-header clause	
	(v0.14)	
R-25	Database.search, global-search UI	T-24
R-26	index/reindex/prune/remove lifecycle; Close	T-24, T-25, T-55
	All (v0.14)	
R-27	BookmarksManager, title/anchor/menu	T-22, T-26, T-48, T-52
	paths; row context menu reordering; block-0	
	anchor/title exception (R-44) (v0.14)	
R-28	PlaceholderStore (with the R-27 title),	T-27, T-44, T-48, T-52
	DocumentSession.setPlaceholder/jumpToPlaceholder/clearPlaceholder,	
	PlaceholderRow and BookmarksPaneModel	
	(first row, 0, Clear Placeholder);	
	hidden-header exclusion (v0.14)	
R-29	ThemeManager, toolbar picker	T-12, T-42
R-30	font-scale step/HUD and scaled renderers	T-11, T-42
R-31	HelpManager.openHelp	T-38
R-32	C-04 @AppStorage keys	T-42
R-33	bin/mdv6	T-03

Spec id	Where realized	Verified by
R-34	Makefile (check-version exact-tag gate, dist chain), build.sh; mermaid.js fetch (v0.14)	T-01, T-02, T-43, T-54
R-35	diagnostic call sites and absence of content logging; WebKit-process clause (v0.14)	T-36, T-54
R-36	ContentLimits before every parser; fallback views	T-13, T-41
R-37	Tests/mdv6Tests, Tests/mdv6RenderTests, .github/workflows/build.yml	swift test, CI
R-38	mdv6/Grammars/swift, mdv6/Grammars/sql, CodeLanguage	T-37
R-39	tools/render-harness, DocumentRenderer, test-docs/mermaid, test-docs/render-cases.json	T-13, T-17, T-19
R-40	DocumentSession.restoreOnLaunch/coldStart, AppModel.startup/register, non-restorable windows (E-30)	T-28
R-41	ContentLimits checks in DocumentSession.readDocument, MDVMermaidPipeline.prepare, MathImageCache.typeset, ImageLoading	T-41
R-42	SidebarViews (chrome views), ChromeModel (rules), DocumentRootView (toolbar, title, strip), ArticleView (rhythm, stripe, find button)	T-44, T-45, T-46, T-47, T-48
R-43	mdv6/Grammars/{cpp,json,lua,opencl,perl,markdown,markdown-inline}, mdv6/Queries/*-highlights.scm, CodeLanguage, CodeRenderer.tsLanguage (embedded inline pass)	T-51
R-44	Frontmatter (frontmatterSpan, frontmatterRows), FrontmatterTableView, ParsedDocument rule 9, View-menu toggle	T-52
R-45	PrintController (pre-pass, block PDFs, container, pagination), File · Print...; <i>verification pending</i> : the print panel and Save as PDF unobserved (locked console, W15); the pipeline is T-53's scripted half	T-53
R-46	MermaidDispatch (C-06.4), MermaidWebView/MermaidWebRenderer (C-06.5), tools/fetch-mermaid.sh (fetch and digest check); <i>verification pending</i> : right-click menu, wheel over a diagram and VoiceOver unobserved on screen (W15); testHostIntegration holds them at unit level	T-13, T-21, T-54
R-47	DocumentSession.closeFile/closeAll, File-menu items, Close All confirmation	T-55
R-48	DocumentSession.stepFile, Navigate-menu items, key monitor	T-56

Spec id	Where realized	Verified by
R-49	ScrollKeyMonitor, EnclosingScrollViewAccessor in DocumentRootView	T-57
R-50	DiffHighlighter (C-05.1), CodePalette diff fields, CodeRenderer dispatch	T-58
R-51	RawHTMLImages (C-22.1), HTMLImageSpec, ImageProviders (block and inline)	T-59
C-01	Info.plist, entitlements, build.sh; mermaid.min.js resource (v0.14)	T-01, T-54
C-02	ParsedDocument.parseBlocks/parseTOC, blockLines/lineCount (rule 8); rule 9 (v0.14)	T-07, T-30, T-39, T-49, T-50, T-52
C-03	Database.search, FTSQuery (rank, path tie-breaks)	T-24
C-04	@AppStorage declarations and invalid-value fallbacks; mdv6_show_frontmatter (v0.14)	T-11, T-42, T-52
C-05	language resolution, highlighting, code cache; C-05.1 (v0.14)	T-06, T-11, T-51, T-58
C-06	MDVMermaidPipeline, diagram theme/repairs; C-06.4/C-06.5 (v0.14); C-06.1 rule 0 (v0.14)	T-13..T-20, T-54
C-07	MathSpec, MathMarkdown, MathSymbols, MathImageCache	T-07, T-08, T-17
C-08	Anchors (bookmarkFingerprint, resolveBookmarkAnchor), Database bookmarks/scroll tables	T-26, T-28
C-09	MDVTheme, ThemeManager.markdownTheme; C-09.1 (v0.14); declared fields (v0.14.1)	T-08, T-10, T-12, T-60
C-10	SmartTypography.swift	T-10
C-11	headingSlug	T-22
C-12	sectionRange, copySection, stripInlineMarkdown; step (0) (v0.14)	T-08, T-22, T-30, T-59
C-13	build.sh; mermaid.js fetch and digest (v0.14)	T-01, T-54
C-14	fallback views, placeholders, beeps, alerts	T-07, T-09, T-13, T-26, T-38, T-41
C-15	HistoryEntry, HistoryManager.save/load	T-25
C-16	ImageLoading.RemoteImageLoader (ephemeral session, header-free, redirect/size/time limits)	T-41
C-17	tools/render-harness/Sources/render-harness/main.swift, DocumentRenderer, RenderMetrics.PixelCompare, test-docs/render-cases.json; web status and --print-pdf (v0.14)	T-13, T-17, T-19, T-43, T-45, T-46, T-53
C-18	SidebarViews (C-18.3–C-18.9), DocumentRootView (C-18.1, C-18.2), ArticleView (C-18.7, C-18.10), ChromeModel (metrics and rules), reference/*.png	T-44, T-47, T-48

Spec id	Where realized	Verified by
C-19	<code>LineCitation.parse/resolve</code> (grammar, range normalisation, line→block), <code>DocumentSession.handleLink</code> (dispatch, scroll, flash)	T-49, T-50
C-20	see R-44	T-52
C-21	see R-45; <code>render-harness --print-pdf</code>	T-53
C-22	see R-51	T-59
I-001	pure render inputs; <code>RemoteImageLoader</code> the only network path; web path and print (v0.14)	T-09, T-13, T-41, T-53, T-54
I-002	<code>ContentLimits</code> before every parser; fallback per renderer	T-13, T-41
I-003	<code>Diagnostics</code> (R-35), <code>RemoteImageLoader</code> (URL-only disclosure); WebKit clause (v0.14)	T-36, T-41, T-54
I-004	cached <code>ParsedDocument</code> block split	T-30
I-005	shared display/raster size function	T-18
I-006	<code>Database</code> connection flags/pragmas	T-33
I-007	whole-row/upsert persistence writes	T-33
I-008	bitmap-backed math rendering	T-32
I-009	pixel-snapped Mermaid math and §7.1 metric	T-17
I-010	<code>TOCHeading.slugText</code> and heading math handling	T-08, T-22
I-011	<code>Vendor/SwiftMath/README.md</code> patch inventory	T-34
I-012	smart-typography exclusions and ordering	T-10
I-013	<code>HistoryManager.add/select</code> ordering	T-25
I-014	<code>ArticleBlockView.blockInset</code> (<code>max(bottom, top)</code>), <code>RhythmMetric</code> , <code>DocumentRenderer.Options.singleView</code>	T-45
I-015	one chrome file (<code>SidebarViews</code>) over theme-independent <code>ChromeModel</code> rules	T-44
I-016	<code>Vendor/MarkdownUI</code> and its <code>README.md</code>	T-62
I-017	<code>PrintController</code> (fixed print theme and scale)	T-53
I-018	<code>ParsedDocument</code> rule 9; the toggle is view-only	T-52
K-01	<code>Package.swift</code> , <code>Info.plist</code> , build checks	T-01
K-02	<code>Info.plist</code> , entitlements, bundle build	T-01
K-03	history/search/bookmark limits	T-24, T-25, T-26
K-04	font and panel clamps	T-11, T-31
K-05	<code>CGrammars</code> , supported-language table	T-06, T-37, T-51
K-06	watcher/timer/mtime/title constants	T-26, T-28, T-29, T-30
K-07	Mermaid view/cache and §7.2 formula; native-only scope (v0.14.1)	T-18, T-21
K-08	Mermaid/math layout constants	T-17, T-19
K-09	fingerprint and FTS tokenizer	T-24, T-26
K-10	<code>MDVTheme</code> defaults	T-11
K-11	<code>Makefile dist</code> chain	T-02, T-43
K-12	<code>build.sh</code> codesign and bundle placement	T-01
K-13	padded article-frame layout	T-18

Spec id	Where realized	Verified by
K-14	<code>ContentLimits</code>	T-41
K-15	bitmap-backed math (<code>MathImageCache.bake</code>)	T-32
K-16	<code>ChromeMetrics</code> , <code>ChromeOpacity</code> ; K-16 band in <code>RhythmAndDisplayMathTests</code>	T-44, T-45, T-48
K-17	<code>PrintController</code> constants	T-53
K-18	<code>ScrollKeyMonitor</code> constants	T-57
K-19	<code>tools/fetch-mermaid.sh</code> pin, <code>MermaidWebPage</code> constants	T-54
E-01	subgraph ownership normalization	T-14
E-02	diagram fallback; the web-path clause (v0.14)	T-13, T-54
E-03	read/decode guards and startup path	T-04, T-28, T-39
E-04	empty-directory path	T-04
E-05	broken local-link handling	T-22
E-06	<code>DocumentSession.handleLink</code> (same-document no-op; cross-file load at top)	T-22
E-07	math delimiter recognition	T-07
E-08	anchor resolution/scroll validity	T-26, T-28
E-09	missing-bookmark handling	T-26
E-10	math fallback	T-07
E-11	<code>ImageProviders</code> placeholders, <code>RemoteImageLoader</code> failures	T-09, T-41
E-12	database error paths	T-33
E-13	sequence-label spacing repairs	T-19
E-14	Mermaid colour normalization	T-15
E-15	xychart sanitization	T-16
E-16	image-only math placement	T-07
E-17	find tint/highlight exclusions	T-23
E-18	sidebar-focused find routing	T-23
E-19	reload selection behavior	T-29
E-20	independent per-window watchers	T-35
E-21	delete/transient-read watcher behavior	T-29
E-22	non-TOC heading behavior	T-22, T-30
E-23	fence/splitter deviations	T-39
E-24	empty global search	T-24
E-25	stale render cancellation	T-13
E-26	key-window routing; <i>Close All</i> exception (v0.14)	T-40, T-55
E-27	missing snapshot/placeholder targets	T-27
E-28	<code>ContentLimits.exceededMessage</code> paths	T-41
E-29	<code>DocumentSession.tocSelectedBlock</code> and its setters/clears	T-44
E-30	<code>DocumentRootView</code> (<code>WindowAccessor</code> marks each window non-restorable)	T-28
E-31	<code>LineCitation.resolve</code> (total resolution: last-block clamp, first-block fallback); #L0 slug fall-through in <code>DocumentSession.handleLink</code>	T-49, T-50
E-32	<code>frontmatterSpan</code> rejection paths	T-52
E-33	<code>PrintController</code> guards and fallbacks	T-53

Spec id	Where realized	Verified by
E-34	<code>MermaidWebView</code> failure and load-refusal paths	T-54
E-35	<code>RawHTMLImages</code> literal paths	T-59
E-36	<i>Close All</i> broadcast to every window	T-55
E-37	<code>ScrollKeys.documentOwnsKey</code>	T-57
E-38	window creation on demand (<code>AppModel.needsWindow</code> , <code>AppEnvironment.onDemand</code>) (open events, O, bookmarks, 0, ?, Dock reopen)	T-61

12. Open questions and decisions to confirm

ID	Decision	Default taken	Alternatives	Affects	Owner / status
D-01	No automated test suite exists today; the product MUST have one.	R-37 added; §9.0 names the target groups and which manual tests migrate to each; the app SHOULD be split into <code>mdv6Core</code> (library) + executable so tests can <code>@testable</code> import it; the harness and diagram corpus move into the repository (R-39).	Keep manual-only; or XCUITest-only.	R-37 , §9, R-36	owner / confirmed v0.1 (2026-09-14)
D-02	Inline math with descenders sits <code>descent</code> points above the baseline (SwiftUI <code>Text(Image)</code> has no baseline hook through MarkdownUI).	Accepted as a known limitation; documented in <code>NOTES.md</code> . v0.14 vendors MarkdownUI for another patch (D-49), so the one-line <code>.baselineOffset</code> fix is now a local change away, but it is not part of I-016 's inventory.	Fork or vendor MarkdownUI to apply <code>.baselineOffset</code> in <code>TextInlineRenderer.renderImage</code> (one-line patch).	R-12	maintainer / confirm
D-03	SwiftMath is vendored (not a package dependency) because of the resource-bundle/codesign conflict.	Vendored with four patches, one font.	Fork on GitHub and depend on the fork; ship more math fonts and expose a font choice.	C-07 , I-011 , K-12	maintainer / confirm

ID	Decision	Default taken	Alternatives	Affects	Owner / status
D-04	Mermaid diagram types the library lacks (<code>timeline</code> , <code>gantt</code> , <code>pie</code> , <code>mindmap</code> , <code>gitGraph</code> , <code>journey</code> , <code>quadrantChart</code>) show the fallback.	Superseded in v0.14 by D-46: they render through a bundled mermaid.js in a <code>WKWebView</code> (R-46), as the original has since <code>a02b2f6</code> ; the fallback remains for failures only.	Implement the simpler ones (<code>pie</code> , <code>timeline</code>) in <code>mdv6</code> on top of the library's renderer primitives; or switch library.	R-10 , R-46 , E-02	maintainer / resolved by the original (v0.14)
D-05	Sequence diagrams do not mirror actor boxes at the bottom, and message labels use the library's muted grey rather than Mermaid's black.	Library defaults kept.	Draw mirrored actors in <code>rasterize</code> ; override the label colour to foreground.	C-06.2	maintainer / confirm
D-06	<code>xychart</code> series names are dropped (legend reads <code>Line n</code>) and front-matter <code>themeCSS</code> (dash patterns, widths) is discarded.	Accept.	Draw the legend in <code>mdv6</code> from the names captured in <code>sanitize</code> .	E-15	maintainer / confirm
D-07	Parallelogram nodes render as rectangles (library has no such shape).	Rectangle.	Draw the slanted shape in <code>mdv6</code> after rendering (node <code>rects</code> are known).	C-06.1 rule 5	maintainer / confirm

ID	Decision	Default taken	Alternatives	Affects	Owner / status
D-08	State-diagram descriptions render as a single multi-line label rather than Mermaid's title compartment + divider.	Single label.	Draw the divider line in mdv6 under the first line.	C-06.1 rule 4	maintainer / confirm
D-09	Document-style Mermaid node fills use the page colour (25 % toward the code background) on light themes.	As stated.	Keep the previous grey (6 % toward foreground); make it a per-theme field.	C-06.3	maintainer / confirm
D-10	History sidebar width is not persisted (inspector width is).	Not persisted.	Persist under <code>mdv6_sidebar_width</code> for symmetry.	R-20 , C-04	maintainer / confirm
D-11	The <code>.txt</code> and <code>.mkd</code> extensions are accepted for drag-and-drop but not for link navigation or directory scans.	As built.	Unify the extension sets (C-02 uses <code>md/markdown/mdown</code> , drop uses five).	R-02 , R-03 , R-19	maintainer / confirm
D-12	The CLI symlink installed by <code>make install</code> points into the checkout (<code>bin/mdv6</code>), while the in-app installer points at <code>Contents/Resources/mdv6</code> .	Two install paths coexist.	Make <code>make install-cli</code> link to the bundled copy too.	R-33 , R-34	maintainer / confirm

ID	Decision	Default taken	Alternatives	Affects	Owner / status
D-13	Bundle version is fixed at 1.0.0 in <code>Info.plist</code> while releases are versioned by git tag.	Tag governs the artefact name only.	Stamp <code>CFBundleShortVersionString</code> from the tag in <code>build.sh</code> release.	K-02 , K-11	maintainer / confirm
D-14	“Load Remote Images” is off by default (privacy).	Off.	On by default like most viewers.	R-16	product / confirmed by README intent
D-15	Which SQL grammar backs R-38 .	DerekStride/tree-sitter-sql (dialect-agnostic, actively maintained, ships <code>highlights.scm</code>); Swift from <code>alex-pinkus/tree-sitter-swift</code> (its <code>parser.c</code> is generated — vendor the generated <code>src/</code> , ~10 MB, not <code>grammar.js</code>).	tenosqlkov/tree-sitter-sql (PostgreSQL-only); per-dialect grammars.	R-38 , K-05	maintainer / confirm
D-16	Files that are not valid UTF-8 (Latin-1 / Windows-1252 Markdown) are refused silently (E-03).	Strict UTF-8 , load aborted, window unchanged.	Decode with <code>U+FFFD</code> replacement; try UTF-8 then ISO-8859-1; show an “unreadable” notice in place.	R-04 , E-03 , T-39	maintainer / confirm
D-17	Fragment targets are only <code>#-###</code> single-line ATX headings, and duplicate slugs resolve to the first heading (no GitHub -1 suffixes).	As built (E-22 , C-11).	Collect h4–h6 and setext headings for slug purposes; generate GitHub’s numeric suffixes.	R-19 , R-21 , C-02 , C-11 , E-22	maintainer / confirm

ID	Decision	Default taken	Alternatives	Affects	Owner / status
D-18	How a reload treats an empty or undecodable file read mid-save.	A failed/undecodable read is ignored (page kept until a later readable event); a zero-byte read is re-read after 500 ms and that read is shown (R-05, E-21).	Always show what was read (as built — blanks the page); never show empty; make the window configurable.	R-05, E-21, T-29	maintainer / confirm
D-19	Multi-window command and open-event routing (R-01, E-26, T-40).	Requirement kept: only the key window acts; fixed in code at v0.7 (F-042): the target <code>NSWindow</code> rides in each notification’s <code>userInfo</code> , <code>NotificationHandlers</code> ignores notifications not addressed to its window, and <code>application(_:open:)</code> targets <code>NSApp.keyWindow</code> .	Declare <code>mdv6</code> single-window: drop <code>O</code> , or document that every window reacts to every command.	R-01, R-18, R-24, R-27, §5.1, E-20, E-26	maintainer / confirm
D-20	Heading-slug hyphen rule (C-11): GitHub-compatible or as built.	GitHub-compatible (every whitespace run $\rightarrow$ -); fixed in code at v0.7 (F-052).	Keep the as-built rule and document it precisely (“a whitespace run after a - emits nothing”).	C-11, I-010, R-19, T-22	maintainer / confirm
D-21	Find: whether the current occurrence is visually distinguished within a block, and whether <code>\$\$</code> blocks are tinted (R-24, E-17).	Current occurrence not distinguished (as built, documented); <code>\$\$</code> blocks tinted (requirement; fixed in code at v0.7, F-048).	Store the source range in <code>SearchMatch</code> and mark the current occurrence; or leave <code>\$\$</code> blocks inline-highlighted and document it.	R-24, E-17, T-23	maintainer / confirm
D-22	<code>migrate()</code> transactionality (§3.3).	Requirement kept (one transaction per migration incl. the bump); fixed in code at v0.7 (F-051).	Reword §3.3 to the as-built idempotent-statement rule.	§3.3, I-007, T-33	maintainer / confirm

ID	Decision	Default taken	Alternatives	Affects	Owner / status
D-23	Undecodable or vanished file, and what an empty file displays (R-04, E-03).	Requirement kept: decode before the history change, abort on failure, previous document kept; an empty file shows an empty article with its row selected; fixed in code at v0.7 (F-062).	Document the as-built behaviour (empty panel, row added); or decode with U+FFFD replacement (see D-16).	R-04, E-03, §3.1, T-39, T-27	maintainer / confirm
D-24	Adding vs. selecting routes (R-01), and snapshots whose row was removed (R-18).	Routes split as built: sidebar row, search hit, $\leftarrow/\rightarrow$ and delete-current-row neither reorder history nor re-index. Snapshots of a removed row MUST be dropped; fixed in code at v0.7 (F-063).	Route every selection through history.add (sidebar reorders on click, re-indexes on every selection); or let $\leftarrow$ re-add the row.	R-01, R-18, R-20, R-26, §3.1, T-24, T-25	maintainer / confirm
D-25	Index rows for files evicted by the 100-entry cap (R-26).	Requirement kept: prune on eviction and at launch; fixed in code at v0.7 (F-064).	Narrow R-26 to swipe-delete only and accept orphan rows.	R-26, I-013, K-03, T-25	maintainer / confirm
D-26	Whether fenced code blocks follow the zoom factor (R-30).	They MUST (fence text = $0.85 \times \text{base} \times \text{scale}$); fixed in code at v0.7 (F-065).	Exempt fences (fixed at $0.85 \times \text{base}$) and say so in R-30/T-11.	R-30, C-05, T-11	maintainer / confirm
D-27	Placeholder or snapshot whose file is missing (E-27).	Beep and no navigation, as E-09; fixed in code at v0.7 (F-074).	Document the as-built silent no-op for 0.	R-28, R-18, E-27, T-27	maintainer / confirm
D-28	Initial history head is unreadable at launch (F-076).	Attempt only the head, retain it, enter EMPTY , and do not scan later rows.	Remove the head; scan to first readable row; show an error page.	R-40, §3.1, E-03, T-28	maintainer / confirm
D-29	Cold-start file argument and unseen restored history (F-077).	The argument pre-empts restoration and creates no snapshot for the unseen head.	Restore the head first and make it reachable with $\leftarrow$.	R-18, R-40, T-28	maintainer / confirm

ID	Decision	Default taken	Alternatives	Affects	Owner / status
D-30	Cross-file bookmark and placeholder snapshot policy (F-078).	Neither route pushes, regardless of whether it changes files.	Push on every file change; push bookmarks but not placeholders.	R-18 , R-27 , R-28 , T-22 , T-27	maintainer / confirm
D-31	Remote-image disclosure and resource ceilings (F-079, F-080).	Isolated unauthenticated session per C-16 ; fixed K-14 byte/dimension ceilings; visible E-11 / E-28 fallback.	Use shared URL loading; permit cookies; rely on available memory; make limits configurable.	R-16 , R-36 , R-41 , C-16 , I-001..I-003 , K-14 , E-11 , E-28 , T-41	maintainer / confirm
D-32	Can VERSION bypass exact-tag release provenance (F-081)?	No; every dist derives the version only from the exact tag.	Keep the override; add a separate non-release packaging target.	R-34 , §5.3, K-11 , T-02 , T-43	maintainer / confirm
D-33	Render-harness CLI, corpus discovery, and comparison tolerance (F-082).	C-17 commands; raw .mmd plus Markdown fences; deterministic manifest; channel threshold 8 and mismatch fraction $q \leq 0.001$.	Raw files only; exact PNG equality; platform-specific goldens without tolerance.	R-39 , C-17 , T-13 , T-17 , T-19	maintainer / confirm
D-34	Local path plus fragment semantics (F-086).	Percent-decode once, load the file, suppress saved-scroll restoration, then match the first slug; unmatched remains at top.	Restore saved position; ignore cross-file fragments; hand them to the system.	R-19 , E-06 , T-22	maintainer / confirm
D-35	Durable anchor normalization (F-087).	Unicode-whitespace split, U+0020 join, locale-independent lowercase without normalization, first 80 grapheme clusters.	ASCII whitespace; NFC normalization; UTF-8 byte truncation.	C-08 , K-09 , T-26	maintainer / confirm
D-36	Equal-rank search order (F-088).	Rank, then case-insensitive path, then binary path.	Rank only; filename; history order.	C-03 , R-25 , T-24	maintainer / confirm
D-37	Zoom tie rounding (F-090).	Round the current factor to one decimal, half away from zero, then apply the step; HUD shows only the result.	Add then round; ties-to-even; show a pre-change HUD.	R-30 , C-04 , T-11	maintainer / confirm

ID	Decision	Default taken	Alternatives	Affects	Owner / status
D-38	Idle CPU acceptance metric (F-091).	K-15 : 5 s warm-up, 30 one-second samples, median $\leq 1\%$, nearest-rank p95 $\leq 3\%$.	Structural bitmap-only check; mean only; no numeric threshold.	I-008 , K-15 , T-32	maintainer / confirm
D-39	Empty in-document find (F-085).	No matches and stepping disabled; whitespace-only non-empty input remains verbatim.	Treat empty as every boundary; trim all input.	R-24 , T-23	maintainer / confirm

ID	Decision	Default taken	Alternatives	Affects	Owner / status
D-40	v0.8 scoped window chrome out (“visual design values live in <code>TYPOGRAPHY.md</code> ” and <code>TYPOGRAPHY.md</code> in turn says the sidebar “isn’t retypeset”. Two independent recreations built from v0.8 (<code>reference/RECREATION-*.png</code>) therefore implemented every behaviour row and produced panes that look nothing like the original (no headers, no active states, no badges, full paths or relative times, a labelled picker), and one of them lost the vertical rhythm and centred display math outright — because no gate ever looked at a rendered page.	Chrome structure is normative (R-42 , C-18 , I-015 , K-16) with reference screenshots checked in; rhythm and single-line display math are pinned (I-014 , C-07.1 , T-45 , T-46); a conformance report MUST carry an observed T-44/T-47 (§9.6).	Keep chrome as implementer’s choice; or ship pixel goldens of the original’s panes and diff against them (rejected: theme and font rendering make pixel diffs brittle across hosts — structure, not pixels, is what was lost).	R-42 , C-18 , I-014 , I-015 , K-16 , E-29 , T-44..T-47	maintainer / confirmed by the v0.8 outcome

ID	Decision	Default taken	Alternatives	Affects	Owner / status
D-41	The original's TOC highlight is <i>choice-driven</i> (the last clicked or landed-on row stays selected; <code>tocSelectedBlock</code> is set only by a choice) rather than tracking the viewport as the reader scrolls.	Specified as built (C-18.8, E-29).	Viewport-tracking selection (the row of the topmost visible heading), which reads better for long documents but was never in the product.	C-18.8, E-29, T-44	maintainer / confirm
D-42	Reference screenshots are normative for structure only, not for pixels.	<code>reference/*.png</code> are cited by C-18 and T-44; pixel values come from <code>TYPOGRAPHY.md/C-09</code> .	Make them pixel goldens (see D-40 alternatives).	C-18.0, I-015	maintainer / confirmed

ID	Decision	Default taken	Alternatives	Affects	Owner / status
D-43	Three affordances of the original that v0.9 omitted, supplied from its screenshots: the hovered-block stripe (the accent rule <code>TYPOGRAPHY.md</code> calls the bookmark-hover stripe), <i>Clear Placeholder</i> on the placeholder row, and the bookmark row's nine-entry context menu (go, reveal, four moves, remove).	Specified as the original shows them (C-18.7, C-18.9, K-16, T-48).	Leave the stripe out (hover already changes the pointer on headings only); reorder by drag only.	C-18.7, C-18.9, R-27, R-28, K-16, T-48	maintainer / confirmed by screenshots
D-44	Line citations (<code>#L...</code>) into the <i>paragraph</i> , not the <i>line</i> .	C-19: a citation resolves to the block containing its start line and flashes that whole block (§3.2 renders one Markdown view per block; C-07.1 rewrites block text and cmark-gfm normalises line breaks, so no line identity survives to the view tree). Line-accurate highlighting inside a block is out of scope, as is a citation whose target is not one of the R-19 Markdown extensions (C-19.5).	Fragment sub-view ranges per line; render the block as one Markdown view with a text-layout callback to locate line runs; treat <code>#L...</code> as a slug and leave it unmatched (the pre-v0.12 behaviour).	R-18, R-19, C-02 rule 8, C-18.8, C-19, E-06, E-22, E-29, E-31, T-22, T-49, T-50	maintainer / confirm

ID	Decision	Default taken	Alternatives	Affects	Owner / status
D-45	Which grammars back the R-43 languages, what a <code>metal</code> fence resolves to, and how a <code>markdown</code> fence is highlighted.	<code>tree-sitter-cpp</code> v0.23.4, <code>tree-sitter-json</code> v0.24.8, <code>tree-sitter-grammars/tree-sitter-lua</code> v0.5.0, <code>lefp/tree-sitter-openssl</code> pinned by commit (the repository has no tag), <code>tree-sitter-perl/tree-sitter-perl</code> v2.0.0 from its release tarball (the tag's tree carries no generated parser), <code>tree-sitter-grammars/tree-sitter-c</code> v0.5.3 for both the block and the inline grammar; <code>metal</code> , <code>c++</code> , <code>cc</code> resolve to the C++ grammar, because Metal Shading Language is C++14-based and the Metal grammars in the wild ship no licence file; <code>markdown</code> is the one language run as two grammars (C-05).	<code>jdc-pub/tree-sitter-Peta</code> (MIT claimed in <code>tree-sitter.json</code> , no LICENSE in the <code>tree-sitter-lua</code> <code>highlights.scm</code> , experimental), <code>save-buffer/GPotoshin</code> Metal grammars (no licence either); a query over the C++ grammar; using the C grammar for OpenCL instead of <code>tree-sitter-markdown</code> (the C grammar mis-parses <code>__global const float *);</code> <code>ikatyang/tree-sitter-markdown</code> (one grammar with inline constructs, but dormant since 2021, no shipped query and a C++ scanner); the block grammar alone (headings and markers coloured, every emphasis, code span and link plain).	Peta , K-05 , C-05	maintainer / confirm

ID	Decision	Default taken	Alternatives	Affects	Owner / status
D-46	Whether to break the <i>no WebView</i> principle for the Mermaid types the native library lacks, and how far to trust the web view.	Break it once, bounded (R-46 , C-06.4 , C-06.5): one hash-pinned mermaid.js 11.4.1, <code>securityLevel: 'strict'</code> , no base URL, the page's own background (no private transparency API), and — tighter than the original , which relied on <code>strict</code> and a nil base URL alone — every navigation and subresource load beyond the generated page refused, so I-001 's no-network rule and I-003 's disclosure rule still hold. The native path keeps the six families it speaks; a natively dispatched source that fails is not retried on the web.	Keep the fallback (D-04 's old default); bundle a JavaScript engine without WebKit (<code>JavaScriptCore</code> has no DOM, and mermaid needs one); render web diagrams to a cached image once and drop the live view.	Principle, §0, R-09 , R-10 , R-35 , R-46 , C-06.1 , C-06.4 , C-06.5 , C-13 , I-001 , I-003 , K-19 , E-02 , E-34 , T-36	maintainer / confirm

ID	Decision	Default taken	Alternatives	Affects	Owner / status
D-47	What a printed page looks like.	The original's choices (C-21): always high-contrast (ink-friendly, whatever the screen theme); type scaled by the ratio of the paper column to the theme's screen column ($s_p \approx 0.586$ on Letter), margins scaled with it, screen zoom ignored; vector text, vector formulas (standalone and inline; the overlay method is informative), diagrams vector from PDFs; breaks between blocks; the s_p formula kept although it yields ~12% more characters per line than the widest screen line (F-170). A formula SwiftMath rejects prints its E-10 fallback as the original does (possibly a bitmap), and T-53 exempts it (F-176). Departure (F-160) : printed blocks keep the screen's inter-block rhythm scaled by s_p , not the original's flat $12 s_p$ gap, which drops heading margins.	Print in the screen theme; print at screen point size (~80 characters per line, paragraphs break differently from the window); honour the zoom; rasterise each page.	R-45 , C-21 , I-017 , K-17	maintainer / confirm

ID	Decision	Default taken	Alternatives	Affects	Owner / status
D-48	Frontmatter recognition and display.	Narrow recognition (byte-0 fence after an optional BOM, exact closers, a YAML shape guard, no TOML guard, and — departing from the original (F-150) — a candidate with zero rows is not a header), a display-only row reduction that never rejects, a properties table styled as a GFM table, default on , and hiding is view-only (C-20, I-018).	Parse with a real YAML/TOML parser and show typed values; recognise only YAML; default off (show the file as written); drop the header from the block list when hidden (renumbers every block).	R-44, C-02 rule 9, C-04, C-20, I-018, E-32	maintainer / confirm
D-49	MarkdownUI as a package dependency or vendored.	Vendored at Vendor/MarkdownUI (v2.4.1, Sources/only) with the resolved-inline-images hook print needs (C-21.4) and two warning-only fixes, inventoried in its README (I-016, T-62); swift-cmark and NetworkImage stay package dependencies.	Keep the package and print inline formulas as bitmaps baked into the paragraph text (the view tree would have to be rebuilt without MarkdownUI for print); fork on GitHub and depend on the fork.	§10, I-016, C-21.4, D-02	maintainer / confirm
D-50	What W closes, and what <i>Close All</i> does to other windows.	W closes the file (the history row, like a tab), W the window, W everything after a confirmation (R-47). <i>Close All</i> puts every window into EMPTY (E-36) — the original acts only in the key window, leaving another window showing a file that no longer has a history row, which R-18's snapshot rule and C-18.5's selection anatomy cannot represent.	W closes the window (the macOS default for a single-document window); <i>Close All</i> acts in the key window only (as built in the original).	R-01, R-47, §3.1, §5.1, E-26, E-36	maintainer / confirm

ID	Decision	Default taken	Alternatives	Affects	Owner / status
D-51	Whether an inline <code></code> with a remote <code>src</code> is gated.	Gated exactly like every other remote image (R-16, C-16, C-22.2). The original's inline provider loads a rewritten tag's <code>src</code> with <code>UIImage(contentsOf:)</code> , which fetches an <code>https</code> URL synchronously and outside C-16 even with <i>Load Remote Images</i> off — this specification does not carry that behaviour.	Carry it as built (a privacy regression against D-14 and D-31).	R-16, R-51, C-16, C-22, I-003	maintainer / confirm
D-52	How the offline harness treats web-dispatched diagrams, and how print is verified.	The harness reports a web-dispatched case as <code>web</code> without rendering it (it hosts no WebKit, and T-13 must stay deterministic and offline); <code>--print-pdf</code> runs the C-21 pipeline to a file and reports block placements, making T-53's vector, pagination and zoom-independence claims scriptable. The original verified print with a temporary environment-variable self-test (<code>MDV_PRINT_TEST</code> , marked for deletion); that hook is not specified.	Host WebKit in the harness and golden web diagrams (brittle across WebKit versions); verify print by eye only.	C-17, T-13, T-53, T-54	maintainer / confirm

ID	Decision	Default taken	Alternatives	Affects	Owner / status
D-53	Next / Previous File, , and keyboard scrolling as specified.	Stepping is a selecting route that pushes a back snapshot like a sidebar click and clamps at the ends (R-48); / are a second binding through a key monitor, since a menu item carries one key equivalent; the document takes scroll keys only when nothing, the window, or a read-only text view has focus, with the K-18 steps (Safari's 7/8 page) (R-49).	Wrap at the ends; step in most-recent order without a snapshot; take scroll keys from any focus.	R-01, R-18, R-48, R-49, K-18, E-37	maintainer / confirm

ID	Decision	Default taken	Alternatives	Affects	Owner / status
D-54	The hovered-block stripe on fenced code and Mermaid blocks. While capturing the §9.7 references (v0.14.2), the original at 68aa008 was seen drawing the 3 pt accent stripe along a hovered <code>``diff</code> block and a hovered <code>``mermaid</code> (gantt) block, beside their hover toolbars; C-18.7 (from D-43, read from the original's earlier screenshots) says fences show no stripe.	Keep C-18.7 as written (no stripe on fences: their own hover chrome already marks them); the §9.7 references were re-captured with the pointer outside the window so they do not contradict it.	Follow the current original: draw the stripe on every hovered block, fences included, and drop T-48's <i>no stripe</i> clause.	C-18.7, K-16, T-48	maintainer / confirm

Revision history

- *v0.14.8 (2026-09-27): §11 and Status** follow the built tree — the v0.14 rows realised by W10–W15, the C-19/E-31/T-49/T-50 markers (stale since W8, e61c60f) removed, *verification pending* on R-01 (open -g activation), R-45 (the print panel) and R-46 (right-click, wheel, VoiceOver on a web diagram), which the W15 observed pass could not see at a locked console; file names corrected to the build's (tools/fetch-mermaid.sh, EnclosingScrollViewAccessor, ScrollKeys.documentOwnsKey). No requirement changed; no id added or renumbered.*
- *v0.14.7 (2026-09-27): F-184, found by the W14 build — R-45 and T-53 (2) required that no image object lie inside an inline formula's rectangle, but SwiftUI rasterizes every image placed in a line of text (measured: even an empty Image(size:) becomes a bitmap), so any ImageRenderer layout must leave an image where an inline formula sits. The vector formula is drawn over a fully transparent 1 × 1-pixel spacer; R-45 and T-53 (2) now except that spacer. No id added or renumbered.*
- *v0.14.6 (2026-09-27): F-183, found by the W13 build — C-13 put the fetched mermaid.min.js at*

Raw HTML images

GitHub sanitises HTML and renders tags, so READMEs reach for them for what Markdown cannot express — most often a sized header image. mdv has no HTML renderer, so RawHTMLImages rewrites each tag into an image reference its own providers understand, carrying the size the tag asked for. Everything else in the HTML is left alone and keeps printing as text.

A sized header, the README pattern



Width, height, and neither

Width only, so the height follows the aspect:



Both, which fit inside the pair without stretching:



A lone height, deriving the width:



No attributes at all: the image's own size, which the column may shrink.



Inline, in the middle of a sentence



This paragraph has one in it — — and the image sits on the text's baseline: its line grows to the image's height, and no text wraps around it, both on screen and in print.

A plain Markdown image inline works the same way:  sits in this sentence, and prints.

Quotes



Single quotes work:



Bare attribute values work too:

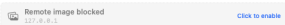
Failures and non-images

As a block of its own, a missing file shows its alt text:

missing file

Inline, a missing file is left out of the line, so these parentheses stay empty: ().

Remote sources go through the same gate as any other remote image. This one is served locally — run `python3 -m http.server 8765 in test-docs/assets/` — so the test never depends on a third-party URL. With View → Load Remote Images off it shows the blocked placeholder; with it on and the server running it loads; with it on and the server stopped it shows the failure placeholder:



What must stay literal

A fenced block keeps its HTML as HTML:

```

```

Inline code keeps it too: ``.

An `<a>` tag is not an image tag and renders as text: `<a href="#">a link</a>`.

Figure 8: Reference — the original, raw-html-images.md up to its departures section (reference/ORIGINAL-RAW-HTML-IMAGES.png) 100

Math

LaTeX math via $\$$ (inline) and $\$\$$ (display), typeset natively with Latin Modern Math. No WebView, no MathJax.

Inline

Einstein said $E = mc^2$, Pythagoras said $a^2 + b^2 = c^2$, and the Greeks said $\alpha, \beta, \gamma, \dots, \omega$. Subscripts like x_i and a_{n+1} ; superscripts like 2^{10} ; both in π^2 .

Functions and operators: $\sin^2 \theta + \cos^2 \theta = 1$, $\log_2 n$, $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$, $\nabla \cdot \vec{F}$.

Sets and logic: $x \in \mathbb{R}$, $A \subset B$, $\forall \epsilon > 0 \exists \delta > 0$, $\neg(p \wedge q) \Leftrightarrow \neg p \vee \neg q$.

Inline fractions and roots sit a touch high — a known limitation, see the notes at the bottom: $\frac{1}{2}$, $\sqrt{2}$, $\sqrt[3]{x}$.

Display

$$\int_0^\infty e^{-x^2} dx = \frac{\sqrt{\pi}}{2}$$
$$\sum_{n=1}^\infty \frac{1}{n^2} = \frac{\pi^2}{6}$$

Maxwell, in a single line: $\nabla \times \mathbf{B} - \frac{1}{c^2} \frac{\partial \mathbf{E}}{\partial t} = \mu_0 \mathbf{J}$

A blank line inside a display block is fine:

$$f(x) = \begin{cases} 1 & \text{if } x \in \mathbb{Q} \\ 0 & \text{otherwise} \end{cases}$$

Matrices:

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} ax + by \\ cx + dy \end{bmatrix}$$

Aligned equations:

$$\begin{aligned} (a + b)^2 &= (a + b)(a + b) \\ &= a^2 + 2ab + b^2 \end{aligned}$$

In other blocks

- A list item with $\alpha + \beta$
- Display math inside a list item:

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

- Nested list with $h\omega$

1. $\int f d\mu$
2. $\oint \mathbf{F} \cdot d\mathbf{r}$

Blockquote with $\Delta S \geq 0$, the second law.

Symbol	Meaning
π	circle constant
e	Euler's number
ϕ	$\frac{1+\sqrt{5}}{2}$

Heading with Σ in it (math scales with the heading)

π at h2 size, $\frac{a}{b}$ too

Beyond SwiftMath's built-ins

Symbols mdv registers itself: $a \geq b$, $a \leq b$, $\prec$, $\succ$, $p \implies q$, $\iint_D f dA$, $x_1, \dots, x_n$, $\emptyset$, $\square$, $\surd$, $\hookrightarrow$, $\nabla^\pm$, $\vec{A}$, $\leq$, $\#$, $\cdot$.

Commands rewritten into ones the parser knows: $\arg\max_x f(x)$, $a \bmod b$, $a \equiv b \pmod n$, $\frac{1}{2}$, $\mathbf{v}$, $x := y$, $a \neq b$, size hints dropped in $(p = h/N)$ and $\{\frac{1}{2}\}$, and $\text{align/equation/multline}$ environments:

$$\begin{aligned} \nabla \cdot \mathbf{E} &= \rho/\epsilon_0 \\ \nabla \cdot \mathbf{B} &= 0 \end{aligned}$$

Boxed: $\boxed{x}$, $\boxed{E = mc^2}$, and in display:

$$M = P \cdot \frac{r(1+r)^n}{(1+r)^n - 1}$$

Still unsupported (shows the source): $\underbrace{a+b}_c$, $\stackrel{a}{\sim} b$, $\substack{a \\ b}$, $\mathbb{A}$.

Things that must NOT become math

- Prices: \$5 and \$10, or a range of \$5-\$10, or \$100/month.
- Escaped dollars: $\$x$ stays literal.
- Inline code: $\$HOME$, $\$x^25$, echo $\$PATH$.
- A fenced block:

```
bash
echo "$HOME costs $5"
export X=$((1 + 1))
```

- An empty pair $\$$ on its own: $\$$

Errors

An unbalanced brace shows the source instead of vanishing: $\frac{a}{}$ and a bogus command in display:

Invalid command $\not$ otacommand
 $\not$ otacommand{x}

Inside Mermaid

Node labels that are a single $\$$ span are typeset; text mixed with math, and edge labels, get the Unicode approximation.



Figure 9: Reference — the original’s print pipeline, `math.md` at the Letter column, before pagination (reference/ORIGINAL-PRINT-MATH.png) 101

`mdv6/mermaid.min.js` and made it a SwiftPM resource, but the file is not committed, and SwiftPM fails on an absent declared resource (and warns on an absent excluded one), so a clean checkout's `swift test` (R-37) could not build. The script and its licence move to `Vendor/mermaid/` (outside every target), `tools/fetch-mermaid.sh` carries the fetch and the digest check, and the application reads the script flat from `Resources/`. No id added or renumbered.

- v0.14.5 (2026-09-27): F-182, found by the W11 build — C-09.1 gave h1/h2 a bottom padding of $0.3\times$ the heading size whether or not the rule is drawn, while the article's own heading pads $0.3\times$ the **body** size and only above a rule; a find match therefore grew an h2 by 7 pt and an h1 by 4 pt under **high-contrast**, contradicting C-09.1's purpose and T-60. The table now states the article heading's padding. No id added or renumbered.
- v0.14.4 (2026-09-27): ninth review applied in full (`SPEC_REVIEW_REPORT.md`, F-173...F-181). P1: F-173 — C-22.2's inline placement excludes math spans, which follow R-12 and D-02; F-174 — R-27, R-28 and C-18.9 carry R-44's block-0 rules (a hidden header is never the anchor; a block-0 bookmark is titled `Frontmatter`), and a document that is only a hidden header makes D and 0 beep (T-52); F-175 — T-53 (1) runs on `rhythm.md`, whose plain prose needs no GFM normalisation; F-176 — T-53 (2) covers the spans SwiftMath accepts, (2b) requires a rejected span's E-10 fallback to be printed, and R-45 and D-47 say a rejected formula may print as a bitmap. P2: F-177 — C-21.2's printed gap is taken under the print theme, and gaps for pairs I-014 does not pin are implementation freedom; F-178 — an on-demand window is created only once the action has a file to load (R-01, E-38, T-61); F-179 — the verbatim header view includes the fence lines (R-24); F-180 — T-59 joins §9.6's observed set, and §9.7 no longer cites a raster as evidence of vector output; F-181 — the front-matter Status* line now states only the current state (per-version narrative stays here). No id added or renumbered.*
- v0.14.3 (2026-09-27): `raw-html-images.md`, rendered by the original, showed that T-59 asserted `alt` text for a missing file whose tag is inline, where C-22.2 (and the original) draw nothing, and that the fixture's remote `<img>` pointed at a URL that now returns HTTP 404. The fixture now has a block-level missing case (shows `alt`) and an inline one (empty parentheses), serves its remote image from `http://127.0.0.1:8765/local.png` (`python3 -m http.server 8765` in `test-docs/assets/`), and ends with a section of the cases where this specification departs from the original (`alt="a > b"`, an inline remote `<img>`, `<img>` in a heading). C-22.2 pins inline placement: on the baseline, the line grows, nothing wraps. T-59 rewritten to match. `reference/ORIGINAL-RAW-HTML-IMAGES.png` added (§9.7). No id added or renumbered.
- v0.14.2 (2026-09-27): the §9.7 reference images captured from the original's build at 68aa008 — `ORIGINAL-FRONTMATTER.png`, `ORIGINAL-GANTT.png` and `ORIGINAL-DIFF.png` as $2\times$ window captures with the pointer outside the window, and `ORIGINAL-PRINT-MATH.png` from the original's `MDV_PRINT_TEST` self-test, rasterised at 144 ppi. The self-test writes the composited block column as one unpaginated page, so §9.7 now states that this image is normative for the printed blocks and not for pagination or rhythm (it shows the flat gap F-160 departs from). `mdv6/mermaid.LICENSE.txt` added (F-165), and C-13 names its sources. D-54 records a discrepancy seen during capture: the original draws the hover stripe on fences, which C-18.7 forbids; C-18.7 is kept pending confirmation. No id renumbered.
- v0.14.1 (2026-09-27): eighth review applied in full (`SPEC_REVIEW_REPORT.md`, F-149...F-172). P0: F-149 — with no document window, an open event, O, a bookmark, 0 or ? creates one window and a Dock click creates one that follows R-40 (R-01, §3.1, new E-38, T-61); F-151 — the web view shows the R-09 block menu, never WebKit's, passes wheel scrolling to the article, never becomes first responder and allows no selection (C-06.5, R-09, T-54). P1: F-153 — R-08 excepts `diff/patch`; F-154 — R-06 and §3.1 **LOADING** persist nothing for a file whose row is being removed; F-155 — E-26 excepts Close All, and E-36 cancels in-flight loads first; F-157 — a blocked inline remote image draws a non-clickable Remote image blocked* text image (R-16, C-22.2); F-156 — R-11 and K-07 bind only the native path, and C-06.5 states the web width rule; F-158 — a hidden header takes no space, is never the D/ 0 anchor, flashes nothing; a shown one takes the stripe, is selectable and is followed by `paragraphBottomSpacing`; a block-0 bookmark is titled `Frontmatter` (R-44); F-159 — a header

with a match always takes the verbatim inline path (R-24); F-150 — a zero-row candidate is not a header (C-20.1, E-32, D-48); F-152 — C-06.1 rule 0 strips the preamble C-06.4 skips; F-160 — printed blocks keep the screen’s rhythm scaled by s_p (C-21.2, K-17, D-47); F-161 — every inline formula must print as vector, the placeholder method is informative, and an unplaced block is an E-33 case (R-45, C-21.4); F-162 — `--print-pdf` accepts `.mmd` and reports per-block kind, images and formula rects, and T-53 is rewritten on that oracle; F-163 — T-52, T-53 (manual), T-54 and T-58 join §9.6’s observed set, and four reference images of the original are specified, not yet captured (§9.7). P2: F-164 — mermaid `logLevel: 'fatal'`, non-inspectable, and R-35/T-36 cover WebKit’s content process; F-165 — `mermaid.LICENSE.txt` committed and bundled (C-01, C-13); F-166 — R-04 strips one leading BOM, and rule 9’s compatibility claim is qualified; F-167 — C-09 declares the five fields C-09.1 uses; F-168 — T-42 covers `mdv6_show_frontmatter`, and the C-17 manifest’s `expect` names `web`; F-169 — D-46 and D-50 *Affects* extended; F-170 — C-21.1’s rationale corrected (the formula is kept) and R-45’s “empty” means no blocks; F-171 — a quoted `>` does not end an `<img>` tag, and C-12 step (0) removes `<img>` tags; F-172 — T-53’s no-flash clause is conditional on the PDF route. New id: E-38. No id renumbered.*

- *v0.14 (2026-09-27): the original’s features after the v0.7 commit, read from its history through 68aa008, specified for this rebuild. New: R-44 / C-20 / C-02 rule 9 / I-018 / E-32 — frontmatter as block 0 with a properties table and View · Show Frontmatter (`mdv6_show_frontmatter`, C-04); R-45 / §2.7 / C-21 / I-017 / K-17 / E-33 — printing (P) with a paper-derived type scale, vector text, formulas and diagrams, and block-boundary pagination; R-46 / C-06.4 / C-06.5 / K-19 / E-34 — Mermaid types the native library lacks drawn by a bundled `mermaid.js` 11.4.1 in a `WKWebView`, with the principle, §0’s trust boundary, I-001 and I-003 amended and E-02 rewritten (D-04 superseded by D-46); R-47 / E-36 — Close File, Close Window, Close All; R-48 — Next / Previous File and ; R-49 / K-18 / E-37 — keyboard scrolling; R-50 / C-05.1 — diff tinting with per-theme diff colours; R-51 / C-22 / E-35 — raw `<img>` tags; C-09.1 — find-highlight typography (R-24); I-016 — vendored MarkdownUI. Amended: R-01 (frontmost document window; background opens), R-09 (native-only style menu and export), R-10, R-16 (inline images), R-24, R-26, R-34 and C-13 (`mermaid.js` fetch and SHA-256 gate), C-01, C-14, C-17 (*web status*, `--print-pdf`), §3.1, §3.2, §5.1, §5.3, §10, T-13, T-21. §9.7 adds T-52...T-61, §9.5 T-62. D-02 and D-04 updated; D-46...D-53 added; D-50 and D-51 depart from the original on purpose (Close All empties every window; an inline remote `<img>` is gated). Every new row is not yet realised* here. No id renumbered.**
- *v0.13.2 (2026-09-25): F-148, reported against a downstream document rendered in `mdv6 (gpu-quicksort-revisited/SPEC.md` §7, K-06): `\lVert/\rVert` name *amsmath*’s open/close variants of *U+2016* — commands *KaTeX*, *MathJax* and *GitHub* all accept — but the vendored *SwiftMath*’s delimiter and symbol tables know only `\Vert/\vert`, so every norm in that document took the E-10 fallback and showed its source. C-07.2 (b) gains `\lVert/\rVert` → `\Vert` and `\lvert/\rvert` → `\vert`, applied after `\not=` → `\neq` and before the size-command removal (statement 7 of the order), so the table is fourteen entries and `\left\lVert ... \right\rVert` rewrites to a delimiter *SwiftMath* accepts. T-07’s corpus carries the norm (*test-docs/math.md*), and the C-07.2 unit and render tests pin the four rewrites, the `\lVertneqq` boundary and the reported expression. No id renumbered.*
- **v0.13.1 (2026-09-24): the spec-model findings applied (F-139...F-147), from the Lean model of this specification (proof_from_spec/). Each finding was a place the document left a cell silent, a pin too loose, or a witness missing; the model’s kernel-checked witnesses are in docs/reviews/SPEC_MODEL_FINDINGS.md. P1: F-139 — C-06.1 rule 3 named 46 CSS colour names and pinned no value, so `#eee/white` could render differently in two conforming builds (the SVG 1.1 keyword table is now normative); F-140 — §3.1’s table was partial, leaving a reachable `EMPTY+delete-row` and `RELOADING+event` cell unstated (both now named, and the table is declared total); F-143 — C-12 removes trailing `#s` and nothing removed the leading ATX marker, so `TOCHHeading.text` kept `#` while T-08 expected the bare heading text (C-02 rule 7 now says the text and slugText come from the heading body); F-146 — C-07.1 pinned a span’s acceptance but not the scan that finds spans, which no §9 test carried (the scan is now stated, so T-07 covers it). P2: F-141 — `stripInlineMarkdown`’s removals had no order and the order is observable; F-142 — “uses its first line only” (rule 7) contradicted the `tocHeadings` comment’s “single-line ATX only”; F-144 — a quote’s direction was*

“chosen from the preceding character” with no mapping; F-145 — C-10 defined a GFM table by example while R-24 defined it exactly; F-147 — “drop the characters from each token” deletes them, so a**b** and **ab** collide (now documented as accepted). No id renumbered; no row removed.

- v0.13 (2026-09-19): seven more highlighted languages (R-43, D-45). New R-43: `cpp` (+ `c++/cc/metal`), `json`, `lua`, `opencl` (+ `cl`), `perl` (+ `pl`) and `markdown` (+ `md/gfm`) fences highlight from six more upstream grammars (seven vendored directories) under `mdv6/Grammars/` — `tree-sitter-cpp` v0.23.4, `tree-sitter-json` v0.24.8, `tree-sitter-grammars/tree-sitter-lua` v0.5.0, `lefp/tree-sitter-opencl` (pinned by commit; the repository has no tag), `tree-sitter-perl` v2.0.0 (release tarball) and `tree-sitter-grammars/tree-sitter-markdown` v0.5.3 (block and inline) — pinned in its README together with the files that are derived rather than verbatim. C++ also backs a `metal` fence, because Metal Shading Language is C++14-based and the Metal grammars in the wild ship no licence file; markdown is the one language run as two grammars, the block grammar’s captures followed by the inline grammar’s over each (`inline`) span (C-05, D-45). K-05, C-05, §9.0, §10 and §11 carry the additions; T-51 proves them. No id renumbered.
- v0.12.1 (2026-09-19): seventh review applied (`SPEC_REVIEW_REPORT.md`, F-126..F-138). P0: F-126 — R-18 no longer claims a back snapshot for a line-citation jump (it contradicted R-19/C-19.4); F-128 — the C-19.1 grammar is `~L([1-9][0-9]*) (?:(?:L)?([1-9][0-9]*))?$`, so #L0 is not a citation and E-06/E-31 agree; F-129 — T-49 cites #L7-L9, which its fixture actually resolves to a paragraph (line 10 is the gap case, line 12 the `past-lineCount` case). P1: F-127 — “slug fragment” (C-11) and “line citation” (C-19) named in R-19 and applied in R-18, C-18.8, E-22, E-29; F-130 — rule 8 resolution is total for a line before the first block (E-31 extended); F-131 — rule 8 owns the half-open `blockLines` convention; F-132 — C-19/E-31/T-49/T-50 marked not yet realised* and §11’s blanket sentence replaced; F-133 — C-19.5 restricts citations to the R-19 in-app extensions; F-134 — T-49’s map assertions became the scripted T-50, and T-49 joins §9.6’s observed set. P2: F-135 — K-06 names the block flash and both of its uses; F-136 — C-19 moved out of the §5.5 chrome section into §4; F-137 — C-19.3’s TOC analogy corrected and D-44’s *Affects* extended; F-138 — T-49 names its empty-document fixture. No id renumbered.
- v0.12 (2026-09-19): GitHub-style line citations. New C-19 (#L14, #L14-L22, #l14-22, reversed ranges) — grammar, range normalisation, line→block resolution, and the **block-granular flash** it produces; D-44 records why the highlight is the containing paragraph, not the line run. C-02 gains rule 8 and the `blockLines/lineCount` fields on `ParsedDocument`, so a line number is addressable at all. R-19 gains the ordered fragment kinds (line citation → C-11 slug → E-06) and states that a line-citation jump is a position move: no back snapshot on a same-document jump, no TOC selection. E-06 narrows (a citation never reaches it), E-31 added for a line that does not exist (past `lineCount`, in a split gap, or #L0); T-22 extended, T-49 added; §11 rows for C-02, C-19, R-19, E-31. No id renumbered.
- v0.11.2 (2026-09-18): F-002, found by the build’s speccheck gate — C-18 was cited by R-42, I-015 and §11 but its §5.5 heading (### 5.5 Window chrome: **C-18** ...) did not declare it; the heading is now ### C-18 Window chrome (§5.5, *normative structure*). Editorial; no id, row or requirement changed.
- v0.11.1 (2026-09-18): F-001, found by the build’s speccheck gate — the §5.4 row | **R-35** (above) | declared R-35 a second time (line 480 vs the §2.6 declaration at line 102); the cross-reference is now un-bolded. Editorial; no id, row or requirement changed.
- v0.11 (2026-09-17): sixth review applied (`SPEC_REVIEW_REPORT.md`, F-114..F-125). P0: E-30 — window-state restoration disabled, one window per launch (R-40, §3.1, E-26, T-28). P1: status markers, §9.0, §10 and the §11 matrix refreshed to the built tree (F-115); View · Show/Hide Inspector* 0 row (F-116); R-19 restated as resolve-then-classify with a T-22 clause (F-117); MDV6_SUPPORT_DIR / MDV6_DEFAULTS_SUITE in §10 and --theme in C-17 (F-118); sidebar has focus defined once in R-24 and used by E-18 / C-18.4 (F-119); C-18.1 colour-scheme rule for the title bar with a T-44 clause (F-120). P2: §9.6 names T-48 (F-121); the collapse chevron placed and made hover-only (F-122); hover fills use `text.opacity` (F-123); overlay precedence (F-124); observed tests on an isolated store (F-125). No id renumbered.*

- *v0.10.1 (2026-09-17): editorial — the Sources* line and §2’s citation note point at the original’s repository (tqbf/mdv) instead of listing its file paths, and state that this repository is a from-scratch rebuild. No row changed.**
- *v0.10 (2026-09-17): three chrome affordances of the original added from its screenshots (D-43): C-18.7 hovered-block stripe (3 pt, accent; not on fences), C-18.9 context menus — Clear Placeholder* on the placeholder row, and Go to Bookmark · Reveal in Finder · Move Up · Move Down · Move to Top · Move to Bottom · Remove Bookmark on a bookmark row; R-28 clear clause; K-16 stripe width; T-48; §11 rows extended. No id renumbered.**
- *v0.9.2 (2026-09-17): F-109 (found by the v0.9 build’s I-015 snapshot under Twilight): C-18.6 and C-18.9 pinned white* text on the accent fill, which is illegible on a theme whose accent is light (Twilight’s cream, Phosphor’s green); the rows now use the page background colour, the rule C-18.8 already states for the TOC row — identical to white on every light theme with a dark accent, so the reference images are unchanged. No id renumbered.**
- *v0.9.1 (2026-09-17): F-107 (found by the v0.9 build’s T-45 run): an ink-row gap cannot equal a layout margin within ± 2 pt — MarkdownUI’s own single view measured Sevilla 40.5/22.5/29.5/18.5 pt against margins 32/14/22/14 and Charcoal 32/18/22.5/15.5 against 26/11/18/11, the excess being the fonts’ leading. I-014, K-16 and T-45 now state the band $v \leq g \leq v + 0.6f$ and the single-view equality within ± 2 pt; the max(bottom, top) combination rule is stated explicitly. No id renumbered; no other row changed.*
- *v0.9 (2026-09-17): chrome-parity uplift after the first two spec-driven recreations (D-40). New: R-42 (window chrome), §5.5 / C-18.0–C-18.10 (window, toolbar, section headers, reveal-on-click search, history rows, search hits, article affordances, TOC rows, bookmark rows, per-block rhythm), I-014 (rhythm independent of view tree), I-015 (theme-independent chrome), K-16 (chrome metrics, rhythm tolerance), E-29 (TOC selection lifecycle), §9.6 / T-44..T-47, D-40..D-42; §5.1 Toolbar row rewritten to the five icon buttons; C-07.1 clarified for single-line own-paragraph \$\$\$...\$\$\$; R-28 now requires the placeholder to be visible as the first bookmarks-pane row (it was set/jump-only, and one recreation showed it nowhere); reference/ added with the original’s screenshots and the two recreations’. No v0.8 row was weakened; §11 marks the new rows not yet realised* for this tree.**
- *v0.8 (2026-09-14): fifth review applied in full (F-076..F-093). P0: startup-head and cold-argument state fixed (R-40/§3.1/E-03/T-28); bookmark/placeholder exceptions ordered in R-18; C-16/K-14/E-28/T-41 define remote disclosure and resource exhaustion; exact-tag release gate and positive release test added; C-17 pins the render harness, corpus, manifest, exits, and pixel metric. P1: I-013 recency, 1 pt raster bound, empty find, cross-file fragments, Unicode fingerprints, FTS tie order, complete preference test, zoom rounding, and CPU protocol pinned. P2: E-21 self-transition and math notation corrected. R-37..R-39/R-41 remain not realised; newly exposed implementation gaps are marked in §11. D-28..D-39 record the defaults awaiting confirmation.*
- *v0.7 (2026-09-14): all ten open defects* fixed in the code, in the same commit as this revision — F-042 (commands and open events addressed to the key window), F-045 (___ stripping), F-048 (\$\$ fences tinted in find), F-051 (migrate() in one transaction with rollback), F-052 (GitHub whitespace-run slug rule), F-062 (decode before any history/selection change; EMPTY keyed on the selection), F-063 (snapshots of removed rows dropped/skipped), F-064 (index pruned on eviction and at launch), F-065 (fenced code follows the zoom factor; cache key), F-074 (0/ $\leftarrow$ / $\rightarrow$ to a missing file beep). Markers removed from R-01, R-04, R-18, R-24, R-26, R-30, §3.1, §3.3, C-05, C-11, C-12, E-03, E-17, E-26, E-27, T-11, T-23, T-25, T-27, T-39, T-40 and §11; D-19..D-27 record the fix. No ids renumbered.**
- *v0.6 (2026-09-14): fourth review applied (F-062..F-075). P0: R-04/E-03 decode-before-add and the empty-file rule, open defect* (F-062, D-23); R-01 split into adding and selecting routes, §3.1 LOADING row split, R-20/R-26 reworded to “added”, R-18 snapshot-drop rule open defect (F-063, D-24). P1: R-26 eviction/launch prune open defect (F-064, D-25); R-30/C-05 fence zoom open defect (F-065, D-26); C-05 prompt-aware fence set (F-066); §7.2 column-width formula, K-13/K-10/T-18 (F-067); C-07.1 host rule, K-08, E-16 (F-068). P2: R-26 scroll-position removal (F-069); C-04 invalid-value*

paragraph, [K-06](#) index gate (F-070); [R-24](#) reload/verbatim clause (F-071); §5.3 release inputs (F-072); §5.2 launcher edge rows (F-073); [E-27](#) and [D-27](#) (F-074); front-matter commit, §3.3 “clear”, §11 [R-40](#) citation, [C-06.1](#) rule 3, [C-07.2](#) \operatornamename, [C-12](#) “TOC heading” (F-075). New ids: [E-27](#), §7.2, [D-23..D-27](#). No ids renumbered.*

- *v0.5 (2026-09-14): third review applied (F-042..F-061). P0: [R-40](#) (launch restores the history head) and the §3.1 **EMPTY/LOADING** entries (F-043); key-window routing kept as the requirement, [E-26/T-40](#) added, [R-01](#) marked open defect* (F-042, [D-19](#)). P1: [R-27](#) title rule (F-044); [C-12](#) `__` rule, open defect (F-045); [C-14](#) narrowed to document-content failures, §5.1 CLI-install row (F-046); [R-24/E-17](#) restated — count on source, highlight on rendered text, tint-by-exclusion, `$$` clause open defect (F-047, F-048, [D-21](#)); R-18 snapshot policy and R-28 placeholder anchor (F-049); §3.1 delete-row transitions, R-26 (F-050); §3.3 migration transaction open defect (F-051, [D-22](#)); C-11 GitHub hyphen rule open defect (F-052, [D-20](#)); R-01 multi-URL and R-03 first-item rules (F-053); R-22 “TOC heading block” and E-22 (F-054). P2: §5.1 enabled/beep cells (F-055); R-35 log inventory (F-056); R-05/E-21/K-06 NoDefer and transient wording (F-057); R-31/§3.3 Help overwrite (F-058); front matter, §10, C-01, §1, §5.3 drift (F-059); R-02 collation and R-26 “added to history” (F-060); C-15 moved after C-13, [D-17/D-18](#) order, revision order, `$\leq$` (F-061). No ids renumbered.**
- *v0.4 (2026-09-14): F-033 (CRLF block splitting) and F-034 (reload on a failed/transient read) fixed in the code; their open defect* markers removed from [C-02](#), [E-21](#), [R-05](#), [T-29](#), [T-39](#) and §11.**
- *v0.3 (2026-09-14): second review applied (F-032..F-041). P0: [R-22](#) rewritten to the as-built heading-click model — the block-selection model it described was removed in [c50817a](#); its traces purged from [R-05](#), §3.1, §5.1, [E-19](#), [I-004](#), [T-30](#), §11. P1: open defect* status introduced in the front matter and §11 (F-036) and applied to the CRLF line-ending rule ([C-02](#) rule 6, F-033) and the delete/transient reload rule ([R-05/E-21](#), F-034); transient-state rule and [D-18](#) (F-037). P2: [K-13](#) drag handles, blank line before §3, §9.0 wording, rule citations by name (F-038..F-041). No ids renumbered.**
- *v0.2 (2026-09-14): all findings of `SPEC_REVIEW_REPORT.md` applied. P0: F-001 (lifecycle vs [E-03](#)). P1: F-002 (Copy Without Prompts* output), F-003 (path-based watcher, [E-21](#)), F-004 (bookmark anchor and title), F-005..F-008, F-010..F-012, F-015 (interaction rules made explicit), F-009 ([C-15](#) history JSON), F-016 ([R-39](#) harness + corpus in-repo), F-017 (§7.1 ink metric). P2: F-013 ([E-22](#), [D-17](#)), F-014 (colour list enumerated), F-018 ([D-16](#)), F-019 ([C-02](#) rules 2–4, [E-23](#)), F-020..F-031 (editorial, notation, [K-13](#) column width, [E-24](#), [E-25](#), [T-38](#), [T-39](#)). No ids renumbered.**
- *v0.1 (2026-09-14): first as-built draft, covering the tree at [a6feb14](#); §3.2 diagram made vertical; [R-37](#) and §9.0 added after [D-01](#) was confirmed (automated suite is a product requirement); [R-38](#) (Swift and SQL highlighting) and [D-15](#) added.*